HJ-Gauss: A Monte-Carlo HJ Reachability Scheme

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Abstract

Backward reachable tubes (BRTs), computed by grid-based level-set solvers for Hamilton-Jacobi (HJ) partial differential equations, furnish principled safety certificates for learned controllers and planning algorithms — at an $O(M^n)$ memory cost (M grid points per n state dimensions) that forecloses systems beyond a handful of dimensions. We dismantle this bottleneck with a frozen-coefficient sampling scheme for the viscous HJ equation: a generalized Cole-Hopf-type transformation, exact for quadratic Hamiltonians and iterated as a Picard quasi-linearization otherwise, reduces the nonlinear PDE to a sequence of linear heat equations whose solutions are Gaussian heat-kernel expectations under the Feynman-Kac formula. The value function and its spatial gradient are then recovered by Monte Carlo roll-outs over Gaussian densities, yielding a grid-free algorithm whose memory scales as $O(N \cdot n)$ in the sample count N — decoupled from any per-dimension discretization. We prove a dimension-independent $O(N^{-1/2})$ finite-sample concentration bound, conditional linear convergence of the Picard iteration with an explicit contraction constant, an explicit Duhamel bound on the quasi-linearization defect separating the scheme's fixed point from the viscous solution, and robustness to Hamiltonian and terminal-cost perturbations; a one-sided thresholding corollary converts the total error budget into a conservative safety certificate with a declared abstention band. Against grid-based level-set references on pursuit-evasion games, the scheme attains relative L^2 errors of 0.03–0.20 in 14–26 seconds per 2D slice on a single CPU; it resolves a 45-dimensional fifteen-rocket game beyond the reach of any grid; and it certifies the safety of murmurations of $\mathbf{100{,}000}$ European starlings (*Sturnus vulgaris*), modeled as 4D aerial Dubins vehicles, through embarrassingly parallel per-bird evaluations of shared flock-level value functions. Our results demonstrate real scalability of HJ reachability safety verification on large-scale multi-agent systems.

1 Introduction

A central challenge in trustworthy machine learning is certifying that a learned controller, neural policy, or planning algorithm will not violate safety constraints at deployment time. This concern appears directly in safe reinforcement learning [Li et al., 2021], model-based RL with learned dynamics [Berkenkamp et al., 2017], and any pipeline where a learned model must operate in a continuous, possibly high-dimensional state space under adversarial or uncertain inputs. Backward reachable (reach-avoid) tubes, or BRTs (BRATs), are the set of all states from which a system is guaranteed to reach (avoid) a target region regardless of disturbances that may affect its behavioral evolution during deployment; when the controller must robustly counter a worst-case disturbance we write RCBRT (RCBRAT), and we use these terms consistently hereafter. They provide a principled certificate for solving *reachability* problems. Computing a BRT amounts to solving a Hamilton-Jacobi-Isaacs (HJI) partial differential equation (PDE) [Mitchell et al., 2005, Lygeros, 2004], typically under two competing inputs.

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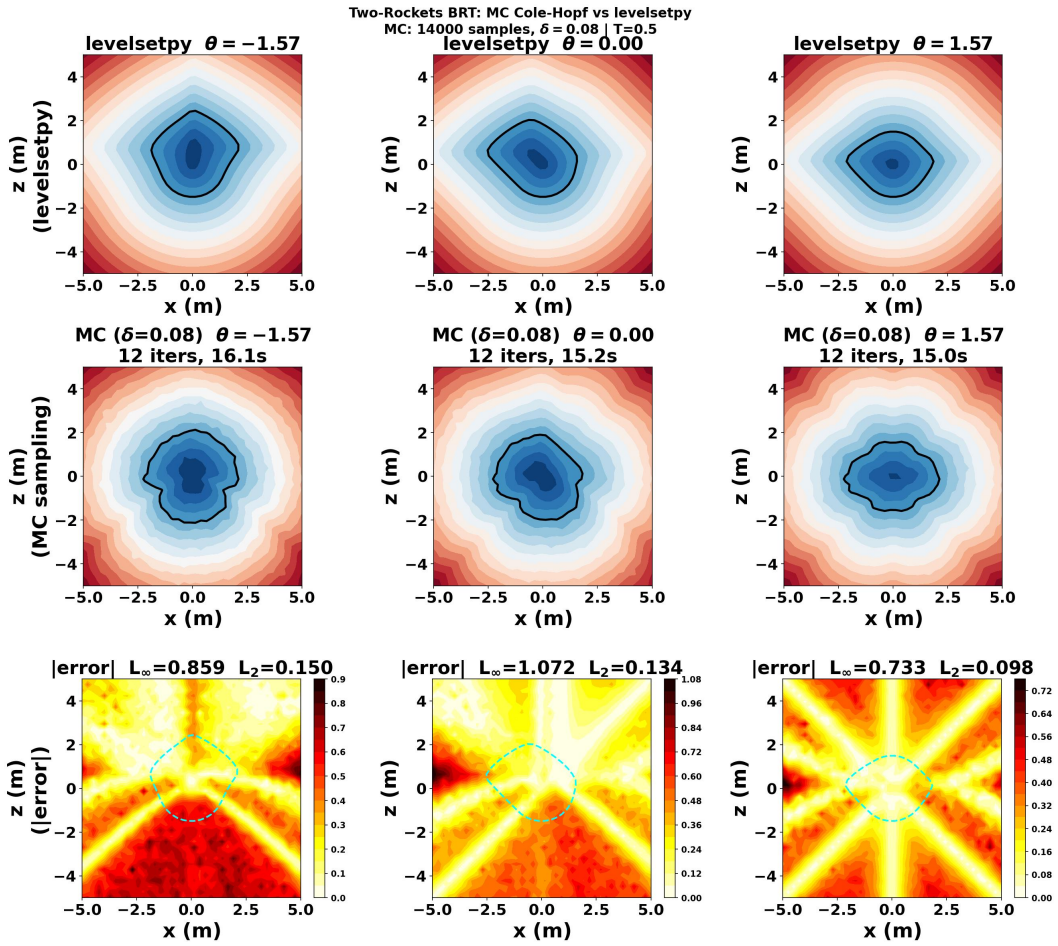


Figure 1: Rockets BRT slices: Grid-based `levelsetpy` from [Molu, 2024b] vs. Monte Carlo (ours). The bottom row shows a heat map of the pointwise errors at various relative vehicle orientations θ; errors are largest near the boundary of the usable part where $|D\boldsymbol{v}^{\delta}|$ is largest.

Central Contribution: Monte-Carlo Trajectory Optimization of Quasi-Linear BRTs

A *backward reachable tube* (BRT) is the set of initial states from which a controlled system is guaranteed to reach a target set within a given time horizon, for all adversarial inputs. Given system dynamics $\dot{\boldsymbol{x}} = f(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w})$ with controller $\boldsymbol{u}$ and disturbance $\boldsymbol{w}$, the BRT is the zero sublevel set of a value function $\boldsymbol{v}^{\delta}$ that solves the viscous HJI PDE (HJI-RCBRT-Visc). Classical grid-based solvers [Mitchell et al., 2005, Molu, 2025] discretize the state space on a grid with M points per dimension; the resulting memory cost is $O(M^n)$, which grows exponentially in the state dimension n. For $n = 6$ with $M = 100$, this already requires 10^{12} grid cells. The proposed method replaces the grid with N Monte Carlo samples of state space trajectories, reducing spatial memory to $O(N \cdot n)$.

The major bottleneck for realizing the solutions to the BRT of complex systems is computational: classical HJ solvers scale as $O(M^n)$ in memory, where M is the number of grid points per dimension and n is the state dimension. While GPU-accelerated implementations [Molu, 2024b, 2025] reduce wall-clock time, they retain the same exponential memory footprint. Self-supervised, PINN-based approaches such as DeepReach [Bansal and Tomlin, 2021] train a neural network to minimize the HJ PDE residual directly via a physics-informed loss, without requiring a reference grid solution. However, the method scales moderately in dimensions (reportedly 9D) and accuracy degrades in the limit of higher-dimensional data. We introduce a fundamentally different approach that replaces grid storage with a sampling scheme so that the memory cost is $O(N \cdot n)$ and the algorithm is grid-free.

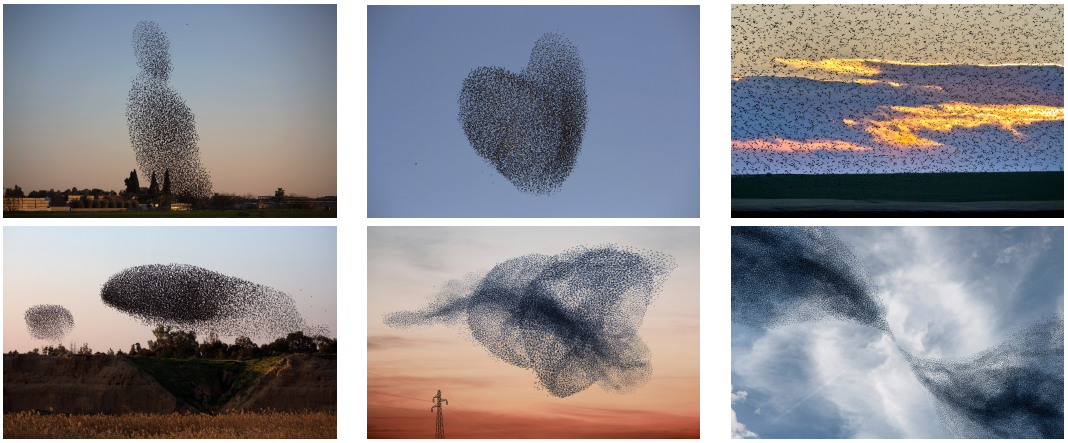


Figure 2: The scalability veracity of our formalism is evaluated on a murmuration of European starlings (*sturnus vulgaris*). From the top-left and clockwise. (i) A starlings flock rises into the air, in a dense structure (Reuters/Amir Cohen). (ii) Starlings migrating over an Israeli village (AP Photo/Oded Balilty). (iii) Starlings feeding on laid seeds in the ground in Romania. (iv) Two flocks of migrating starlings (Menahem Kahana/AFP/Getty Images). (v) A concentric conical formation of starlings (Courtesy of The Gathering Site.). (vi) Splitting and joining of a flock of starlings.

Relation to grid-free and sampling-based HJ solvers. Three research families border ours, and the boundaries deserve drawing. *Hopf- and Lax-Oleinik-formula methods* [Darbon and Osher, 2016, Chow et al., 2017, Kirchner et al., 2018] also dispense with grids, evaluating the value pointwise by convex optimization; they are exact where they apply, but the Hopf formula demands a convex (or state-independent) Hamiltonian, whereas the state-dependent, nonconvex game Hamiltonians of reachability fall outside that class — our quasi-linearization trades their exactness for this generality, at the quantified price of the defect bound in Appendix C. *System decomposition* [Chen et al., 2018] conquers dimension by splitting self-contained subsystems and is exact when the coupling structure permits; our sampler is indifferent to coupling structure and so complements decomposition precisely where it fails. *Path-integral control* [Kappen, 2005, Theodorou et al., 2010] is the stochastic sibling of our transformation: there, the same logarithmic substitution linearizes the stochastic HJB equation under a noise-control duality, with the diffusion supplied by process noise; here, the viscosity δ is an analysis parameter of a deterministic worst-case game, and the frozen coefficient $\boldsymbol{c}$ plays the duality's role. Finally, *sample-based stochastic reachability* [Summers and Lygeros, 2010, Lesser et al., 2013] certifies probabilistic reach-avoid for stochastic systems; we certify adversarial (worst-case) reachability, a different guarantee altogether.

We consider first-order nonlinear scalar HJ PDEs of the form

$$\boldsymbol{v}_t + \boldsymbol{H}(t;\boldsymbol{x}, D\boldsymbol{v}) = 0 \text{ in } \Omega \times (0,T], \quad \boldsymbol{v}(0;\boldsymbol{x}) = \boldsymbol{g}(0;\boldsymbol{x}) \text{ on } \bar{\Omega} \times \{t=0\} \tag{HJ-IVP}$$

where the state $\boldsymbol{x}$ belongs to the open set $\Omega \subseteq \mathbb{R}^n$; $\boldsymbol{v}_t$ denotes the partial time derivative of $\boldsymbol{v}(t;\boldsymbol{x})$; $\boldsymbol{H}(t;\boldsymbol{x}, D\boldsymbol{v})$ is the Hamiltonian, continuously defined on $\mathbb{R}^n$, with $D\boldsymbol{v}$ representing the spatial gradient; and $\boldsymbol{g}(\boldsymbol{x})$ being bounded and uniformly continuous (BUC) on $\mathbb{R}^n$.

Fixing a viscosity parameter $\delta > 0$, Crandall and Lions [Crandall and Lions, 1984] introduced the *vanishing viscosity* solution, $\boldsymbol{v}^\delta$, of (HJ-IVP) i.e. ,

$$\boldsymbol{v}^\delta_t + \boldsymbol{H}(t;\boldsymbol{x}, D\boldsymbol{v}^\delta) = \frac{\delta}{2}\Delta \boldsymbol{v}^\delta \text{ in } \Omega \times (0,T]; \quad \boldsymbol{v}^\delta(0;\boldsymbol{x}) = \boldsymbol{g}(0;\boldsymbol{x}) \text{ on } \bar{\Omega} \times \{t=0\} \tag{HJ-Visc}$$

such that (HJ-Visc) satisfies the uniform convergence bound

$$\sup_{t\in(0,T]} \sup_{\boldsymbol{x}\in\mathbb{R}^n} \left|\boldsymbol{v}(t;\boldsymbol{x}) - \boldsymbol{v}^\delta(t;\boldsymbol{x})\right| \leq k\sqrt{\delta} \tag{1}$$

for a constant $k > 0$. *Our key insight is that* (HJ-Visc) *can be reduced, via a generalized Cole-Hopf-type transformation, to a sequence of linear heat equations admitting explicit Gaussian heat-kernel solutions.* This yields a locally linearized PDE. The scheme is *not* an exact Cole-Hopf reduction

for general Hamiltonians; it is an iterative (Picard) approximation in which the nonlinear coefficient is frozen at each step and updated after the linear solution is found. The exact reduction holds for the quadratic case $\boldsymbol{H} = \frac{1}{2}|p|^2$, where the frozen coefficient is constant and the residual vanishes identically.

Connections to ML. Beyond reachability, the proposed scheme connects to several active ML research directions. (i) *Safe RL and policy verification*: certifying that a learned policy satisfies safety constraints requires computing BRTs for the closed-loop system; the memory bottleneck of grid-based solvers is the primary obstacle to safe RL in high-dimensional spaces [Li et al., 2021, Berkenkamp et al., 2017]. (ii) *Model-based RL*: planning algorithms that learn a dynamics model and then verify safety via HJ reachability are limited to low-dimensional state spaces [Bansal and Tomlin, 2021]. (iii) *Diffusion-based methods*: the Gaussian heat-kernel representation derived here is structurally similar to the nonconvex optimizers for retrieving global minima of differentiable objectives [Chaudhari et al., 2018, Heaton et al., 2024], which may be extended to Stein score-based diffusion generative models.

Contributions. Our contributions are as follows. (i) We derive a generalized Cole-Hopf-type transformation that reduces the nonlinear viscous HJ equation to a sequence of linear heat equations with Gaussian heat-kernel solutions, and we characterize exactly when the reduction is exact and what residual it incurs otherwise (Lemma B.1). (ii) We construct a sampling-based algorithm (Algorithm 1) with $O(N \cdot n)$ memory complexity that scales with sample count rather than discretization dimension, a viable alternative to grid-based $O(M^n)$ approaches in high dimensions. (iii) We prove (a) a finite-sample concentration bound of $O(N^{-1/2})$ for the Monte Carlo value estimator, independent of state dimensions (Theorem 2.5), (b) conditional linear convergence of the Picard iteration with explicit contraction constant (Theorem 2.12), and (c) a Duhamel bound on the quasi-linearization defect separating the iteration's fixed point from the viscous solution (Theorem C.4). (iv) We convert the total error budget into a one-sided conservative safety certificate with a declared abstention band (Corollary 2.14). (v) We validate on two-player pursuit-evasion games achieving L^2_{rel} errors of 0.03–0.20 against grid-based references in 12–15 iterations, on 45-dimensional multi-agent problems where grid-based discretization is computationally prohibitive, and on the safety certification of 100,000-bird starling murmurations (§ D.1).

A logical reading of this work consists in the following: background work and notations are in Appendix A. Section 2 derives the quasi-linearization of (HJ-Visc), states the algorithm, and establishes the theoretical guarantees. Section 3 presents numerical results and Section 4 concludes the paper. All proofs appear in Appendix B. Appendix C establishes a rigorous analysis of the error bounds, convergence rates, and robustness properties of the quasi-linearized Cole-Hopf transformation and Feynman-Kac approximation of the HJ scheme. Lastly, additional numerical experiments are provided in Appendix D.

2 Transformation of the HJ PDEs

We now transform the nonlinear viscous HJ PDE (HJ-Visc) into a linearized form and extract its solution via Gaussian heat-kernel expectations. All proofs are in Appendix B.

HJ Sign and Time Conventions

Throughout this paper we adopt the backward-reachability viscosity-solution convention,

$$\boldsymbol{v}_t + \boldsymbol{H}(t; \boldsymbol{x}, D\boldsymbol{v}) = 0, \tag{2}$$

where the Hamiltonian is defined via the zero-sum differential game formulation of Appendix A,

$$\boldsymbol{H}(t; \boldsymbol{x}, p) = \max_{\boldsymbol{u}} \min_{\boldsymbol{w}} \langle p, f(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) \rangle. \tag{3}$$

Time convention: t denotes the *backward horizon* (time-to-go); the target data is posed at $t = 0$ and the tube is grown over $t \in (0, T]$. Under this convention (HJ-Visc) is a well-posed forward parabolic problem and the heat kernel of its linearization at horizon t has covariance $\delta t\, I_n$, degenerating to the target data as $t \downarrow 0$. Negative signs appearing in the Hamiltonian expressions in Appendix A arise from the backward-time transformation [Mitchell, 2004] and the min-over-max structure of reachability propagation. This convention is standard in reachability literature [Mitchell et al., 2005] and is consistent with the viscosity solution.

2.1 Exact Reduction and Quasi-Linearization: Two Cases

Let us first draw the distinction that governs the interpretation of our contribution.

Exact case ($\boldsymbol{H} = \frac{1}{2}|p|^2$). When the Hamiltonian is purely quadratic in the co-state, setting $\boldsymbol{c} = 1/\delta$ (a constant) and $\boldsymbol{\omega}^\delta := \exp(-\boldsymbol{v}^\delta/\delta)$ is an *exact* Cole-Hopf transformation: $\boldsymbol{\omega}^\delta$ satisfies the homogeneous heat equation $\boldsymbol{\omega}^\delta_t = (\delta/2)\Delta\boldsymbol{\omega}^\delta$ with no residual. In this case what follows below is exact.

Quasi-linearization. For a general Hamiltonian $\boldsymbol{H}(t; \boldsymbol{x}, p)$, define the spatially-varying coefficient,

$$\boldsymbol{c}(t; \boldsymbol{x}) = \frac{2}{\delta} \cdot \boldsymbol{H}^\delta / |D\boldsymbol{v}^\delta|^2, \tag{4}$$

where $\boldsymbol{H}^\delta := \boldsymbol{H}(t; \boldsymbol{x}, D\boldsymbol{v}^\delta)$. The reader deserves candor on what this transformation does and does not achieve. With $\boldsymbol{c}$ spatially varying, $\boldsymbol{\omega}^\delta := \exp(-\boldsymbol{c}\boldsymbol{v}^\delta)$ applied to (HJ-Visc) yields a heat equation plus a residual $\boldsymbol{R} = \boldsymbol{R}_{\text{alg}} + \boldsymbol{R}_{\text{der}}$: the *algebraic* part is annihilated exactly by the choice (4), whilst the *derivative* part — built from $\boldsymbol{c}_t$, $D\boldsymbol{c}$, and $\Delta\boldsymbol{c}$ — survives every algebraic choice, and no freezing of the value of $\boldsymbol{c}$ removes the derivatives of $\boldsymbol{c}$ (Lemma B.1, Appendix B). Our algorithm is therefore a *Picard quasi-linearization*, a successive quadratic matching in the spirit of Bellman: at iteration k, the coefficient $\boldsymbol{c}^{(k)}$ is *frozen* at the current iterate, the linear heat equation is solved *exactly* for $\boldsymbol{\omega}^{(k+1)}$ — equivalently, the viscous HJ equation with the locally-matched quadratic surrogate Hamiltonian $\tilde{\boldsymbol{H}}^{(k)} = \frac{\delta}{2}\boldsymbol{c}^{(k)}|p|^2$ — the value $\boldsymbol{v}^{(k+1)}$ is recovered, the gradient $D\boldsymbol{v}^{(k+1)}$ is updated, and $\boldsymbol{c}^{(k+1)}$ is recomputed. The iteration's limit is the fixed point of this surrogate solve map, *not*, in general, the viscous solution itself; Theorem C.4 (Appendix C) bounds the gap between the two by the Duhamel norm of the discarded residual, and the bound degrades, as it must, where $\boldsymbol{c}$ varies rapidly — precisely where our pointwise errors concentrate in Fig. 1. There is no joint solve of $\boldsymbol{c}$ with the unknown iterate; the apparent circularity in (4) is resolved by the frozen-coefficient interpretation. This distinction is stated in Remark 2.4 and is the basis of the central Algorithm 1.

2.2 HJ Linearization

Throughout this section we replace $\boldsymbol{H}(t; \boldsymbol{x}, D\boldsymbol{v}^\delta)$ with $\boldsymbol{H}^\delta$ for brevity and we write $\boldsymbol{g}(\boldsymbol{x})$ for the terminal cost $\boldsymbol{g}(0; \boldsymbol{x})$. With $\boldsymbol{c}^{(k)}$ frozen, the transformation,

$$\boldsymbol{\omega}^\delta := \exp(-\boldsymbol{c}\boldsymbol{v}^\delta), \tag{Linear-Trans}$$

reduces (HJ-Visc) to a heat equation under the frozen-coefficient approximation (see Proposition 2.1). *For general nonlinear Hamiltonians, this transformation induces a residual term and should be interpreted as a quasi-linearization*, i.e. , the approximation is exact only when $\boldsymbol{H}^\delta$ is quadratic in p.

Proposition 2.1 (Heat-equation solve under frozen coefficient)**.** *With $\boldsymbol{c}^{(k)}$ frozen at the current iterate, the linear step of Algorithm 1 defines the transformed variable $\boldsymbol{\omega}^\delta$ as the solution of the heat initial-value problem below. When $\boldsymbol{H} = \frac{1}{2}|p|^2$ and $\boldsymbol{c} = 1/\delta$, the identification $\boldsymbol{\omega}^\delta = \exp(-\boldsymbol{c}\boldsymbol{v}^\delta)$ is*

exact; for general Hamiltonians it discards the derivative residual of Lemma B.1 in Appendix B. Thus $\boldsymbol{\omega}^\delta$ *satisfies,*

$$\boldsymbol{\omega}_t^\delta - \frac{\delta}{2}\Delta\boldsymbol{\omega}^\delta = 0 \text{ in } \Omega \times (0,T], \quad \boldsymbol{\omega}^\delta = \exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right) \text{ on } \bar{\Omega} \times \{t=0\}. \tag{Linear-HJ}$$

The solution to (Linear-HJ) *admits the following unique, explicit Green's convolution representation,*

$$\boldsymbol{\omega}^\delta(t;\boldsymbol{x}) = \frac{1}{(\sqrt{2\pi\delta t})^n}\int_\Omega \exp\left(-\frac{|\boldsymbol{x}-\boldsymbol{y}|^2}{2\delta t}\right)\exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right)d\boldsymbol{y}, \quad \boldsymbol{x}\in\Omega,\ t\in(0,T], \tag{5}$$

which, via the Feynman-Kac formula, can equivalently be written as a Gaussian expectation,

$$\boldsymbol{\omega}^\delta(t;\boldsymbol{x}) = \mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}\left[\exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right)\right]. \tag{6}$$

Lemma 2.2 (Smoothed HJ Solution)**.** *Under the frozen-coefficient interpretation of* (4) *(or exactly when* $\boldsymbol{H} = \frac{1}{2}|p|^2$*), the solution to the smoothed Hamilton-Jacobi equation* (HJ-Visc) *is,*

$$\boldsymbol{v}^\delta(t;\boldsymbol{x}) = -\frac{1}{\boldsymbol{c}^{(k)}}\log\left\{\frac{1}{(\sqrt{2\pi\delta t})^n}\int_\Omega \exp\left(-\frac{|\boldsymbol{x}-\boldsymbol{y}|^2}{2\delta t}\right)\exp\left(-\boldsymbol{c}^{(k)}\boldsymbol{g}(\boldsymbol{y})\right)d\boldsymbol{y}\right\}, \tag{7}$$

in $\Omega\times(0,T]$*, where* $\boldsymbol{c}^{(k)}$ *is the frozen coefficient at iteration* k*. This can equivalently be written as,*

$$\boldsymbol{v}^\delta(t;\boldsymbol{x}) = -\frac{1}{\boldsymbol{c}^{(k)}}\cdot\log\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}\left[\exp\left(-\boldsymbol{c}^{(k)}\boldsymbol{g}(\boldsymbol{y})\right)\right]. \tag{8}$$

Lemma 2.3 (Spatial gradients of the HJ solution)**.** *Under the same frozen-coefficient assumption as Lemma 2.2, the spatial gradient of the value function* $\boldsymbol{v}^\delta(t;\boldsymbol{x})$ *admits the form,*

$$D\boldsymbol{v}^\delta = \frac{1}{t\cdot\delta\cdot\boldsymbol{c}^{(k)}}\left(\boldsymbol{x} - \frac{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}\left[\boldsymbol{y}\cdot\exp\left(-\boldsymbol{c}^{(k)}\boldsymbol{g}(\boldsymbol{y})\right)\right]}{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}\left[\exp\left(-\boldsymbol{c}^{(k)}\boldsymbol{g}(\boldsymbol{y})\right)\right]}\right), \tag{9}$$

where $\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)$*, i.e. , with mean* $\boldsymbol{x}$ *and covariance* $\delta t I_n$*.*

Remark 2.4 (Exact reduction vs. quasi-linearization). Lemmas 2.2 and 2.3 are exact when $\boldsymbol{H} = \frac{1}{2}|p|^2$, in which case $\boldsymbol{c} = 1/\delta$ is a constant and the residual $\boldsymbol{R}(t;\boldsymbol{x})$ in the linearized equation vanishes identically (see Appendix B). For general Hamiltonians, $\boldsymbol{c}^{(k)}$ depends on the unknown $D\boldsymbol{v}^\delta$; we resolve this by Algorithm 1, whereupon $\boldsymbol{c}^{(k)}$ is formed from the previous iterate, frozen, and updated after each linear solve. The formulas in both lemmas apply at each iteration with $\boldsymbol{c}^{(k)}$ in place of $\boldsymbol{c}$.

2.3 Relations to Backward Reachable Sets/Tubes

Since we now have the derivatives that constitute the viscous HJ equation,we can rewrite (HJI-RCBRT) as

$$\boldsymbol{v}_t^\delta(t;\boldsymbol{x}) + \min\{0,\boldsymbol{H}^\delta\} = \frac{\delta}{2}\Delta\boldsymbol{v}^\delta, \quad \boldsymbol{v}^\delta(0;\boldsymbol{x}) = \boldsymbol{g}(0;\boldsymbol{x}), \tag{HJI-RCBRT-Visc}$$

where the viscosity acts on the whole equation rather than inside the $\min$ — the regularization smooths the solution, not the freezing decision of the tube, so that the solution to (HJI-RCBRT-Visc) is from integrating the quantities given in Lemma 2.3. This is tantamount to sampling from the kernels of the given Gaussian densities. The invariant set $\mathcal{L}_0$ obtained at T is

$$\mathcal{L}_0(T) = \{\boldsymbol{x}\in\Omega \,|\, \boldsymbol{v}^\delta(0;\boldsymbol{x}) \le 0\}, \tag{10}$$

and the RCBRT is as given in (HJI-RCBRT). Similarly, the viscous version of the *reach-avoid BRT* i.e. (HJI-RCBRAT) can be represented as

$$\min\{\boldsymbol{v}_t^\delta(t;\boldsymbol{x}) + \boldsymbol{H}^\delta - \frac{\delta}{2}\Delta\boldsymbol{v}^\delta, \boldsymbol{g}(t;\boldsymbol{x}) - \boldsymbol{\ell}(t;\boldsymbol{x})\} \le 0, \quad \boldsymbol{v}^\delta(0;\boldsymbol{x}) = \boldsymbol{g}(0;\boldsymbol{x}).$$

(HJI-RCBRAT-Visc)

Algorithm 1 Quasi-Linearization Algorithm Cole-Hopf

Fix: $\epsilon > 0$, $\boldsymbol{v}^{(0)}(t;\boldsymbol{x}) = g(\boldsymbol{x})$, $\boldsymbol{c}^{(0)} = \frac{2\,\boldsymbol{H}(t;\boldsymbol{x},D\boldsymbol{g})}{\delta\,|D\boldsymbol{g}|^2}$.

For $k = 0, 1, 2, \ldots$**:**

1. Freeze $\boldsymbol{c}^{(k)}$ at the current iterate.
2. Solve the heat equation $\boldsymbol{\omega}_t = \frac{\delta}{2}\Delta\boldsymbol{\omega}$ with initial data $\boldsymbol{\omega}^{(k)}(0;\boldsymbol{x}) = e^{-c^{(k)}g(\boldsymbol{x})}$.
3. Recover $\boldsymbol{v}^{(k+1)} = -(1/\boldsymbol{c}^{(k)})\log\boldsymbol{\omega}^{(k+1)}$.
4. Update $D\boldsymbol{v}^{(k+1)}$ and $\boldsymbol{c}^{(k+1)} = \frac{2\boldsymbol{H}(t;\boldsymbol{x},D\boldsymbol{v}^{(k+1)})}{\delta|D\boldsymbol{v}^{(k+1)}|^2}$.
5. Check convergence: $\|\boldsymbol{v}^{(k+1)} - \boldsymbol{v}^{(k)}\|/\|\boldsymbol{v}^{(k)}\| < \varepsilon$.

2.4 Weighted Importance Sampling of Reachable Sets

While we may not recover the exact value function system, the mean and variance of its derivatives can be evaluated with the kernel expectations of Lemma 2.2, and 2.3. Previously, self-supervised physics-informed neural network (PINN) approaches such as DeepReach [Bansal and Tomlin, 2021] train neural networks to minimize the HJ PDE residual directly without requiring external supervision or a reference grid solution. However, neural network-based approximation methods encounter inherent scaling limitations; DeepReach scales only to moderate dimensions (reported to 9D/10D in [Bansal and Tomlin, 2021]), and approximation accuracy degrades as dimension increases. ***Our sampling-based scheme addresses this bottleneck by decoupling the memory cost from dimensionality.***

The estimators of Lemmas 2.2 and 2.3 are ratios of exponential-weight expectations, and their failure mode in high dimensions is not subtle: when $\boldsymbol{c}\,\boldsymbol{g}$ is large over the bulk of the kernel's support, the denominator $\mathbb{E}[e^{-\boldsymbol{c}\boldsymbol{g}}]$ is carried by a rare event, the weights collapse onto a few samples, and the ratio's variance explodes. The remedy is to move the samples to where the weight lives. We *tilt* the proposal: in place of $\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{x}, \sigma^2 I_n)$, $\sigma^2 \triangleq \delta t$, we draw $\boldsymbol{y} \sim q_\theta \triangleq \mathcal{N}(\boldsymbol{x}+\theta, \sigma^2 I_n)$ and reweight by the exact density ratio $w_\theta(\boldsymbol{y}) = \exp\big((|\theta|^2 - 2\langle \boldsymbol{y}-\boldsymbol{x}, \theta\rangle)/(2\sigma^2)\big)$, which leaves every expectation unbiased. The zero-variance proposal is proportional to the integrand $\varphi_{\boldsymbol{x}}(\boldsymbol{y})e^{-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})}$ itself; its Laplace (first-order) Gaussianization fixes the shift,

$$\theta^\star = -\sigma^2\,\boldsymbol{c}^{(k)}\,D\boldsymbol{v}^{(k)}(t;\boldsymbol{x}), \tag{11}$$

i.e. , the samples are pushed one preconditioned gradient step along the descent direction of the running value iterate — the previous Picard iterate hands the sampler exactly the drift it needs, at no extra cost. The health of the resulting self-normalized estimators is monitored by the effective sample size $\mathrm{ESS} \triangleq (\sum_i \tilde{w}_i)^2 / \sum_i \tilde{w}_i^2$ with $\tilde{w}_i \triangleq w_\theta(\boldsymbol{y}_i)e^{-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y}_i)}$: an ESS near N certifies the expectations; an ESS collapse flags exactly the weight degeneracy described above. The tilted counterparts of Lemmas 2.2 and 2.3 are stated as Corollary B.6 in Appendix B. The quasi-linearization, essentially a Picard fixed-point iterative scheme, is stated in Algorithm 1. The heat-kernel expectation is estimated via Monte Carlo, while the resulting log-sum-exp estimator for the value function enjoys the sample guarantee given in §2.5.

2.5 Sampling Complexity and Convergence Guarantees

In this sub-section, we analyze the sample complexity of the scheme with coefficient $\boldsymbol{c}$ frozen and then generate a conditional linear-convergence guarantee as a fixed-point iteration on a finite collection of evaluation states.

Theorem 2.5 (Finite-sample concentration of the frozen-coefficient value estimator)**.** *Fix phase $(t;\boldsymbol{x}) \in (0,T]\times\Omega$ and a frozen coefficient $\boldsymbol{c} > 0$ (the case $\boldsymbol{c} < 0$ and the removable zero at $\boldsymbol{c} = 0$ are handled in Remark 2.11). Let $\sigma \triangleq \sqrt{\delta t}, \boldsymbol{\zeta} \sim \mathcal{N}(\boldsymbol{x}, \delta t I_n)$, and assume that the terminal cost is bounded, i.e. there exist constants $g_{\min} \le g_{\max}$ such that $g_{\min} \le \boldsymbol{g}(\boldsymbol{\zeta}) \le g_{\max}$, $\forall\,\boldsymbol{\zeta} \in \Omega$.*

Let $Z \triangleq \exp(-\boldsymbol{c}\,\boldsymbol{g}(\boldsymbol{\zeta}))$, $\mu \triangleq \mathbb{E}[Z]$, $\boldsymbol{v}_{\boldsymbol{c}}(t;\boldsymbol{x}) \triangleq -\frac{1}{\boldsymbol{c}}\log\mu$. Given i.i.d. samples $\boldsymbol{\zeta}_1, \ldots, \boldsymbol{\zeta}_N \sim \mathcal{N}(\boldsymbol{x}, \delta t I_n)$, let $Z_i \triangleq \exp(-\boldsymbol{c}\,\boldsymbol{g}(\boldsymbol{\zeta}_i))$, $\bar{Z}_N \triangleq \frac{1}{N}\sum_{i=1}^N Z_i$, and $\hat{\boldsymbol{v}}_{\boldsymbol{c},N}(t;\boldsymbol{x}) \triangleq -\frac{1}{\boldsymbol{c}}\log\bar{Z}_N$. Let $\alpha \triangleq e^{-\boldsymbol{c}g_{\max}}$, $\beta \triangleq e^{-\boldsymbol{c}g_{\min}}$. Then $\alpha \le Z_i \le \beta$ almost surely, and for all $\varepsilon > 0$,

$$\mathbb{P}\left(|\hat{\boldsymbol{v}}_{\boldsymbol{c},N}(t;\boldsymbol{x}) - \boldsymbol{v}_{\boldsymbol{c}}(t;\boldsymbol{x})| \ge \varepsilon\right) \le 2\exp\left(-\frac{2N\mu^2(1-\exp(-\boldsymbol{c}\varepsilon))^2}{(\beta-\alpha)^2}\right). \tag{12}$$

Remark 2.6 (Tightness via Bernstein and Jensen bounds). The Hoeffding bound in (12) is asymptotically tight but ignores variance. When $\mathrm{Var}(Z) \ll (\beta-\alpha)^2/4$, a Bernstein tail bound yields strictly tighter concentration. Additionally, Corollary 2.7 uses the loose lower bound $\mu \geq \alpha = e^{-\boldsymbol{c}g_{\max}}$; when a sharper estimate (e.g., via Jensen, $\mu \geq e^{-\boldsymbol{c}\mathbb{E}[\boldsymbol{g}(\boldsymbol{\zeta})]}$) is available, the required sample size can be substantially reduced.

Corollary 2.7 (Explicit sample size independent of μ)**.** *In particular, since $\mu \geq \alpha$, it is sufficient to choose*

$$N \geq \frac{(\beta-\alpha)^2}{2\alpha^2(1-\exp(-\boldsymbol{c}\varepsilon))^2}\log\left(\frac{2}{\alpha}\right) \tag{13}$$

to guarantee $\mathbb{P}\left(|\widehat{\boldsymbol{v}}_{\boldsymbol{c},N}(t;\boldsymbol{x}) - \boldsymbol{v}_{\boldsymbol{c}}(t;\boldsymbol{x})| \geq \varepsilon\right) \leq \alpha$.

Assumption 2.8. Let $V = \mathbb{R}^M$ with the sup norm $\|v\|_\infty = \max_{1\leq m\leq M}|v_m|$, where $v_m \approx \boldsymbol{v}(t;\boldsymbol{x}_m)$. Assume the following hold on a closed admissible set $\mathcal{A} \subset V$.

1. There exists $G > 0$ such that $|\boldsymbol{g}(\boldsymbol{x})| \leq G$ for all $\boldsymbol{x} \in \mathbb{R}^n$.
2. There exist constants $0 < m_0 \leq P_*$ and $0 < c_{\min} \leq c_{\max} < \infty$ such that for every $v \in \mathcal{A}$ and each m,
$$m_0 \leq |p_m(v)| \leq P_*, \qquad c_{\min} \leq (\Gamma(v))_m \leq c_{\max}. \tag{14}$$
3. The Hamiltonian is Lipschitz in the co-state on the ball $|p| \leq P_*$: there exist constants $H_*, L_H > 0$ such that
$$|\boldsymbol{H}(t;\boldsymbol{x}_m,p)| \leq H_*, \qquad |\boldsymbol{H}(t;\boldsymbol{x}_m,p) - \boldsymbol{H}(t;\boldsymbol{x}_m,q)| \leq L_H|p-q| \tag{15}$$
for all $|p|, |q| \leq P_*$ and all m.
4. The reconstruction map is Lipschitz on $\mathcal{A}$: there exists $L_D > 0$ such that
$$\|\mathcal{G}(v) - \mathcal{G}(w)\|_\infty \leq L_D\|v-w\|_\infty, \qquad \forall v, w \in \mathcal{A}. \tag{16}$$
5. The admissible set is invariant under Λ, and the quantity
$$q \triangleq \frac{2G}{c_{\min}} \cdot \frac{2L_D}{\delta}\left(\frac{L_H}{m_0^2} + \frac{2H_*P_*}{m_0^4}\right) \tag{17}$$
satisfies $0 \leq q < 1$.

Remark 2.9 (Remark on Assumption 2.8). Note that item 2 of Assumption 2.8 excludes regions where the value-gradient degenerates, such as flat reachable interiors or singular shocks. We do not regard this as a defect of scope so much as a statement of *where the hypotheses are needed*: BRT membership is decided on the band $\mathcal{N}_\eta \triangleq \{(t;\boldsymbol{x}) : |\boldsymbol{v}^\delta(t;\boldsymbol{x})| \leq \eta\}$ about the zero level set — the usable part of Isaacs [Isaacs, 1999] — and for nondegenerate games the gradient is bounded away from zero there, so Assumption 2.8 is a hypothesis on $\mathcal{N}_\eta$, not on all of Ω. Off the band, we run the scheme with the regularized coefficient of Lemma 2.10 below, at a price the lemma quantifies. Furthermore, under item 5, the contraction constant q may become unbounded in the inviscid limit, i.e. as $\delta \to 0$ and/or $m_0 \to 0$. In practice, numerical stabilization mechanisms such as viscosity regularization, coefficient clipping, or solution "smearing" [Osher and Fedkiw, 2004] may be introduced to maintain stability of the discrete integration scheme.

Reachability Hamiltonians: homogeneity, sign, and the flat interior. A structural honesty is owed here. The game Hamiltonian $\boldsymbol{H}(t;\boldsymbol{x},p) = \max_{\boldsymbol{u}}\min_{\boldsymbol{w}}\langle p, f(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w})\rangle$ is positively 1-homogeneous in the co-state and changes sign across the barrier; the exact quadratic case $\boldsymbol{H} = \frac{1}{2}|p|^2$ is therefore a motivating limit rather than a member of the reachability problem class. Two consequences follow for the coefficient (4): it scales as $\boldsymbol{c} \sim C_{\boldsymbol{H}}/(\delta|D\boldsymbol{v}^\delta|)$ and so grows without bound in flat regions, and it vanishes (and flips sign) where the optimal dynamics run tangent to the level set. The first pathology is cured by regularizing the denominator; the next lemma shows the cure costs a *uniform*, quantified price for exactly the Hamiltonian class at hand.

Lemma 2.10 (Regularized coefficient as a uniform Hamiltonian perturbation)**.** *Let $\boldsymbol{H}$ be positively 1-homogeneous in p with $C_{\boldsymbol{H}} \triangleq \sup_{t,\boldsymbol{x},|q|=1} |\boldsymbol{H}(t;\boldsymbol{x},q)| < \infty$, and fix $\eta > 0$. Running Algorithm 1 with the regularized coefficient,*

$$\boldsymbol{c}_\eta(t;\boldsymbol{x}) \triangleq \frac{2}{\delta} \cdot \frac{\boldsymbol{H}^\delta}{|D\boldsymbol{v}^\delta|^2 + \eta}, \tag{18}$$

is equivalent to running it, unregularized, for the perturbed Hamiltonian $\boldsymbol{H}_\eta \triangleq \boldsymbol{H}\,|D\boldsymbol{v}^\delta|^2/(|D\boldsymbol{v}^\delta|^2 + \eta)$, which satisfies,

$$|\boldsymbol{H}_\eta - \boldsymbol{H}|_\infty \le \frac{C_{\boldsymbol{H}}\sqrt{\eta}}{2}, \tag{19}$$

uniformly on $\Omega \times [0,T]$ — including the flat interior where $D\boldsymbol{v}^\delta \to 0$. Consequently, by Theorem C.11, the induced value-function perturbation is at most $TC_{\boldsymbol{H}}\sqrt{\eta}/2$.

Remark 2.11 (Sign and the removable zero of the coefficient). The concentration bound of Theorem 2.5 is stated for $\boldsymbol{c} > 0$; for $\boldsymbol{c} < 0$ it holds verbatim with $\alpha \triangleq \min\{e^{-\boldsymbol{c}g_{\max}}, e^{-\boldsymbol{c}g_{\min}}\}$ and β the corresponding maximum. The zero of $\boldsymbol{c}$ on the barrier is a *removable* singularity of the estimator: as $\boldsymbol{c} \to 0$, $-\frac{1}{\boldsymbol{c}} \log \mathbb{E}[e^{-\boldsymbol{c}\boldsymbol{g}}] \to \mathbb{E}[\boldsymbol{g}]$, i.e. the log-sum-exp recovery degenerates gracefully to the Gaussian mean of the terminal data — which is the correct pure-diffusion limit, since $\boldsymbol{H} \to 0$ reduces (HJ-Visc) to the heat equation. In implementation we evaluate the estimator in its $\log 1p$-stable form and clip $|\boldsymbol{c}|$ to $[c_{\min}, c_{\max}]$, the clipping being one more perturbation of the kind Lemma 2.10 prices.

Theorem 2.12 (Contraction convergence of Algorithm 1)**.** *Fix evaluation states $\boldsymbol{x}_1, \dots, \boldsymbol{x}_M \in \Omega$ and a time $t \in (0,T]$. Let $\mathcal{G} : V \to (\Omega)^M$ denote a stable deterministic gradient reconstruction operator, which may arise from smoothing, interpolation, kernel regression, or variance-controlled Monte Carlo estimation. Further, write $p_m(v) \triangleq (\mathcal{G}(v))_m$. For $v \in V$, define the coefficient-update map by $(\Gamma(v))_m \triangleq 2\boldsymbol{H}(t;\boldsymbol{x}_m, p_m(v))/(\delta|p_m(v)|^2)$, $m = 1, \dots, M$, and define the frozen-coefficient heat-kernel map as*

$$(\Phi(c))_m \triangleq -\frac{1}{c_m} \log \mathbb{E}_{\boldsymbol{\zeta}\sim\mathcal{N}(\boldsymbol{x}_m, \delta t I_n)} \left[\exp\left(-c_m \boldsymbol{g}(\boldsymbol{\zeta})\right)\right]. \tag{20}$$

Let $\Lambda \triangleq \Phi \circ \Gamma$ so that Algorithm 1 is the iteration $v^{(k+1)} = \Lambda(v^{(k)})$.

Then Λ is a contraction on $\mathcal{A}$. Consequently, Λ has a unique fixed point $v^* \in \mathcal{A}$, and for every initial iterate $v^{(0)} \in \mathcal{A}$ the sequence generated by Algorithm 1 converges linearly to v^* with

$$\|v^{(k+1)} - v^*\|_\infty \le q\|v^{(k)} - v^*\|_\infty \le q^{k+1}\|v^{(0)} - v^*\|_\infty. \tag{21}$$

Moreover,

$$\|\Gamma(v^{(k)}) - \Gamma(v^*)\|_\infty \le \frac{2L_D}{\delta}\left(\frac{L_H}{m_0^2} + \frac{2H_* P_*}{m_0^4}\right)\|v^{(k)} - v^*\|_\infty,$$
$$\|v^{(k+1)} - v^{(k)}\|_\infty \le q^k\|v^{(1)} - v^{(0)}\|_\infty, \tag{22}$$

and the a posteriori error estimate

$$\|v^{(k)} - v^*\|_\infty \le \frac{q}{1-q}\|v^{(k)} - v^{(k-1)}\|_\infty \tag{23}$$

holds for every $k \ge 1$.

Remark 2.13. Theorem 2.12 is a convergence result for the *frozen-coefficient numerical map* implemented by Algorithm 1. It shows conditional linear convergence under explicit regularity, nondegeneracy, and contraction hypotheses. It does not claim unconditional global convergence for arbitrary Hamiltonians, nor does it identify the fixed point with the exact solution of the original nonlinear HJ PDE: the two differ by the *quasi-linearization defect* $\varepsilon_{\mathrm{QL}}$, which we bound in Appendix C.

To connect the discrete algorithm to the exact viscosity solution $\boldsymbol{v}$ of the original HJ PDE, see Theorem C.9 in Appendix C, which combines the iteration error (Theorem C.1), Monte Carlo error (Theorem C.6), the quasi-linearization defect (§C.2), and the viscosity approximation error (Theorem C.8) to establish the total error bound,

$$\|\hat{\boldsymbol{v}}^{K,N,\delta} - \boldsymbol{v}\|_\infty \le C_1\rho^K + \frac{C_2}{\sqrt{N}}\sqrt{\log(1/\delta_p)} + \varepsilon_{\mathrm{QL}} + C_3\sqrt{\delta}. \tag{24}$$

This decomposition shows how all four error sources scale with iterations K, samples N, and viscosity δ; only the first three shrink with computational effort — the defect $\varepsilon_{\mathrm{QL}}$ is a property of the surrogate and is reduced only by the fidelity of the quasi-linearization itself.

A safety certificate must not merely be accurate on average; it must never admit an unsafe state. The error budget (24) converts directly into a one-sided guarantee by thresholding with margin,

Corollary 2.14 (Conservative safety certification)**.** *Let* $E(K,N,\delta,\delta_p) \triangleq C_1\rho^K + \frac{C_2}{\sqrt{N}}\sqrt{\log(1/\delta_p)} + \varepsilon_{\mathrm{QL}} + C_3\sqrt{\delta}$ *be the total error budget of* (24)*, and adopt the convention that a state is* unsafe *at time* t *iff* $\boldsymbol{v}(t;\boldsymbol{x}) \le 0$*. Declare* $\boldsymbol{x}$ SAFE *only if* $\hat{\boldsymbol{v}}^{K,N,\delta}(t;\boldsymbol{x}) > E$*,* UNSAFE *only if* $\hat{\boldsymbol{v}}^{K,N,\delta}(t;\boldsymbol{x}) < -E$*, and* UNDETERMINED *otherwise. Then, with probability at least* $1-\delta_p$*, no unsafe state is certified* SAFE *and no safe state is certified* UNSAFE*; all classification error is confined to the declared band* $|\hat{\boldsymbol{v}}^{K,N,\delta}| \le E$ *of width* $2E$ *about the reachability boundary. (Proof in Appendix C.)*

The certificate errs on refusal, never on admission: shrinking K^{-1}, N^{-1}, and δ narrows the undetermined band, whilst the defect $\varepsilon_{\mathrm{QL}}$ sets its floor. This is, in our view, the honest currency in which a sampling-based reachability method should quote its accuracy — worst-case sign correctness with an explicit abstention region — rather than an L^2 average that flatters the interior and forgives the boundary.

2.6 Notes on Monte Carlo and Viscosity Approximation Errors

The standard MC estimator for (HJ-Visc) converges at rate $O(1/\sqrt{N})$ where N is the number of samples. The variance depends on $\boldsymbol{c}$ and the range of $\boldsymbol{g}$: when $\boldsymbol{c}\,\|\boldsymbol{g}\|_\infty \gg 1$ (small δ), the exponential weights become concentrated and the effective sample size shrinks. Importance sampling with a tilted proposal (e.g., Laplace approximation around the mode of $e^{-\boldsymbol{cg}}$) can reduce variance. By [Crandall et al., 1984], $\|\boldsymbol{v}^\delta - \boldsymbol{v}\|_\infty \le k\sqrt{\delta}$ where $\boldsymbol{v}$ is the inviscid viscosity solution. Thus, smaller δ improves the approximation quality but increases the MC variance (the weights become overly peaked). This creates a fundamental bias-variance trade-off controlled by δ.

3 Numerical Results

All experiments run on a single CPU core of an Intel Core i7-14700K processor (20 physical cores, 33.8 MiB L3 cache per core, base clock 5.6 GHz) with 31 GiB RAM running Ubuntu 22.04. The processor supports SIMD vectorization (AVX2, SSE4.2) which JAX and PyTorch leverage for batch operations. All reported metrics are averaged across $N_{\text{runs}} = 5$ independent trials with different random seeds; tables and figures report mean values with standard deviation error bars where applicable. The safety analysis on European starlings is summarized in §3.4, with the complete treatment in Appendix D.1.

3.1 Territorial Defense: A Game of Two Rockets on a Plane

We adopt the rockets launch problem (see Fig. 12) of Dreyfus [Dreyfus, 1966] and cast it as a terminal differential game between two identical rockets, $\boldsymbol{P}$ and $\boldsymbol{E}$, on an $(x-z)$ cross-section of a Cartesian plane in [Molu, 2024a, 2025]. The game terminates when *capture* occurs, i.e. , the distance $\|\boldsymbol{PE}\|$ becomes less than a certain prespecified (scalar) quantity. The states of $\boldsymbol{P}$ and $\boldsymbol{E}$ are denoted as (x_p, x_e) respectively, driven by thrusts (u_p, u_e) in the (x,z)-plane. The motion of $\boldsymbol{P}$ relative to $\boldsymbol{E}$'s along the $(\boldsymbol{x}-\boldsymbol{z})$ plane includes the relative orientation, the control input, shown in Fig. 12 as $\theta = u_p - u_e$. Full derivation of the scheme is provided in Appendix D.3.1.

Scope. The proposed method is not intended to outperform structured grid solvers in low-dimensional settings ($n \le 4$), where highly-optimized implementations exist; rather, its advantage emerges as dimensionality increases and grid storage becomes prohibitive.

3.2 Dimensions Scaling: A 15 Rockets System in a Pursuit-Evasion Game

To validate the scalability benefits of Algorithm 1, we consider a 15-rocket multi-pursuer single-evader game with state dimension $n = 45$; this dimension is intractable for grid-based solvers. State is $\boldsymbol{x} = (x_1, y_1, \theta_1, \ldots, x_{15}, y_{15}, \theta_{15}) \in \mathbb{R}^{45}$, where each agent $i \in \{1,\ldots,15\}$ has position (x_i, y_i), heading θ_i, and control input $u_i \in [-1,+1]$ regulating turn rate. Dynamics are:

$$\dot{x}_i = a_i\cos(\theta_i), \quad \dot{y}_i = a_i\sin(\theta_i), \quad \dot{\theta}_i = u_i, \tag{25}$$

Table 1: 15-agent multi-pursuer single-evader game with three speed regimes on $n = 45$.

Case	a_{evader}	a_{pursuers}	Iterations	$\varepsilon(k)$	Wall-clock (s)
1. (Evader faster)	2.0	1.0	15	0.003121	27.6
2. (Equal speed)	1.0	1.0	15	0.002454	30.0
3. (Pursuers faster)	1.0	2.0	15	0.003462	27.6

with forward speeds $a_i \in \mathbb{R}_{>0}$. The evader (agent 15) seeks escape; capture occurs when any of 14 pursuers reach distance $r_{\text{capture}} \leq 1.5$ ft. Target set: $\phi(\boldsymbol{x}) = \min_{i \leq 14} \|\boldsymbol{x}_i^{\text{pos}} - \boldsymbol{x}_{15}^{\text{pos}}\|_2 - r_{\text{capture}}$.

We test three speed regimes to examine relative agent capabilities on the reachable set, (1.) evader speed $a_{\text{evader}} = 2.0$ ft/s, pursuer speeds $a_{\text{pursuers}} = 1.0$ ft/s; evader advantage; (2.) all agents equal speed $a = 1.0$ ft/s; balanced game; (3.) evader speed $a_{\text{evader}} = 1.0$ ft/s, pursuer speeds $a_{\text{pursuers}} = 2.0$ ft/s; pursuer advantage.

This $n = 45$ problem is computationally inaccessible to grid-based reachability solvers. A 101-point grid would require $101^{45} \approx 10^{90}$ cells, exceeding all conceivable storage. Our method operates at $O(N \cdot n) \approx 7.2$ MB per iteration, where the memory cost is independent of the state dimension exponent. The quasi-linear Picard iteration (Algorithm 1) computes the relative change in value function between successive iterations; Table 1 reports the final relative residual, $\varepsilon = \|\boldsymbol{v}^{(k+1)} - \boldsymbol{v}^{(k)}\|/\|\boldsymbol{v}^{(k)}\|$ as a measure of iteration convergence.

Table 1 shows the results for 15-rocket pursuit-evasion across three speed regimes. Wall-clock times are reported on a 16-core Intel i7-14700K. The residual is the final relative L^2 change between Picard iterates. All three cases converge in 15 iterations with residuals below 10^{-2}, demonstrating stability across different game parameters. Memory consumption remains constant at 7.2 MB per iteration; wall-clock variance reflects Monte Carlo sampling fluctuations. We are candid about what this table does and does not establish: at $n = 45$ no grid-based reference exists, so the table reports what can be measured without one — iteration stability, memory, and wall-clock — and demonstrates *scalability*, not certified accuracy. The per-sample cost is itself modest: for control-affine dynamics with box-constrained inputs, the min-max Hamiltonian evaluates in closed form (the optimizers reduce to sign structures), so each iteration costs $O(N \cdot n)$ arithmetic alongside the $O(N \cdot n)$ memory.

3.3 Experimental Setup and Quantitative Results: Pursuit-Evasion Games

Viscosity parameter selection. The viscosity parameter δ controls the degree of smoothing in the Cole-Hopf transformation. By Crandall-Lions theory, the approximation error scales as $O(\sqrt{\delta})$, so smaller δ yields more accurate solutions but requires higher Monte Carlo sample counts to maintain accuracy. In our experiments, we empirically set $\delta = 0.08$–0.1, which balances approximation error $O(\sqrt{\delta}) \approx 0.28$ with manageable variance in the log-sum-exp estimator. The choice $\delta \in [0.05, 0.2]$ is robust across all benchmark problems; practitioners should adjust based on the required accuracy-to-computation trade-off.

Frozen-coefficient bias. The quasi-linearization freezes the coefficient $\boldsymbol{c}^{(k)}$ — formed from the previous iterate's gradient, initialized at $D\boldsymbol{g}$ — over each Picard step, introducing a systematic bias that is largest near the zero level set where $\boldsymbol{c}$ varies most rapidly. Theorem 2.12 guarantees convergence to a fixed point of the frozen-coefficient map, not to the true viscous solution; the gap between the two is the quasi-linearization defect ε_{QL}, bounded in Theorem C.4 by the Duhamel norm of the discarded residual, and proportional there to the variation rate L_c of the converged coefficient and to how well the surrogate gradient tracks the true one. Empirically, this bias manifests as larger L^∞ errors near the reachability boundary where the gradient is most active — the theory and the heat maps of Fig. 1 indict the same region; the method mitigates it through the viscosity smoothing and iterative refinement of the frozen coefficient.

Numerical realization: We discretize the spatial and temporal domains as,

$$(x, z) \in (-100, +100], \quad \theta \in (-\pi/2, \pi/2], \quad t \in (0, 1], \tag{26}$$

with 1,000 temporal grid points and spatial step $\boldsymbol{x}_k - \boldsymbol{x}_{k-1} = 10^{-2}$.

Table 2: Pursuit-evasion games: Monte Carlo vs. `LevelSetPy` error metrics on 2D (x, z) slices. Both CPU-based; MC uses $N = 14{,}000$ samples per quasi-linear iteration; $\delta = 0.08$ throughout.

System	θ (rad)	L^∞	L^2_{rel}	MC time (s)	Iters
Rockets	$-\pi/2$	0.844	0.147	14.7	12
	0	1.044	0.134	13.9	12
	$\pi/2$	0.740	0.099	14.0	12
Dubins	$-\pi/2$	1.139	0.197	25.5	15
	0	0.821	0.032	24.9	15
	$\pi/2$	1.163	0.196	24.7	15

We set the target set as the ℓ_2-ball $\ell(\boldsymbol{x}) = \sqrt{x^2 + z^2}$ and run Algorithm 1 over the time range $(0, 1]$ with Dirichlet boundary conditions,

$$\boldsymbol{v}^\delta(0; \boldsymbol{x}) = \boldsymbol{g}(0; \boldsymbol{x}) \triangleq 0. \tag{27}$$

Spatial gradients (co-states) are computed via (8) and (9); the Hamiltonian is evaluated iteratively using $N = 14{,}000$ Monte Carlo samples per iteration according to Algorithm 1. The results are seen in Fig. 1.

Convergence. Algorithm 1 stabilizes within 20 iterations across all three benchmark systems, with the relative residual $\|\boldsymbol{v}^{(k+1)} - \boldsymbol{v}^{(k)}\|/\|\boldsymbol{v}^{(k)}\|$ decaying over the early iterations before settling at a floor of 0.14–0.22. Both phases are predicted by the theory, and we are precise about which theorem owns which phase: the initial decay is the geometric contraction of Theorem 2.12; the floor is *not* a failure of the contraction but the Monte Carlo noise floor of Theorem 2.5 — each iteration draws fresh samples, so even at the fixed point successive iterates differ by the $O(C_{\text{eff}}/\sqrt{N})$ sampling perturbation, and the residual sequence stagnates at that level rather than at zero. The floor is therefore problem-dependent through the weight variance (set by $\boldsymbol{c}\|\boldsymbol{g}\|_\infty$ and N), and indeed it differs across our experiments: the 45-dimensional game of §3.2 settles near 0.003 (Table 1), whilst the 3D benchmarks — whose pointwise errors concentrate where the coefficient turns over sharply near the usable-part boundary (Fig. 1) — settle one order of magnitude higher.

BRT geometry. Errors concentrate near the zero level-set boundary where $|\nabla v^\delta|$ is maximal, as expected in frozen-coefficient approximations [Crandall et al., 1992].

Table 2 reports point-wise error metrics (L^∞ and L^2_{rel}) for 2D slices on the 3D Dubins and Rockets pursuit-evasion benchmarks across three representative heading angles, evaluated on a uniform 40×40 grid with reference from `LevelSetPy`'s 45^3 solution interpolated to the same points. The L^∞ (point-wise maximum) error represents worst-case divergence, useful where outlier errors matter. The relative RMS error, L^2_{rel} weights by solution magnitude and captures overall accuracy. A caution on interpretation is owed here, and we state it plainly: the Crandall-Lions bound $O(\sqrt{\delta}) \approx 0.283$ [Crandall et al., 1992] controls the sup-norm distance between the *inviscid* and *viscous* solutions, whereas the table's entries measure our Monte Carlo output against a *grid* reference in different norms — the two numbers live in different currencies, and the smallness of L^2_{rel} (at $\theta = 0$, both systems achieve $L^2_{\text{rel}} < 0.04$) should be read as evidence that the sampling and iteration errors are well-controlled, not as beating the viscosity bound. The L^∞ errors (~ 0.8–1.2) are larger and driven by pointwise deviations near the zero level-set boundary where $|\nabla v^\delta|$ is maximal, as expected in frozen-coefficient approximations [Crandall et al., 1992] and as Theorem C.4 predicts through the coefficient-variation constant L_c. For certification purposes, the honest report is that of Corollary 2.14: sign-correctness outside an explicit abstention band about the boundary, with the band width set by the total error budget; the L^2 averages flatter the interior, and the L^∞ figures are dominated by exactly the band the corollary declares undetermined. The 3D isosurface computation over 15,625 sample points completes in 125 seconds (rockets) and 75.1 seconds (Dubins) on a single CPU, compared to 1.1 seconds (rockets) and 3.2 seconds (Dubins) for the dense $45^3 = 91{,}125$-point grids. The zero level-set boundary is extracted via marching cubes [Lorensen and Cline, 1987]. The wall-clock overhead of the Monte Carlo method reflects the sample count ($N = 14{,}000$ per iteration) necessary to balance variance in the Cole-Hopf estimator against quasi-linearization residuals, a tradeoff that becomes favorable in higher dimensions where grid storage becomes prohibitive.

Scalability. Algorithm 1 requires a memory footprint per iteration of $O(N \cdot n)$ for N samples per n state dimensions. With $N = 14{,}000$ and $n = 3$, each iteration allocates approximately 0.6 MB for samples and intermediate values. In contrast, the dense 45^3 `LevelSetPy` grid requires approximately 1.5 MB for the value function and gradient storage. Each 2D θ-slice completes in $14 - 26$ seconds on a single core, fully parallelizable across the CPU's 20 cores. This reduces aggregate wall-clock time proportionally with available core count and makes the method well-suited to parallel architectures (GPUs, HPC clusters) where grid-based methods become memory-prohibitive in dimensions $n \geq 5$. A clarification against a natural misreading: the *solve* is grid-free — no discretization of the state space is ever stored or marched — whilst the uniform grids that appear in this section (40×40 slices, $15{,}625$ isosurface points) are *evaluation* grids, used only to visualize the tube and to measure errors against the grid-based reference at common points.

3.4 Safety Certification at Population Scale: Starling Murmurations

As a stress test of the claim that safety certification need inherit neither the curse of dimensionality nor the curse of agent cardinality, we certify the collective behaviors of murmurations of European starlings (*Sturnus vulgaris*), modeled as 4D aerial Dubins vehicles under attack by ringed predators (Fig. 2). We are explicit about the problem structure: each flock is resolved by its *own* 4D value function and the murmuration safe set is their union — we solve many coupled low-dimensional games, *not* a joint $400{,}000$-dimensional PDE — so certifying the population reduces to one embarrassingly parallel per-bird evaluation of already-computed value functions. Three findings carry the section. First, *scale*: because the sampler is grid-free, the value-function solve is independent of the bird count; re-running the seven-predator cordon scenario for $100{,}000$ birds across ten flocks completes the value solve in 73.8 s and returns a certified safe fraction of 87.71% — within 0.3 percentage points of the $7{,}998$-bird run — with memory pinned at $O(N \cdot n)$. Second, *topology as a safety instrument*: the certificates recover the field-documented repertoire of collective behaviors as topological events of the reachable set — a defensive *cordon* appears as an annular safe set ($\beta_1 = 1$, $\chi = 0$) enclosing a protected core, and its collapse registers as a clean $\beta_1 : 1 \to 0$ transition at backward time $\tau \approx 1.29$; flock fragmentation registers as a rising component count. Third, *operational meaning*: the triple $(\chi, \beta_1, n_{\text{comp}})$ compresses the safety posture of the population into three integers per time step, telling an operator not merely *that* safety is being lost but *how*. The full experimental protocol, topology extraction, per-scenario tables, and discussion appear in Appendix D.1.

4 Conclusion

We have presented a quasi-linearized, frozen-coefficient sampling scheme for the viscous Hamilton-Jacobi PDE arising in safety analysis of dynamical systems. By applying a generalized Cole-Hopf-type transformation — exact when $\boldsymbol{H} = \frac{1}{2}|p|^2$ and iteratively approximated via Picard quasi-linearization for general Hamiltonians — the nonlinear HJ equation is reduced to a sequence of linear heat equations whose solutions are Gaussian heat-kernel expectations. The value function and its spatial gradient are recovered from these expectations via Monte Carlo sampling, yielding a grid-free algorithm with memory cost $O(N \cdot n)$ rather than the $O(M^n)$ cost of classical grid-based solvers.

Theorem 2.5 provides a finite-sample concentration bound for the frozen-coefficient Monte Carlo estimator, establishing a standard $O(N^{-1/2})$ error rate under explicit boundedness conditions on the terminal cost. Theorem 2.12 establishes conditional linear convergence of the Picard fixed-point iteration under Lipschitz and nondegeneracy assumptions, with an explicit contraction constant and a posteriori error estimate. Neither result claims unconditional global convergence for arbitrary Hamiltonians; the gap between the iteration's fixed point and the viscous solution is the quasi-linearization defect, which Theorem C.4 bounds by the Duhamel norm of the discarded residual, and Corollary 2.14 converts the resulting total error budget into a one-sided conservative safety certificate with a declared abstention band. Numerical experiments on three benchmark reachability problems i.e. a 3D rocket pursuit-evasion game, a 3D Dubins two-car game, and a double integrator plant (Appendix D) confirm stabilization within 20 Picard iterations, with the residual floor set by the Monte Carlo sampling noise; the scheme further scales to a 45-dimensional multi-agent game and to the certification of $100{,}000$-bird starling murmurations (§ D.1). The method's failure modes are documented with equal candor in **??**: discrete bang-bang switching structures defeat the frozen-coefficient scheme, and we chart the boundary of applicability there.

The connection to safe reinforcement learning, policy certification, and model-based control with learned dynamics suggests several directions for future work. These include adaptive importance sampling to reduce variance in high dimensions (§2.4), extension to systems with stochastic dynamics, high-dimensional accuracy validation against decoupled synthetic benchmarks with known solutions, and tighter integration with deep learning pipelines for scalable safety certification of learned policies.

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A Background and Preliminaries.

This section provides a context for the contribution of this work.

We first introduce the notations that are commonly used throughout the rest of this article. Reachable sets within the context of two person games [Isaacs, 1999] and their accompanying "viscous" terminal HJ PDE [Evans and Souganidis, 1984] are then introduced. This is followed by the HJ-Isaacs (HJI) PDEs commonly used to characterize reachable sets.

A.1 Notations and Terminologies

Conventions: Upper-case and lower-case bold font Roman letters are matrices and vectors, respectively; calligraphic letters are sets. Time variables e.g. t_0, t, τ, T will always be real positive numbers. Unless otherwise stated, Calligraphic letters are sets. The n-dimensional Euclidean space is $\mathbb{R}^n$; the set of natural numbers is $\mathbb{N}$. The natural log of T is denoted $\log T$. The state $\boldsymbol{x}$ belongs to the open set $\Omega \subseteq \mathbb{R}^n$. The HJ value function has a terminal value defined on the boundary of Ω, denoted $\partial\Omega$. and we denote by $\text{int}\,\Omega$ the interior of Ω. The closure of Ω is $\bar{\Omega}$. We let $\delta\Omega\,(:= \bar{\Omega}\backslash \text{int}\,\Omega)$ be the boundary of Ω. The dot product is signified by $\langle\cdot,\cdot\rangle$. We will use D to denote the gradient of a function $\boldsymbol{x}$ of n variables and $\nabla \boldsymbol{w} = (\boldsymbol{w}_{x_1}, \cdots, \boldsymbol{w}_{x_n}, \boldsymbol{w}_t) := (D\boldsymbol{w}, \boldsymbol{w}_t)$ as the full gradient of a function of $n+1$ variables, $\boldsymbol{w} = \boldsymbol{w}(\boldsymbol{x}, t)$. Where clarity is necessary, we may write $D_y w$ to mean the spatial derivative of w with respect to the vector $\boldsymbol{y}$.

The unique solution $\xi \in \mathbb{R}^n$ of the dynamical system $\dot{\boldsymbol{x}}(\tau) = f(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w})$, influenced by a pursuing player $\boldsymbol{P}$ (with control $\boldsymbol{w} \in \mathcal{W} \subseteq \mathbb{R}^p$) and its evading pair $\boldsymbol{E}$ (with control $u \in \mathcal{U} \subseteq \mathbb{R}^m$), results in system trajectories $\xi_{\boldsymbol{x},\tau}^{\boldsymbol{u},\boldsymbol{w}}$ that are obtained by optimizing a scalar-valued payoff functional, $\boldsymbol{\ell}(\xi_{\boldsymbol{x},t}^{\boldsymbol{u},\boldsymbol{w}}(\tau))$ i.e.

$$\mathcal{J}(\tau; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) = \min_{\tau \in [t,T]} \boldsymbol{\ell}(\xi_{\boldsymbol{x},t}^{\boldsymbol{u},\boldsymbol{w}}(\tau)) \qquad \text{(Payoff function)}$$

within a two-person differential game for a fixed $T \geq \tau > t$, where $\boldsymbol{\ell}$ can be a signed distance to the (boundary of the) target set/region, for example. The controllers belong in compact sets which are measurable functions i.e. $\bar{\mathcal{U}} \equiv \boldsymbol{u} : [t,T] \to \mathcal{U}$, $\bar{\mathcal{W}} \equiv \boldsymbol{w} : [t,T] \to \mathcal{W}$.

For high-dimensional murmuration of bird-agent systems (introduced in §D.1), we work in a multi-agent system context where individual agents self-organize into phases or regions $\mathcal{S}$, which are in turn members of a union of multiple regions $\mathcal{C}$. Note that every $\mathcal{S} \subseteq \mathcal{C}$ and all members of $\mathcal{C}$ are disjoint from one another i.e. $\mathcal{S}_i \cup \mathcal{S}_j = \emptyset$ for any $i \neq j$. The total number of elements in $\mathcal{S}$ is denoted $[\mathcal{S}]$.

A.2 Dynamic Programming and Two-Person Games.

The formal relationships between the dynamic programming (DP) optimality condition for the *value* in differential two-person zero-sum games, and the solutions to PDEs that solve "min-max" or "max-min" type nonlinearity (the Isaacs' equation) were presented in [Isaacs, 1999]. Essentially, Isaacs' claim was that if the *value* functions are smooth enough, then they solve certain first-order partial differential equations (PDE) problems with "max-min" or "min-max"-type nonlinearity. However, the DP value functions are seldom regular enough to admit a solution in the classical sense. "Weaker" solutions, on the other hand [Lions, 1982, Evans and Souganidis, 1984, Crandall et al., 1984, Crandall and Majda, 1980b], provide generalized "viscosity" solutions to HJ PDEs under relaxed regularity conditions; these viscosity solutions are not necessarily differentiable anywhere in the state space, and the only regularity prerequisite in the definition is continuity [Crandall and Lions, 1983]. However, wherever they are differentiable, they satisfy the upper and lower values of HJ PDEs (discussed in § A.3) in a classical sense. Thus, they lend themselves well to many real-world problems existing at the interface of discrete, continuous, and hybrid systems [Lygeros, 2004, Osher and Sethian, 1988, Mitchell, 2020, Evans and Souganidis, 1984, Mitchell et al., 2005].

Matter-of-factly, viscosity solutions to *Cauchy-type*[1] HJ Equations are highly useful in backward reachability analysis [Mitchell et al., 2005]. For a state $\boldsymbol{x} \in \Omega$ and a fixed time t: $0 \leq t < T$, suppose that the set of all controls for players $\boldsymbol{P}$ and $\boldsymbol{E}$ are respectively

$$\bar{\mathcal{U}} \equiv \{\boldsymbol{u} : [t,T] \to \mathcal{U} | \boldsymbol{u} \text{ measurable}, \mathcal{U} \in \mathbb{R}^m\}, \quad \bar{\mathcal{W}} \equiv \{\boldsymbol{w} : [t,T] \to \mathcal{V} | \boldsymbol{v} \text{ measurable}, \mathcal{W} \subset \mathbb{R}^p\}. \tag{A.1}$$

[1] Cauchy-type HJ equations are time-dependent versions of the HJ PDE.

We are concerned with the differential equation,

$$\dot{\boldsymbol{x}}(\tau) = f(\tau, \boldsymbol{x}(\tau), \boldsymbol{u}(\tau), \boldsymbol{w}(\tau)), \;\; \boldsymbol{x}(t) = \boldsymbol{x}, \;\; T \leq \tau \leq t \tag{A.2}$$

where $f(\tau, \cdot, \cdot, \cdot)$ and $\boldsymbol{x}(\cdot)$ are bounded and Lipschitz continuous. This bounded Lipschitz continuity property assures uniqueness of the system response $\boldsymbol{x}(\cdot)$ to controls $\boldsymbol{u}(\cdot)$ and $\boldsymbol{v}(\cdot)$ [Evans and Souganidis, 1984]. Associated with (A.2) is the payoff functional

$$\boldsymbol{\ell}(\boldsymbol{u}, \boldsymbol{w}) = \boldsymbol{\ell}(t; \boldsymbol{x}, \boldsymbol{u}(\cdot), \boldsymbol{w}(\cdot)) \equiv \int_t^T l(\tau, \boldsymbol{x}(\tau), \boldsymbol{u}(\tau), \boldsymbol{w}(\tau)) d\tau + \boldsymbol{g}(\boldsymbol{x}(T)), \tag{A.3}$$

where $\boldsymbol{g}(\cdot) : \mathbb{R}^n \to \mathbb{R}$ satisfies

$$|\boldsymbol{g}(\boldsymbol{x})| \leq k_1, \quad |\boldsymbol{g}(\boldsymbol{x}) - \boldsymbol{g}(\hat{\boldsymbol{x}}) | \leq k_1 |\boldsymbol{x} - \hat{\boldsymbol{x}}| \tag{A.4}$$

and $l : [0, T] \times \mathbb{R}^n \times \mathcal{U} \times \mathcal{W} \to \mathbb{R}$ is bounded and uniformly continuous, with

$$| l(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) | \leq k_2, \; | l(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) - l(t; \hat{\boldsymbol{x}}, \boldsymbol{u}, \boldsymbol{w}) | \leq k_2 \, | \, \boldsymbol{x} - \hat{\boldsymbol{x}} | \tag{A.5}$$

for constants k_1, k_2 and all $0 \leq t \leq T$, $\hat{\boldsymbol{x}}$, $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{u} \in \mathcal{U}$ and $\boldsymbol{w} \in \mathcal{W}$. We call T the *terminal time* (it may be infinity!) and the integral, when it does not depend on the control laws, is the *performance index*. The evader's goal is to maximize the payoff (A.3) and pursuer's goal is to minimize it.

A.3 Upper and Lower Values of the Differential Game.

Suppose that the pursuer's mapping strategy (starting at t) is $\beta : \bar{\mathcal{U}}(t) \to \bar{\mathcal{W}}(t)$ provided for each $t \leq \tau \leq T$ and $\boldsymbol{u}, \hat{\boldsymbol{u}} \in \bar{\mathcal{U}}(t)$; then $\boldsymbol{u}(\bar{t}) = \hat{\boldsymbol{u}}(\bar{t})$ a.e. on $t \leq \bar{t} \leq \tau$ implies $\beta[\boldsymbol{u}](\bar{t}) = \beta[\hat{\boldsymbol{u}}](\bar{t})$ a.e. on $t \leq \bar{t} \leq \tau$. The differential game's lower value for a solution $\boldsymbol{x}(t)$ that solves (A.2) for $\boldsymbol{u}(t)$ and $\boldsymbol{v}(t) = \beta[\boldsymbol{u}](\cdot)$ is

$$\boldsymbol{v}^-(t; \boldsymbol{x}) = \inf_{\beta \in \mathcal{B}(t)} \sup_{\boldsymbol{u} \in \mathcal{U}(t)} \boldsymbol{\ell}(\boldsymbol{u}, \beta[\boldsymbol{u}]) \triangleq \inf_{\beta \in \mathcal{B}(t)} \sup_{\boldsymbol{u} \in \mathcal{U}(t)} \int_t^T l(\tau, \boldsymbol{x}(\tau), \boldsymbol{u}(\tau), \beta[\boldsymbol{u}](\tau)) d\tau + \boldsymbol{g}\left(\boldsymbol{x}(T)\right). \tag{A.6}$$

Similarly, suppose that the evader's mapping strategy (starting at t) is $\alpha : \bar{\mathcal{W}}(t) \to \bar{\mathcal{U}}(t)$ provided for each $t \leq \tau \leq T$ and $\boldsymbol{w}, \hat{\boldsymbol{w}} \in \bar{\mathcal{W}}(t)$; then $\boldsymbol{w}(\bar{t}) = \hat{\boldsymbol{w}}(\bar{t})$ a.e. on $t \leq \bar{t} \leq \tau$ implies $\alpha[\boldsymbol{w}](\bar{t}) = \alpha[\hat{\boldsymbol{w}}](\bar{t})$ a.e. on $t \leq \bar{t} \leq \tau$. The differential game's upper value for a solution $\boldsymbol{x}(t)$ that solves (A.2) for $\boldsymbol{u}(t) = \alpha[\boldsymbol{w}](\cdot)$ and $\boldsymbol{w}(t)$ is

$$\boldsymbol{v}^+(t; \boldsymbol{x}) = \sup_{\alpha \in \mathcal{A}(t)} \inf_{\boldsymbol{w} \in \mathcal{W}(t)} \boldsymbol{\ell}(\alpha[\boldsymbol{w}], \boldsymbol{w}) \triangleq \sup_{\alpha \in \mathcal{A}(t)} \inf_{w \in \mathcal{W}(t)} \int_t^T l(\tau, \boldsymbol{x}(\tau), \alpha[\boldsymbol{w}](\tau), \boldsymbol{w}(\tau)) d\tau + \boldsymbol{g}\left(\boldsymbol{x}(T)\right). \tag{A.7}$$

These non-local PDEs ((A.6) and (A.7)) are hardly smooth throughout the state space so that they lack classical solutions even for smooth Hamiltonian and boundary conditions. However, these two values are "viscosity" (generalized) solutions [Lions, 1982, Crandall and Lions, 1983] of the associated HJ-Isaacs (HJI) PDE, i.e. solutions which are *locally Lipschitz* in $\Omega \times [0, T]$, and with at most first-order partial derivatives in the Hamiltonian.

A.4 Viscosity Solution of HJ-Isaac's Equations.

For any optimal control problem a value function is constructed based on the optimal cost (or payoff) of any input phase $(\boldsymbol{x}, T)$. In reachability analysis, typically this is defined using a terminal cost function $g(\cdot) : \mathbb{R}^n \to \mathbb{R}$ that satisfies

$$|\boldsymbol{g}(\boldsymbol{x})| \leq k, \;\; |\boldsymbol{g}(\boldsymbol{x}) - \boldsymbol{g}(\hat{\boldsymbol{x}}) | \leq k |\boldsymbol{x} - \hat{\boldsymbol{x}}| \tag{A.8}$$

for constant k and all $T \leq t \leq 0$, $\hat{\boldsymbol{x}}$, $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{u} \in \mathcal{U}$ and $\boldsymbol{w} \in \mathcal{W}$. The zero sublevel set of $\boldsymbol{g}(\boldsymbol{x})$ i.e.

$$\mathcal{L}_0 = \{\boldsymbol{x} \in \bar{\Omega} \,|\, \boldsymbol{g}(\boldsymbol{x}) \leq 0\}, \tag{A.9}$$

Lemma A.1. *The lower value $\boldsymbol{v}^-$ in (A.6) is the viscosity solution to the lower Isaac's equation*

$$\boldsymbol{v}_t^- + \boldsymbol{H}^-(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},D\boldsymbol{v}^-) = 0,\ t \in [0,T]\ \boldsymbol{x} \in \mathbb{R}^n, \quad \boldsymbol{v}^-(T;\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{x}(T)), \quad \boldsymbol{x} \in \mathbb{R}^m \tag{A.10}$$

with lower Hamiltonian,

$$\boldsymbol{H}^-(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},p) = \max_{u\in\mathcal{U}} \min_{w\in\mathcal{W}} \langle \boldsymbol{f}(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w}),p\rangle. \tag{A.11}$$

where p, the co-state, is the spatial derivative of $\boldsymbol{v}^-$ w.r.t $\boldsymbol{x}$.

Lemma A.2. *The upper value $\boldsymbol{v}^+$ in (A.7) is the viscosity solution of the upper Isaac's equation*

$$\boldsymbol{v}_t^+ + \boldsymbol{H}^+(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},D\boldsymbol{v}^+) = 0,\ t \in [0,T]\,,\ \boldsymbol{x} \in \mathbb{R}^n, \quad \boldsymbol{v}^+(T;\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{x}(T)), \quad \boldsymbol{x} \in \mathbb{R}^n \tag{A.12a}$$

with upper Hamiltonian,

$$\boldsymbol{H}^+(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},p) = \min_{\boldsymbol{w}\in\mathcal{W}} \max_{\boldsymbol{u}\in\mathcal{U}} \langle \boldsymbol{f}(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w}),p\rangle, \tag{A.13}$$

with p being appropriately defined.

Corollary A.3. *(i) $\boldsymbol{v}^- \leq \boldsymbol{v}^+$ over $(t \in [0,T]\ \boldsymbol{x} \in \mathbb{R}^n)$ (ii) if for all $t \in [0,T]\,, (\boldsymbol{x},p)\ \in \mathbb{R}^n$, the minimax condition is satisfied i.e. $\boldsymbol{H}^+(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},p) = \boldsymbol{H}^-(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},p)$, then $\boldsymbol{v}^- \equiv \boldsymbol{v}^+$.*

A.5 Reachability for Systems Verification.

Reachability analysis is one of many verification methods that allows us to reason about (control-affine) dynamical systems. The verification problem may consist in finding a *set of reachable states* that lie along the trajectory of the solution to a first order nonlinear partial differential equation that originates from some initial state $\boldsymbol{x}_0 = \boldsymbol{x}(0)$ up to a specified time bound, $t = t_f$. *From a set of initial and unsafe state sets, the time-bounded safety verification problem is to determine if there is an initial state and a particular time within the bound that the solution to HJI PDE enters the unsafe set*.

Reachability could be analyzed in a (i) *forward* sense, whereupon system trajectories are examined to determine if they enter certain states from an *initial set*; (ii) *backward* sense, whereupon system trajectories are examined to determine if they enter certain *target sets*; (iii) *reach set* sense, in which they are examined to see if states reach a set at a *particular time*; or (iv) *reach tube* sense, in which they are evaluated that they reach a set at a point *during a time interval*.

Backward reachability consists in avoiding an unsafe set of states under the worst-possible disturbance at all times; relying on nonanticipative control strategies. Backward reachable sets (BRS) and backward reachable tubes (BRTs) are popularly analyzed in a game of two vehicles with non-stochastic dynamics [Merz, 1972]. Such BRTs possess discontinuity at cross-over points (which exist at edges) on the surface of the tube, and may be non-convex. Therefore, treating the end-point constraints under these discontinuity characterizations need careful consideration and analysis when switching control laws if the underlying PDE does not have continuous partial derivatives (we discuss this further in § 2).

A.6 Robustly Controlled Backward Reachable Set and Tube

Suppose that the goal of $\boldsymbol{P}$ is to drive the system into a user-specified target region $\mathcal{L}_0(\tau)$ within $\tau \leq T$ time steps of playing the game; while $\boldsymbol{E}$ simultaneously seeks to prevent this from happening. The invariant set $\mathcal{L}_0$ obtained at T, i.e.

$$\mathcal{L}_0(T) = \{\boldsymbol{x} \in \mathbb{R}^n \,|\, \boldsymbol{v}(0;\boldsymbol{x}) \leq 0\} \tag{Target-Set}$$

is "*robustly controlled*" for the "*distance-to-target-set*" cost $\boldsymbol{g}(0;\boldsymbol{x})$, with constraints $|\boldsymbol{g}(0;\boldsymbol{x})| \leq k,\ |\boldsymbol{v}(0;\boldsymbol{x}) - \boldsymbol{v}(t;\hat{\boldsymbol{x}})\ | \leq k|\boldsymbol{x} - \hat{\boldsymbol{x}}|$ where $k > 0$ is a constant, and all $-T \leq t \leq 0$, $\{\hat{\boldsymbol{x}},\ \boldsymbol{x} \in \mathbb{R}^n\}$[2]. The distance to $\mathcal{L}_0(\tau)$ is typically found by optimizing $\boldsymbol{\ell}(\boldsymbol{x},t)$ as in (Payoff function). The HJI PDE (A.6) and Hamiltonian (A.11) for this set admit the form

$$\boldsymbol{v}_t(t;\boldsymbol{x}) + \min\{0, \boldsymbol{H}\,(t;\boldsymbol{x}, D\boldsymbol{v}(t;\boldsymbol{x}))\} = 0, \quad \boldsymbol{v}(0;\boldsymbol{x}) = \boldsymbol{g}(0;\boldsymbol{x}). \tag{HJI-RCBRT}$$

[2]Time is reversed in BRT computational scenarios.

For the safety problem setup in (A.6), the corresponding *robustly controlled backward reachable* ***tube*** (RCBRT) [Mitchell, 2020] on $(0, T]$ is the closure of the open set,

$$\mathcal{L}([-T, 0], \mathcal{L}_0) = \{\boldsymbol{x} \in \mathbb{R}^n \,|\, \exists\, \beta \in \mathcal{B}(t) \forall\, \boldsymbol{u} \in \mathcal{U}(t), \qquad \exists\, \bar{\tau} \in [-T, 0], \boldsymbol{\xi}(\bar{\tau}) \in \mathcal{L}_0\}, \tag{Target-Tube}$$

Read: The set of states from which there exists a strategy of $\boldsymbol{P}$ such that, for all controls of $\boldsymbol{E}$, the resulting trajectory *reaches and remains in the target set* within the interval $[-T, 0]$. Following Lemma 2 of [Mitchell et al., 2005], the states in the reachable set admit the following properties w.r.t the value function $\boldsymbol{v}$,

$$\boldsymbol{x}(t) \in \mathcal{L}(\cdot) \implies \boldsymbol{v}(t; \boldsymbol{x}) \leq 0, \;\; \boldsymbol{v}(t; \boldsymbol{x}) \leq 0 \implies \boldsymbol{x}(t) \in \mathcal{L}(\cdot). \tag{Zero Levelset}$$

In *backward reach avoid tubes*, the agent must avoid the unsafe region at all times. We can write (HJI-RCBRT) a *robustly controlled backward reach-avoid tube* (RCBRAT) as,

$$\min\{\boldsymbol{v}_t(t; \boldsymbol{x}) + \boldsymbol{H}\left(t; \boldsymbol{x}, D\boldsymbol{v}\right), \boldsymbol{g}(t; \boldsymbol{x}) - \boldsymbol{\ell}(t; \boldsymbol{x})\} \leq 0, \quad \boldsymbol{v}(0; \boldsymbol{x}) = \boldsymbol{g}(0; \boldsymbol{x}). \tag{HJI-RCBRAT}$$

B HJ PDE Linearization

In this section, we construct the Cole-Hopf-like linearization of the viscous Hamilton-Jacobi (HJ) PDE and propose a sampling machinery we use in building our results. We show that the transformation is *exact* only when the Hamiltonian is quadratic in the co-state i.e. $\boldsymbol{H} = \frac{1}{2}\langle \boldsymbol{p}^\top, \boldsymbol{p}\rangle$, and we derive the precise residual it incurs for general Hamiltonians. We then formulate the quasi-linearization iterative algorithm for computing (HJI-RCBRT-Visc) and (HJI-RCBRAT-Visc). We finish this appendix with the correct Gaussian expectation formulas for the value function and its spatial gradient.

B.1 The viscous HJ Equation's solution

In these proofs that follow, we will express the solutions to the viscous HJ equation as the logarithm of the expectation of Gaussian kernels that parameterize the state space. Let us construct the spatial and time derivatives to the solution to the viscous intial-value-HJ problem (HJ-Visc). We begin with the exact computation. Let us be exact about what survives the transformation: with a spatially-varying coefficient, no choice of exponent annihilates the nonlinearity outright, and the lemma below records precisely what remains.

Lemma B.1 (Exact residual of the generalized Cole-Hopf transformation)**.** *Let $\boldsymbol{v}^\delta \in C^{1,2}$ solve (HJ-Visc), let $\boldsymbol{c} \in C^{1,2}\left(\bar{\Omega}\times[0,T]\right)$, and set $\boldsymbol{\omega}^\delta \triangleq \exp(-\boldsymbol{c}\,\boldsymbol{v}^\delta)$. Then, writing $\boldsymbol{v}$ for $\boldsymbol{v}^\delta$ and suppressing arguments,*

$$\boldsymbol{\omega}_t^\delta - \frac{\delta}{2}\Delta\boldsymbol{\omega}^\delta = \boldsymbol{\omega}^\delta \underbrace{\left[\boldsymbol{c}\Big(\boldsymbol{H} - \frac{\delta}{2}\boldsymbol{c}\,|D\boldsymbol{v}|^2\Big)\right]}_{\text{algebraic residual, } \boldsymbol{R}_{\text{alg}}} + \boldsymbol{\omega}^\delta \underbrace{\left[-\boldsymbol{c}_t\boldsymbol{v} + \frac{\delta}{2}(\Delta\boldsymbol{c})\boldsymbol{v} - \frac{\delta}{2}|D\boldsymbol{c}|^2\boldsymbol{v}^2 + \delta\,(1-\boldsymbol{c}\boldsymbol{v})\,D\boldsymbol{c}\cdot D\boldsymbol{v}\right]}_{\text{derivative residual, } \boldsymbol{R}_{\text{der}}}. \tag{B.1}$$

Proof of Lemma B.1. Write $\varphi \triangleq \boldsymbol{c}\,\boldsymbol{v}$ so that $\boldsymbol{\omega}^\delta = e^{-\varphi}$. Then,

$$\boldsymbol{\omega}_t^\delta = -\varphi_t\,\boldsymbol{\omega}^\delta, \qquad D\boldsymbol{\omega}^\delta = -D\varphi\,\boldsymbol{\omega}^\delta, \qquad \Delta\boldsymbol{\omega}^\delta = \left(|D\varphi|^2 - \Delta\varphi\right)\boldsymbol{\omega}^\delta, \tag{B.2}$$

with $\varphi_t = \boldsymbol{c}_t\boldsymbol{v} + \boldsymbol{c}\boldsymbol{v}_t$, $D\varphi = \boldsymbol{v}\,D\boldsymbol{c} + \boldsymbol{c}\,D\boldsymbol{v}$, and $\Delta\varphi = \boldsymbol{v}\,\Delta\boldsymbol{c} + 2\,D\boldsymbol{c}\cdot D\boldsymbol{v} + \boldsymbol{c}\,\Delta\boldsymbol{v}$. Substituting these into $\boldsymbol{\omega}_t^\delta - \frac{\delta}{2}\Delta\boldsymbol{\omega}^\delta$ and eliminating $\boldsymbol{v}_t = \frac{\delta}{2}\Delta\boldsymbol{v} - \boldsymbol{H}$ via (HJ-Visc),

$$\begin{aligned}\boldsymbol{\omega}_t^\delta - \frac{\delta}{2}\Delta\boldsymbol{\omega}^\delta &= \boldsymbol{\omega}^\delta\left[-\boldsymbol{c}_t\boldsymbol{v} - \boldsymbol{c}\boldsymbol{v}_t + \frac{\delta}{2}\left(\boldsymbol{v}\Delta\boldsymbol{c} + 2D\boldsymbol{c}\cdot D\boldsymbol{v} + \boldsymbol{c}\Delta\boldsymbol{v}\right) - \frac{\delta}{2}\left|\boldsymbol{v}D\boldsymbol{c} + \boldsymbol{c}D\boldsymbol{v}\right|^2\right] \\ &= \boldsymbol{\omega}^\delta\left[\boldsymbol{c}\boldsymbol{H} - \frac{\delta}{2}\boldsymbol{c}^2|D\boldsymbol{v}|^2 - \boldsymbol{c}_t\boldsymbol{v} + \frac{\delta}{2}(\Delta\boldsymbol{c})\boldsymbol{v} - \frac{\delta}{2}|D\boldsymbol{c}|^2\boldsymbol{v}^2 + \delta(1-\boldsymbol{c}\boldsymbol{v})D\boldsymbol{c}\cdot D\boldsymbol{v}\right],\end{aligned} \tag{B.3}$$

where the $\boldsymbol{c}\Delta\boldsymbol{v}$ terms cancel and we expanded $|\boldsymbol{v}D\boldsymbol{c}+\boldsymbol{c}D\boldsymbol{v}|^2 = \boldsymbol{v}^2|D\boldsymbol{c}|^2 + 2\boldsymbol{v}\boldsymbol{c}\,D\boldsymbol{c}\cdot D\boldsymbol{v} + \boldsymbol{c}^2|D\boldsymbol{v}|^2$. Grouping the first two terms as $\boldsymbol{R}_{\text{alg}}$ and the remainder as $\boldsymbol{R}_{\text{der}}$ gives (B.1). □

Two observations organize everything that follows. First, the *algebraic* residual $\boldsymbol{R}_{\text{alg}}$ vanishes identically upon the choice,

$$\boldsymbol{c}(t;\boldsymbol{x}) = \tfrac{2}{\delta}\cdot\boldsymbol{H}^\delta/|D\boldsymbol{v}^\delta|^2, \tag{B.4}$$

which is precisely the coefficient (4) of the main text. Second, the *derivative* residual $\boldsymbol{R}_{\text{der}}$ is beyond the reach of any algebraic choice: it is built from $\boldsymbol{c}_t, D\boldsymbol{c}, \Delta\boldsymbol{c}$, and no freezing of the value of $\boldsymbol{c}$ removes the derivatives of $\boldsymbol{c}$. The frozen-coefficient step of Algorithm 1 therefore does *not* transform (HJ-Visc) into a heat equation; rather, it *defines* the next iterate as the exact solution of the linear heat initial-value problem, and $\boldsymbol{R}_{\text{der}}$ — discarded at each step — is exactly the defect whose Duhamel norm we bound in Theorem C.4 (§ C.2).

Corollary B.2 (Exactness in the quadratic case)**.** *The full residual in (B.1) vanishes identically iff $\boldsymbol{R}_{\text{alg}} \equiv 0$ and $\boldsymbol{c}$ is constant in $(t;\boldsymbol{x})$. In particular, when $\boldsymbol{H} = \frac{1}{2}\langle \boldsymbol{p}, \boldsymbol{p}\rangle$, the choice (B.4) gives $\boldsymbol{c} = 1/\delta = \text{const}$, both residuals vanish, and $\boldsymbol{\omega}^\delta = \exp(-\boldsymbol{v}^\delta/\delta)$ satisfies the heat equation exactly — the classical Cole-Hopf reduction.*

Proof of Proposition 2.1. With $\boldsymbol{c}^{(k)}$ frozen at the current iterate, the algorithm step is, by construction, the initial value problem,

$$\boldsymbol{\omega}_t^\delta - \frac{\delta}{2}\Delta\boldsymbol{\omega}^\delta = 0 \text{ in } \Omega\times(0,T], \quad \boldsymbol{\omega}^\delta(0;\boldsymbol{x}) = \exp\left(-\boldsymbol{c}^{(k)}(0;\boldsymbol{x})\,\boldsymbol{g}(\boldsymbol{x})\right) \text{ on } \bar{\Omega}\times\{t=0\}, \quad \text{(B.5)}$$

whose solution is unique and bounded for bounded continuous data. By Lemma B.1 and Corollary B.2, this step is an exact transformation of (HJ-Visc) when $\boldsymbol{H} = \frac{1}{2}|\boldsymbol{p}|^2$; for general Hamiltonians it discards the derivative residual $\boldsymbol{R}_{\text{der}}$ of (B.1), and is to be read as one step of the Picard quasi-linearization whose fixed-point defect is quantified in § C.2. Equation (B.5) is the classical Cole-Hopf-transformed equation [Evans, 2022] of the Eikonal version of (HJ-Visc)[3].

It follows that the unique bounded solution of (B.5) is given by the Green's function convolution,

$$\boldsymbol{\omega}^\delta(t;\boldsymbol{x}) = \frac{1}{(\sqrt{2\pi\delta t})^n}\int_\Omega \exp\left(-\frac{1}{2\delta t}\mid \boldsymbol{x}-\boldsymbol{y}\mid^2\right)\exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right)d\boldsymbol{y}, \quad \text{(B.6a)}$$

$$\triangleq \mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}\left[\exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right)\right] := \mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}\left[\exp\left(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y})\right)\right] \quad \text{(B.6b)}$$

where (B.6a) is the standard heat kernel convolution and the kernel has been rewritten in (B.6b), via the Feynman-Kac formula, as a Gaussian density with mean $\boldsymbol{x}$ and covariance $\delta t I_n$. Note that $\boldsymbol{\omega}^\delta(t;\boldsymbol{x})$ in (B.6) is valid in $\Omega\times[0,T]$, whereas it is 0 on the boundary $\partial\Omega\times[0,T]$. □

Proof of Lemma 2.2. Going by the transformation (Linear-Trans), we can write

$$\boldsymbol{v}^\delta = -(\delta/2)\frac{|D\boldsymbol{v}^\delta|^2}{\boldsymbol{H}(t;\boldsymbol{x},D\boldsymbol{v}^\delta)}\log\boldsymbol{\omega}^\delta := -\frac{1}{\boldsymbol{c}}\log\boldsymbol{\omega}^\delta, \quad \text{(B.7)}$$

so that the unique bounded solution to the initial-value (HJ-Visc) i.e. (7) becomes

$$\boldsymbol{v}^\delta(t;\boldsymbol{x}) = -\frac{1}{\boldsymbol{c}}\cdot\log\left\{\frac{1}{(\sqrt{2\pi\delta t})^n}\int_\Omega\exp\left(-\frac{1}{2\delta t}\cdot|\boldsymbol{x}-\boldsymbol{y}|^2\right)\exp\left(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y})\right)\right\}, \quad \text{(B.8a)}$$

$$\triangleq -\frac{1}{\boldsymbol{c}}\cdot\log\left\{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}\left[\exp\left(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y})\right)\right]\right\}. \quad \text{(B.8b)}$$

Corollary B.3. *For the backward reachability problems* (HJI-RCBRT-Visc) *and* (HJI-RCBRAT-Visc) *over time horizon* $[0,T]$*, the solution is given by the log-sum-exp identity with samples from the correct backward-time covariance:*

$$\boldsymbol{v}^\delta(t;\boldsymbol{x}) = \frac{-1}{\boldsymbol{c}}\log\frac{1}{N}\sum_{i=1}^N\exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y}_i)\right), \quad \textit{with } \boldsymbol{y}_i \overset{iid}{\sim} \mathcal{N}(\boldsymbol{x},\delta t I_n). \quad \text{(B.9)}$$

A fortiori, we have the solution to the viscous HJ equation as the log of the expectation of a Gaussian density with mean $\boldsymbol{x}$ and covariance $\delta t I_n$. □

B.2 Spatial Gradient of the HJ Payoff

Proof of Lemma 2.3. From (B.6a) observe,

$$D\boldsymbol{\omega}^\delta(t;\boldsymbol{x}) = -\frac{1}{(\sqrt{2\pi\delta t})^n}\int_\Omega\frac{(\boldsymbol{x}-\boldsymbol{y})}{\delta t}\exp\left(-\frac{\mid\boldsymbol{x}-\boldsymbol{y}\mid^2}{2\delta t}\right)\exp\left(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y})\right)d\boldsymbol{y}, \quad \text{(B.10a)}$$

$$= -\frac{1}{\delta t}\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}\left[(\boldsymbol{x}-\boldsymbol{y})\exp\left(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y})\right)\right]. \quad \text{(B.10b)}$$

Inspecting (B.7), we may write

$$D\boldsymbol{v}^\delta(t;\boldsymbol{x}) = -\frac{1}{\boldsymbol{c}}D[\log\boldsymbol{\omega}^\delta(t;\boldsymbol{x})] = -\frac{1}{\boldsymbol{c}}\frac{D\boldsymbol{\omega}^\delta(t;\boldsymbol{x})}{\boldsymbol{\omega}^\delta(t;\boldsymbol{x})}$$

$$= \frac{1}{\delta\cdot t\cdot\boldsymbol{c}}\cdot\frac{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}\left[(\boldsymbol{x}-\boldsymbol{y})\exp(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y}))\right]}{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}\left[\exp(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y}))\right]}, \quad \text{(B.11)}$$

$$\triangleq \frac{1}{t\cdot\delta\cdot\boldsymbol{c}}\cdot\left(\boldsymbol{x}-\frac{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}\left[\boldsymbol{y}\exp(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y}))\right]}{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}\left[\exp(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y}))\right]}\right). \quad \text{(B.12)}$$

□

[3] For the Eikonal version of (HJ-Visc), we set $\boldsymbol{H}(D\boldsymbol{v}(\boldsymbol{x})) = 0$ in Ω, and $\boldsymbol{v}(0;\boldsymbol{x}) = 0$ on $\partial\Omega$.

Corollary B.4 (Log-sum-exp estimator for the value function)**.** *For* (HJI-RCBRT-Visc) *and* (HJI-RCBRAT-Visc)*, the terminal cost argument transforms to* $\boldsymbol{g}(\boldsymbol{x} + \sqrt{\delta t}\boldsymbol{y}_i)$*. For numerical stability the* log-sum-exp *identity gives,*

$$\boldsymbol{v}^{\delta}(t;\boldsymbol{x}) = \frac{-1}{\boldsymbol{c}^{(k)}} \log \frac{1}{N} \sum_{i=1}^{N} \exp\Big(-\boldsymbol{c}^{(k)} \boldsymbol{g}\Big(\boldsymbol{x} + \sqrt{\delta t}\boldsymbol{y}_i\Big)\Big)\,, \tag{B.13}$$

with $\boldsymbol{y}_i \overset{iid}{\sim} \mathcal{N}(\boldsymbol{0}, I_n)$*.*

Corollary B.5 (Monte Carlo gradient estimator)**.** *With samples* $\boldsymbol{s}_i = \boldsymbol{x} + \sqrt{\delta t}\boldsymbol{y}_i$*,* $\boldsymbol{y}_i \overset{iid}{\sim} \mathcal{N}(\boldsymbol{0}, I_n)$*, the importance-weighted estimator for* (9) *is,*

$$D\boldsymbol{v}^{\delta}(t;\boldsymbol{x}) = \frac{1}{t \cdot \delta \cdot \boldsymbol{c}^{(k)}} \left(\boldsymbol{x} - \frac{\frac{1}{N}\sum_{i=1}^{N} \boldsymbol{s}_i \cdot \exp\big(-\boldsymbol{c}^{(k)} \boldsymbol{g}(\boldsymbol{s}_i)\big)}{\frac{1}{N}\sum_{i=1}^{N} \exp\big(-\boldsymbol{c}^{(k)} \boldsymbol{g}(\boldsymbol{s}_i)\big)} \right). \tag{B.14}$$

Corollary B.6 (Tilted-proposal estimators)**.** *Fix a shift* $\theta \in \mathbb{R}^n$ *and let* $\sigma^2 \triangleq \delta t$*. Draw* $\boldsymbol{y}_i \overset{iid}{\sim} q_\theta \triangleq \mathcal{N}(\boldsymbol{x} + \theta, \sigma^2 I_n)$ *and set the density-ratio and total weights,*

$$w_\theta(\boldsymbol{y}) \triangleq \exp\left(\frac{|\theta|^2 - 2\langle \boldsymbol{y} - \boldsymbol{x}, \theta \rangle}{2\sigma^2} \right), \qquad \tilde{w}_i \triangleq w_\theta(\boldsymbol{y}_i)\, \exp\Big(-\boldsymbol{c}^{(k)} \boldsymbol{g}(\boldsymbol{y}_i)\Big)\,. \tag{B.15}$$

Then the tilted value and gradient estimators,

$$\hat{\boldsymbol{v}}_\theta^{\delta}(t;\boldsymbol{x}) = -\frac{1}{\boldsymbol{c}^{(k)}} \log \left(\frac{1}{N} \sum_{i=1}^{N} \tilde{w}_i \right), \tag{B.16}$$

$$\widehat{D\boldsymbol{v}}_\theta^{\delta}(t;\boldsymbol{x}) = \frac{1}{t \cdot \delta \cdot \boldsymbol{c}^{(k)}} \left(\boldsymbol{x} - \frac{\sum_{i=1}^{N} \boldsymbol{y}_i\, \tilde{w}_i}{\sum_{i=1}^{N} \tilde{w}_i} \right), \tag{B.17}$$

are consistent for (8) *and* (9) *respectively, for* every θ*; the inner average of* (B.16) *is unbiased for* $\mathbb{E}_{\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{x}, \sigma^2 I_n)}[e^{-\boldsymbol{c}^{(k)} \boldsymbol{g}(\boldsymbol{y})}]$*. The first-order Laplace choice* $\theta^\star = -\sigma^2 \boldsymbol{c}^{(k)} D\boldsymbol{v}^{(k)}(t;\boldsymbol{x})$ *of* (11) *Gaussianizes the zero-variance proposal.*

Proof of Corollary B.6. Unbiasedness is the change of measure: with $\varphi_{\boldsymbol{x}}$ the $\mathcal{N}(\boldsymbol{x}, \sigma^2 I_n)$ density, $w_\theta = \varphi_{\boldsymbol{x}}/q_\theta$ by direct computation of the two exponents, so $\mathbb{E}_{q_\theta}[w_\theta e^{-\boldsymbol{c}\boldsymbol{g}}] = \int \varphi_{\boldsymbol{x}}(\boldsymbol{y}) e^{-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})} d\boldsymbol{y} = \mathbb{E}_{\varphi_{\boldsymbol{x}}}[e^{-\boldsymbol{c}\boldsymbol{g}}]$, and likewise for the numerator of (B.17); the ratio and the logarithm preserve consistency by the continuous mapping theorem. For the shift, the zero-variance proposal is proportional to the integrand $\varphi_{\boldsymbol{x}}(\boldsymbol{y}) e^{-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})} \propto \exp\left(-|\boldsymbol{y} - \boldsymbol{x}|^2/2\sigma^2 - \boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right)$; linearizing $\boldsymbol{g}$ about $\boldsymbol{x}$ and completing the square in the exponent gives $\mathcal{N}\left(\boldsymbol{x} - \sigma^2 \boldsymbol{c}\, D\boldsymbol{g}(\boldsymbol{x}), \sigma^2 I_n\right)$, and replacing $D\boldsymbol{g}$ by the running gradient iterate $D\boldsymbol{v}^{(k)}$ yields (11). □

B.3 Sampling Complexity Analysis

We now analyze the sampling complexity of Algorithm 1.

Proof of Theorem 2.5. Recall that,

$$Z \triangleq \exp(-\boldsymbol{c}\, \boldsymbol{g}(\boldsymbol{\zeta})), \quad \mu \triangleq \mathbb{E}[Z], \quad \boldsymbol{v}_{\boldsymbol{c}}(t;\boldsymbol{x}) \triangleq -\frac{1}{\boldsymbol{c}} \log \mu. \tag{B.18}$$

For i.i.d. samples $\boldsymbol{\zeta}_1, \ldots, \boldsymbol{\zeta}_N \sim \mathcal{N}(\boldsymbol{x}, \delta t I_n)$, $Z_i \triangleq \exp(-\boldsymbol{c}\, \boldsymbol{g}(\boldsymbol{\zeta}_i))$, so that $\bar{Z}_N \triangleq \frac{1}{N}\sum_{i=1}^{N} Z_i$, and $\hat{\boldsymbol{v}}_{\boldsymbol{c},N}(t;\boldsymbol{x}) \triangleq -\frac{1}{\boldsymbol{c}} \log \bar{Z}_N$.

Let $\alpha \triangleq e^{-\boldsymbol{c} g_{\max}}$, $\beta \triangleq e^{-\boldsymbol{c} g_{\min}}$. Since $g_{\min} \le \boldsymbol{g}(\boldsymbol{\zeta}) \le g_{\max}$ and $\boldsymbol{c} > 0$, monotonicity of the exponential implies

$$e^{-\boldsymbol{c} g_{\max}} \le e^{-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{\zeta})} \le e^{-\boldsymbol{c} g_{\min}} \tag{B.19}$$

almost surely. By definition of α and β, this is exactly $\alpha \le Z \le \beta$ almost surely. Hence each Z_i is bounded in $[\alpha, \beta]$.

Value residuals, $\bar{Z}_N$. Since $\boldsymbol{v}_{\boldsymbol{c}}(t;\boldsymbol{x}) = -\frac{1}{\boldsymbol{c}}\log\mu$ and $\hat{\boldsymbol{v}}_{\boldsymbol{c},N}(t;\boldsymbol{x}) = -\frac{1}{\boldsymbol{c}}\log\bar{Z}_N$, we find that

$$\hat{\boldsymbol{v}}_{\boldsymbol{c},N}(t;\boldsymbol{x}) - \boldsymbol{v}_{\boldsymbol{c}}(t;\boldsymbol{x}) = -\frac{1}{\boldsymbol{c}}\log\left(\bar{Z}_N/\mu\right). \tag{B.20}$$

Hence, $|\hat{\boldsymbol{v}}_{\boldsymbol{c},N}(t;\boldsymbol{x}) - \boldsymbol{v}_{\boldsymbol{c}}(t;\boldsymbol{x})| \geq \varepsilon$ implies that $\left|\log\left(\bar{Z}_N/\mu\right)\right| \geq \boldsymbol{c}\varepsilon$, or that $\bar{Z}_N \geq \mu\exp(\boldsymbol{c}\varepsilon)$ or $\bar{Z}_N \leq \mu\exp(-\boldsymbol{c}\varepsilon)$. Now, we can convert the log-scale deviation in the viscous HJ value $|\hat{v}_{c,N} - v_c| \geq \varepsilon$ into two linear-scale deviation events on the sample mean $\bar{Z}_N$, as

$$\begin{aligned}\mathbb{P}\left(|\hat{\boldsymbol{v}}_{\boldsymbol{c},N}(t;\boldsymbol{x}) - \boldsymbol{v}_{\boldsymbol{c}}(t;\boldsymbol{x})| \geq \varepsilon\right) \leq{}& \mathbb{P}\left(\bar{Z}_N - \mu \geq \mu(\exp(\boldsymbol{c}\varepsilon) - 1)\right) \\ &+ \mathbb{P}\left(\bar{Z}_N - \mu \leq -\mu(1 - \exp(-\boldsymbol{c}\varepsilon))\right).\end{aligned} \tag{B.21}$$

Hoeffding bounds for tails. We now employ Hoeffding's concentration inequality to bound how far the sample mean, $\bar{Z}_N$ can deviate from the true mean in the presence of the bounded variables $\hat{\boldsymbol{v}}_{\boldsymbol{c},N}(t;\boldsymbol{x}), \boldsymbol{v}_{\boldsymbol{c}}(t;\boldsymbol{x})$.

Because $Z_1, \ldots, Z_N$ are i.i.d. and each lies in $[\alpha, \beta]$, Hoeffding's inequality implies that for every $s > 0$,

$$\mathbb{P}(\bar{Z}_N - \mu \geq s) \leq \exp\left(-\frac{2Ns^2}{(\beta-\alpha)^2}\right), \qquad \mathbb{P}(\bar{Z}_N - \mu \leq -s) \leq \exp\left(-\frac{2Ns^2}{(\beta-\alpha)^2}\right). \tag{B.22}$$

Applying the first inequality with $s_+ \triangleq \mu(\exp(\boldsymbol{c}\varepsilon) - 1)$ and the second with $s_- \triangleq \mu(1 - \exp(-\boldsymbol{c}\varepsilon))$, we find that

$$\mathbb{P}\left(\bar{Z}_N - \mu \geq \mu(\exp(\boldsymbol{c}\varepsilon) - 1)\right) \leq \exp\left(-\frac{2N\mu^2(\exp(\boldsymbol{c}\varepsilon) - 1)^2}{(\beta-\alpha)^2}\right), \tag{B.23a}$$

$$\mathbb{P}\left(\bar{Z}_N - \mu \leq -\mu(1 - \exp(-\boldsymbol{c}\varepsilon))\right) \leq \exp\left(-\frac{2N\mu^2(1 - \exp(-\boldsymbol{c}\varepsilon))^2}{(\beta-\alpha)^2}\right). \tag{B.23b}$$

Union bound and simplification. Factoring the previous bounds (B.23) into (B.21), we must have

$$\begin{aligned}\mathbb{P}\left(|\hat{\boldsymbol{v}}_{\boldsymbol{c},N}(t;\boldsymbol{x}) - \boldsymbol{v}_{\boldsymbol{c}}(t;\boldsymbol{x})| \geq \varepsilon\right) \leq{}& \exp\left(-\frac{2N\mu^2(\exp(\boldsymbol{c}\varepsilon) - 1)^2}{(\beta-\alpha)^2}\right) \\ &+ \exp\left(-\frac{2N\mu^2(1 - \exp(-\boldsymbol{c}\varepsilon))^2}{(\beta-\alpha)^2}\right).\end{aligned} \tag{B.24}$$

Since $\exp(\boldsymbol{c}\varepsilon) - 1 \geq 1 - \exp(-\boldsymbol{c}\varepsilon)$ for every $\varepsilon > 0$, the first exponential is no larger than the second. Therefore

$$\mathbb{P}\left(|\hat{\boldsymbol{v}}_{\boldsymbol{c},N}(t;\boldsymbol{x}) - \boldsymbol{v}_{\boldsymbol{c}}(t;\boldsymbol{x})| \geq \varepsilon\right) \leq 2\exp\left(-\frac{2N\mu^2(1 - e^{-\boldsymbol{c}\varepsilon})^2}{(\beta-\alpha)^2}\right). \tag{B.25}$$

A fortiori, this proves the claim of Theorem 2.5. □

Proof of Corollary 2.7. Since $\alpha \leq Z$ almost surely, taking expectations yields $\mu = \mathbb{E}[Z] \geq \alpha$. Hence it is enough to require

$$2\exp\left(-\frac{2N\alpha^2(1 - e^{-\boldsymbol{c}\varepsilon})^2}{(\beta-\alpha)^2}\right) \leq \alpha. \tag{B.26}$$

Taking logarithms and solving for N gives

$$N \geq \frac{(\beta-\alpha)^2}{2\alpha^2(1 - e^{-\boldsymbol{c}\varepsilon})^2}\log\frac{2}{\alpha}, \tag{B.27}$$

which proves the second claim. □

Proof of Theorem 2.12. We proceed in several steps.

Step 1: Lipschitz continuity of the frozen-coefficient solve map Φ. Fix an index m and define

$$\psi_m(s) \triangleq -\frac{1}{s}\log M_m(s), \qquad M_m(s) \triangleq \mathbb{E}_{\boldsymbol{\zeta}\sim\mathcal{N}(\boldsymbol{x}_m,\delta t I_n)}\left[\exp\left(-s\boldsymbol{g}(\boldsymbol{\zeta})\right)\right]. \tag{B.28}$$

Then $(\Phi(c))_m = \psi_m(c_m)$. Since $M_m(s) > 0$,

$$\frac{d}{ds}(\psi_m(s)) = \frac{1}{s^2}\log M_m(s) - \frac{1}{s}\frac{M_m'(s)}{M_m(s)}. \tag{B.29}$$

Differentiating under the expectation gives

$$\frac{d}{ds}(M_m(s)) = \mathbb{E}_{\boldsymbol{\zeta}\sim\mathcal{N}(\boldsymbol{x}_m,\delta t I_n)}\left[-\boldsymbol{g}(\boldsymbol{\zeta})\exp\left(-s\boldsymbol{g}(\boldsymbol{\zeta})\right)\right]. \tag{B.30}$$

Thus,

$$-\frac{M_m'(s)}{M_m(s)} = \frac{\mathbb{E}\left[\boldsymbol{g}(\cdot)e^{-s\boldsymbol{g}(\cdot)}\right]}{\mathbb{E}\left[e^{-s\boldsymbol{g}(\cdot)}\right]}, \tag{B.31}$$

which is a weighted expectation of $\boldsymbol{g}$; and possesses absolute value of at most G by assumption 1. In addition, $|\boldsymbol{g}| \le G$ implies that

$$e^{-sG} \le M_m(s) \le e^{sG}. \tag{B.32}$$

We can therefore write $|\log M_m(s)| \le sG$. Substituting these two bounds into the derivative formula yields

$$|\psi_m'(s)| \le \frac{1}{s^2}(sG) + \frac{1}{s}G = \frac{2G}{s} \le \frac{2G}{c_{\min}} \tag{B.33}$$

for every $s \in [c_{\min}, c_{\max}]$. By the mean-value theorem,

$$|\psi_m(c_m) - \psi_m(\widetilde{c}_m)| \le \frac{2G}{c_{\min}}|c_m - \widetilde{c}_m|. \tag{B.34}$$

Taking the maximum over m, we must have

$$\|\Phi(c) - \Phi(\widetilde{c})\|_\infty \le \frac{2G}{c_{\max}}\|c - \widetilde{c}\|_\infty. \tag{B.35}$$

Step 2: Lipschitz continuity of the coefficient-update map Γ. Fix m and write $p \triangleq p_m(v)$, $q \triangleq p_m(w)$. Then

$$|(\Gamma(v))_m - (\Gamma(w))_m| = \frac{2}{\delta}\left|\frac{\boldsymbol{H}(t;\boldsymbol{x}_m,p)}{|p|^2} - \frac{\boldsymbol{H}(t;\boldsymbol{x}_m,q)}{|q|^2}\right|. \tag{B.36}$$

Add and subtract $\boldsymbol{H}(t;\boldsymbol{x}_m,q)/|p|^2$ to obtain

$$\left|\frac{\boldsymbol{H}(t;\boldsymbol{x}_m,p)}{|p|^2} - \frac{\boldsymbol{H}(t;\boldsymbol{x}_m,q)}{|q|^2}\right| \le \frac{|\boldsymbol{H}(t;\boldsymbol{x}_m,p) - \boldsymbol{H}(t;\boldsymbol{x}_m,q)|}{|p|^2} + |\boldsymbol{H}(t;\boldsymbol{x}_m,q)|\left|\frac{1}{|p|^2} - \frac{1}{|q|^2}\right|. \tag{B.37}$$

By assumption (ii), $|p|, |q| \ge m_0$, so that $|p|^2, |q|^2 \ge m_0^2$. Using assumption (iii), the first term is bounded by

$$\frac{|\boldsymbol{H}(t;\boldsymbol{x}_m,p) - \boldsymbol{H}(t;\boldsymbol{x}_m,q)|}{|p|^2} \le \frac{L_H}{m_0^2}|p - q|. \tag{B.38}$$

For the second term, note that

$$\left|\frac{1}{|p|^2} - \frac{1}{|q|^2}\right| = \frac{\left||q|^2 - |p|^2\right|}{|p|^2|q|^2} \le \frac{\left||q| - |p|\right|(|q| + |p|)}{m_0^4} \le \frac{2P_*}{m_0^4}|p - q|, \tag{B.39}$$

where the final inequality uses $|p|, |q| \le P_*$ from assumption (ii). Using $|\boldsymbol{H}(t;\boldsymbol{x}_m,q)| \le H_*$ from assumption (iii), we conclude that

$$\left|\frac{\boldsymbol{H}(t;\boldsymbol{x}_m,p)}{|p|^2} - \frac{\boldsymbol{H}(t;\boldsymbol{x}_m,q)}{|q|^2}\right| \le \left(\frac{L_H}{m_0^2} + \frac{2H_*P_*}{m_0^4}\right)|p - q|. \tag{B.40}$$

Therefore

$$|(\Gamma(v))_m - (\Gamma(w))_m| \le \frac{2}{\delta}\left(\frac{L_H}{m_0^2} + \frac{2H_*P_*}{m_0^4}\right)|p-q|. \tag{B.41}$$

Finally, by assumption (iv),

$$|p-q| \le \|\mathcal{G}(v) - \mathcal{G}(w)\|_\infty \le L_D\|v-w\|_\infty. \tag{B.42}$$

Taking the maximum over m yields

$$\|\Gamma(v) - \Gamma(w)\|_\infty \le \frac{2L_D}{\delta}\left(\frac{L_H}{m_0^2} + \frac{2H_*P_*}{m_0^4}\right)\|v-w\|_\infty. \tag{B.43}$$

Step 3: $\Lambda = \Phi \circ \Gamma$ is a contraction. Combining (B.35) and (B.43), we obtain

$$\|\Lambda(v) - \Lambda(w)\|_\infty = \|\Phi(\Gamma(v)) - \Phi(\Gamma(w))\|_\infty \le \frac{2G}{c_{\min}}\|\Gamma(v) - \Gamma(w)\|_\infty, \tag{B.44}$$

$$\le \frac{2G}{c_{\min}} \cdot \frac{2L_D}{\delta}\left(\frac{L_H}{m_0^2} + \frac{2H_*P_*}{m_0^4}\right)\|v-w\|_\infty = q\|v-w\|_\infty. \tag{B.45}$$

Assumption (v) states that $q < 1$, hence Λ is a contraction on $\mathcal{A}$.

Step 4: Existence, uniqueness, and linear convergence. Since $\mathcal{A}$ is closed in the Banach space $(V, \|\cdot\|_\infty)$ and invariant under Λ, Banach's fixed-point theorem applies. Therefore there exists a unique $v^* \in \mathcal{A}$ such that $\Lambda(v^*) = v^*$. Moreover, for every $v^{(0)} \in \mathcal{A}$, the sequence generated by $v^{(k+1)} = \Lambda(v^{(k)})$ converges to v^* and satisfies

$$\|v^{(k+1)} - v^*\|_\infty \le q\|v^{(k)} - v^*\|_\infty \le q^{k+1}\|v^{(0)} - v^*\|_\infty. \tag{B.46}$$

This proves the first convergence statement.

Step 5: Convergence of the coefficients. Applying (B.43) with $w = v^*$ gives

$$\|\Gamma(v^{(k)}) - \Gamma(v^*)\|_\infty \le \frac{2L_D}{\delta}\left(\frac{L_H}{m_0^2} + \frac{2H_*P_*}{m_0^4}\right)\|v^{(k)} - v^*\|_\infty. \tag{B.47}$$

Step 6: Residual decay. Since $v^{(k+1)} = \Lambda(v^{(k)})$ and Λ is a contraction,

$$\|v^{(k+1)} - v^{(k)}\|_\infty = \|\Lambda(v^{(k)}) - \Lambda(v^{(k-1)})\|_\infty \le q\|v^{(k)} - v^{(k-1)}\|_\infty. \tag{B.48}$$

Applying this recursively yields

$$\|v^{(k+1)} - v^{(k)}\|_\infty \le q^k\|v^{(1)} - v^{(0)}\|_\infty. \tag{B.49}$$

Step 7: A posteriori error estimate. Because $v^{(k)} \to v^*$, we may write

$$v^* - v^{(k)} = \sum_{j=k}^{\infty}(v^{(j+1)} - v^{(j)}). \tag{B.50}$$

Taking sup norms and using the triangle inequality,

$$\|v^* - v^{(k)}\|_\infty \le \sum_{j=k}^{\infty}\|v^{(j+1)} - v^{(j)}\|_\infty \le \sum_{j=k}^{\infty} q^{j-k}\|v^{(k+1)} - v^{(k)}\|_\infty \tag{B.51}$$

$$= \frac{1}{1-q}\|v^{(k+1)} - v^{(k)}\|_\infty. \tag{B.52}$$

Using once more that $\|v^{(k+1)} - v^{(k)}\|_\infty \le q\|v^{(k)} - v^{(k-1)}\|_\infty$, we conclude that

$$\|v^{(k)} - v^*\|_\infty \le \frac{q}{1-q}\|v^{(k)} - v^{(k-1)}\|_\infty. \tag{B.53}$$

A fortiori, this completes the proof. □

C Error Bounds, Convergence Rates, and Robustness

This appendix provides a rigorous analysis of the error bounds, convergence rates, and robustness properties of the quasi-linearized Cole-Hopf transformation scheme for viscous Hamilton-Jacobi PDEs. We establish theoretical guarantees that justify the numerical method proposed in the main text.

Justification and Impact. While Theorem 2.12 establishes that the discrete algorithm is a contraction with linear convergence rate q, it addresses only the *quasi-linearization error*, *with assumptions on the exact heat-kernel evaluation and exact gradients*.

This section extends the result of Theorem 2.12 by quantifying the following error sources viz.,

- **quasi-linearization defect**: the distance between the fixed point of the frozen-coefficient map and the viscous HJ solution (§C.2);
- **Monte Carlo sampling error**: replacing the exact expectations with finite-sample estimates;
- **viscosity approximation error**: bounding the difference between the regularized and inviscid HJ solutions; and
- **robustness to model uncertainty** i.e. perturbations in the Hamiltonian and terminal cost.

We introduce a new theorem (Theorem C.9) that combines these errors via triangle inequality, revealing a fundamental **bias-variance tradeoff** controlled by the viscosity parameter δ: smaller δ reduces viscosity bias but amplifies Monte Carlo variance, and vice versa. This tradeoff may guide parameters selection during numerical optimization and explains why the optimal choice scales as $\delta \sim N^{-1/3}$, yielding a slower but more scalable convergence rate $O(N^{-1/6})$ than standard Monte Carlo $O(N^{-1/2})$. The robustness theorems further establish that the algorithm is stable under small model perturbations, making it suitable for real-world applications where exact dynamics and terminal costs are unavailable.

C.1 Convergence Analysis of the Iterative Scheme

We now establish convergence of the quasi-linearized iteration to the fixed point of the frozen-coefficient operator. We are deliberate about what is, and what is not, claimed: the limit of the iteration is the fixed point $\boldsymbol{v}^\star$ of the surrogate solve map, and its identification with the viscous solution $\boldsymbol{v}^\delta$ of (HJ-Visc) is *not* asserted here; the distance between the two — the quasi-linearization defect — is quantified separately below and enters the total error budget of Theorem C.9 as an explicit term.

Theorem C.1 (Convergence of Quasi-Linearization to the Frozen-Coefficient Fixed Point)**.** *Let $\{\boldsymbol{v}^{(k)}\}_{k=0}^{\infty}$ be the sequence generated by Algorithm 1 starting from $\boldsymbol{v}^{(0)} = \boldsymbol{\ell}$, and let $\mathcal{A}$ be a closed, $\mathcal{T}$-invariant subset of the Banach space $\big(C_b(\bar{\Omega} \times [0,T]), |\cdot|_\infty\big)$ on which the nondegeneracy and Lipschitz hypotheses of Assumption 2.8 hold. Assume that,*

(1) $\boldsymbol{H}(t;\boldsymbol{x},p)$ is C^2 in $(\boldsymbol{x},p)$ and convex in p;

(2) $\boldsymbol{\ell}(\boldsymbol{x})$ is C^2 with compact support;

(3) the contraction constant $\rho = q$ of (17) satisfies $\rho < 1$ on $\mathcal{A}$ (in particular, δ is not taken so small that Assumption 2.8, item 5, is violated).

Then $\mathcal{T}$ admits a unique fixed point $\boldsymbol{v}^\star \in \mathcal{A}$, and the sequence $\{\boldsymbol{v}^{(k)}\}$ converges geometrically to $\boldsymbol{v}^\star$:

$$|\boldsymbol{v}^{(k)} - \boldsymbol{v}^\star|_\infty \le \kappa \rho^k |\boldsymbol{v}^{(0)} - \boldsymbol{v}^\star|_\infty, \tag{B.54}$$

for some constant $\kappa > 0$.

Proof. The proof establishes that the iteration is a contraction mapping on $\mathcal{A}$.

Step 1: Fixed-point formulation. Define the operator $\mathcal{T} : \mathcal{A} \to \mathcal{A}$ by

$$(\mathcal{T}\boldsymbol{v})(t;\boldsymbol{x}) := -\frac{1}{c[\boldsymbol{v}]} \log \mathbb{E}_{y \sim \mathcal{N}(\boldsymbol{x}, \delta t I_n)} \left[\exp(-c[\boldsymbol{v}] \cdot \boldsymbol{\ell}(y))\right], \tag{B.55}$$

where $c[\boldsymbol{v}] := \frac{2}{\delta}\frac{\boldsymbol{H}(t;\boldsymbol{x},D\boldsymbol{v})}{|D\boldsymbol{v}|^2}$. [4] One step of Algorithm 1 is exactly $\boldsymbol{v}^{(k+1)} = \mathcal{T}\boldsymbol{v}^{(k)}$: the coefficient is frozen at the current iterate, the linear heat equation is solved exactly by the Gaussian expectation, and the value is recovered by the logarithmic inverse. We emphasize that the viscous solution $\boldsymbol{v}^\delta$ of (HJ-Visc) is, in general, *not* a fixed point of $\mathcal{T}$: the frozen-coefficient solve discards the derivative terms of c, so $\mathcal{T}\boldsymbol{v}^\delta$ differs from $\boldsymbol{v}^\delta$ by the Duhamel integral of the residual (§ C.2).

Step 2: Contraction property. For any $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathcal{A}$,

$$|\mathcal{T}\boldsymbol{v}_1 - \mathcal{T}\boldsymbol{v}_2|_\infty = \sup_{t,\boldsymbol{x}} \left| \frac{1}{c_1}\log \mathbb{E}[\exp(-c_1\boldsymbol{\ell})] - \frac{1}{c_2}\log \mathbb{E}[\exp(-c_2\boldsymbol{\ell})] \right| \le \rho|\boldsymbol{v}_1 - \boldsymbol{v}_2|_\infty, \quad \text{(B.56)}$$

where $c_i = c[\boldsymbol{v}_i]$. The inequality follows in two moves: first, the map $c \mapsto -\frac{1}{c}\log \mathbb{E}[e^{-c\boldsymbol{\ell}}]$ is Lipschitz in c with constant $2G/c_{\min}$ on $c \in [c_{\min}, c_{\max}]$, $|\boldsymbol{\ell}| \le G$ (mean value theorem applied to the log-sum-exp functional, as in the proof of Theorem 2.12); second, the coefficient functional $\boldsymbol{v} \mapsto c[\boldsymbol{v}]$ is Lipschitz with constant $\frac{2L_D}{\delta}\left(\frac{L_H}{m_0^2} + \frac{2H_*P_*}{m_0^4}\right)$ under Assumption 2.8. Their composition gives $\rho = q$ of (17), and hypothesis (3) asserts $\rho < 1$.

Step 3: Application of the Banach fixed-point theorem. The set $\mathcal{A}$ is a closed subset of the complete space $(C_b, |\cdot|_\infty)$ and is $\mathcal{T}$-invariant by hypothesis, hence complete; since $\mathcal{T}$ is a ρ-contraction on it, the sequence $\boldsymbol{v}^{(k+1)} = \mathcal{T}\boldsymbol{v}^{(k)}$ converges geometrically to the unique fixed point $\boldsymbol{v}^\star \in \mathcal{A}$ as in (B.54). □

Remark C.2 (What the fixed point is, and is not). Theorem C.1 is a statement about the numerical map that Algorithm 1 actually implements. The limit $\boldsymbol{v}^\star$ solves the implicit relation $\boldsymbol{v}^\star = \mathcal{T}\boldsymbol{v}^\star$, i.e. , the viscous HJ equation with the Hamiltonian replaced by its quadratic quasi-linearization at $\boldsymbol{v}^\star$ itself. When $\boldsymbol{H} = \frac{1}{2}|p|^2$ the surrogate is the Hamiltonian and $\boldsymbol{v}^\star = \boldsymbol{v}^\delta$ exactly; for general Hamiltonians the two differ by the quasi-linearization defect $\varepsilon_{\mathrm{QL}} \triangleq |\boldsymbol{v}^\star - \boldsymbol{v}^\delta|_\infty$, which we bound in § C.2 and carry, undiluted, into the total error bound of Theorem C.9.

Corollary C.3 (Convergence Rate)**.** *Under the conditions of Theorem C.1, the number of iterations K required to achieve $|\boldsymbol{v}^{(K)} - \boldsymbol{v}^\star|_\infty \le \varepsilon$ is bounded by*

$$K \le \left\lceil \frac{\log(\varepsilon/\kappa) - \log|\boldsymbol{v}^{(0)} - \boldsymbol{v}^\star|_\infty}{\log \rho} \right\rceil = O\left(\log \frac{1}{\varepsilon}\right). \quad \text{(B.57)}$$

Proof of Corollary C.3. From Theorem C.1, the iterates satisfy $|\boldsymbol{v}^{(k+1)} - \boldsymbol{v}^\star|_\infty \le \rho|\boldsymbol{v}^{(k)} - \boldsymbol{v}^\star|_\infty$ for $\rho < 1$. By induction,

$$|\boldsymbol{v}^{(K)} - \boldsymbol{v}^\star|_\infty \le \rho^K|\boldsymbol{v}^{(0)} - \boldsymbol{v}^\star|_\infty. \quad \text{(B.58)}$$

To achieve $|\boldsymbol{v}^{(K)} - \boldsymbol{v}^\star|_\infty \le \varepsilon$, we require,

$$\rho^K|\boldsymbol{v}^{(0)} - \boldsymbol{v}^\star|_\infty \le \varepsilon \quad \Longrightarrow \quad K \ge \frac{\log(\varepsilon/|\boldsymbol{v}^{(0)} - \boldsymbol{v}^\star|_\infty)}{\log \rho}. \quad \text{(B.59)}$$

Since $\rho < 1$, we have $\log \rho < 0$, so the minimum integer K satisfying this is,

$$K = \left\lceil \frac{\log \varepsilon - \log|\boldsymbol{v}^{(0)} - \boldsymbol{v}^\star|_\infty}{\log \rho} \right\rceil = O(\log(1/\varepsilon)). \quad \text{(B.60)}$$

□

C.2 The Quasi-Linearization Defect

The gap between the fixed point $\boldsymbol{v}^\star$ of Theorem C.1 and the viscous solution $\boldsymbol{v}^\delta$ of (HJ-Visc) is the price of freezing the coefficient. We quantify it here as the Duhamel norm of the discarded residual of Lemma B.1; the resulting bound $\varepsilon_{\mathrm{QL}}$ enters Theorem C.9 as the fourth error source.

[4] Note that $c[\boldsymbol{v}]$ is the *coefficient functional*: the operator that maps a solution function $\boldsymbol{v}$ to its frozen coefficient. At a point $(t;\boldsymbol{x})$, this evaluates to $c[\boldsymbol{v}](t;\boldsymbol{x})$, which coincides with the pointwise notation $c(t;\boldsymbol{x})$ in § 2. The functional notation emphasizes the dependence of the entire operator $\mathcal{T}$ on the solution function.

Theorem C.4 (Fixed-point defect of the frozen-coefficient scheme)**.** *Let $\boldsymbol{v}^\star$ be the fixed point of Theorem C.1 with converged coefficient $\boldsymbol{c}^\star \triangleq c[\boldsymbol{v}^\star] \in C^{1,2}\left(\bar{\Omega} \times [0,T]\right)$, and let $\boldsymbol{v}^\delta \in C^{1,2}$ solve (HJ-Visc). Assume,*

(1) coefficient bounds: $c_{\min} \le \boldsymbol{c}^\star \le c_{\max}$ and $\max\{|\boldsymbol{c}^\star_t|_\infty, |D\boldsymbol{c}^\star|_\infty, |\Delta \boldsymbol{c}^\star|_\infty\} \le L_c$;

(2) value bounds: $|\boldsymbol{v}^\star|_\infty, |\boldsymbol{v}^\delta|_\infty \le G_v$ and $m_0 \le |D\boldsymbol{v}^\star| \le P_*$, $|D\boldsymbol{v}^\delta| \le P_*$;

(3) the Hamiltonian bounds of Assumption 2.8, item (3).

Then the quasi-linearization defect obeys,

$$\varepsilon_{\mathrm{QL}} \;=\; |\boldsymbol{v}^\star - \boldsymbol{v}^\delta|_\infty \;\le\; \frac{T}{c_{\min}}\, e^{2c_{\max}G_v}\, \left(\bar{\boldsymbol{R}}_{\mathrm{alg}} + \bar{\boldsymbol{R}}_{\mathrm{der}}\right), \tag{B.61}$$

where the algebraic and derivative residual ceilings are,

$$\begin{aligned}
\bar{\boldsymbol{R}}_{\mathrm{alg}} &\le c_{\max}\left(L_H + \frac{2H_*P_*}{m_0^2}\right)|D\boldsymbol{v}^\delta - D\boldsymbol{v}^\star|_\infty,\\
\bar{\boldsymbol{R}}_{\mathrm{der}} &\le L_c\, G_v\left(1+\frac{\delta}{2}\right) + \frac{\delta}{2}\, L_c^2\, G_v^2 + \delta\,(1 + c_{\max}G_v)\, L_c\, P_*.
\end{aligned} \tag{B.62}$$

Proof. Set $\boldsymbol{\omega}^\star \triangleq \exp(-\boldsymbol{c}^\star \boldsymbol{v}^\star)$ and $\tilde{\boldsymbol{\omega}} \triangleq \exp(-\boldsymbol{c}^\star \boldsymbol{v}^\delta)$. By the fixed-point property $\boldsymbol{v}^\star = \mathcal{T}\boldsymbol{v}^\star$, the function $\boldsymbol{\omega}^\star$ *is* the heat-kernel expectation of (B.55); i.e. it solves the homogeneous heat equation $\boldsymbol{\omega}_t - \frac{\delta}{2}\Delta\boldsymbol{\omega} = 0$ with initial data $\exp(-\boldsymbol{c}^\star(0;\cdot)\,\boldsymbol{\ell})$. By Lemma B.1 applied with coefficient $\boldsymbol{c}^\star$ and value $\boldsymbol{v}^\delta$, the function $\tilde{\boldsymbol{\omega}}$ solves the same equation with a source,

$$\tilde{\boldsymbol{\omega}}_t - \frac{\delta}{2}\Delta\tilde{\boldsymbol{\omega}} = \tilde{\boldsymbol{\omega}}\, B, \qquad B \triangleq \underbrace{\boldsymbol{c}^\star\left(\boldsymbol{H}(t;\boldsymbol{x},D\boldsymbol{v}^\delta) - \frac{\delta}{2}\boldsymbol{c}^\star|D\boldsymbol{v}^\delta|^2\right)}_{B_{\mathrm{alg}}} + \underbrace{\boldsymbol{R}_{\mathrm{der}}\left(\boldsymbol{c}^\star, \boldsymbol{v}^\delta\right)}_{B_{\mathrm{der}}}, \tag{B.63}$$

and with the *same* initial data, since $\boldsymbol{v}^\delta(0;\cdot) = \boldsymbol{v}^\star(0;\cdot) = \boldsymbol{\ell}$. The difference $w \triangleq \tilde{\boldsymbol{\omega}} - \boldsymbol{\omega}^\star$ therefore solves $w_t - \frac{\delta}{2}\Delta w = \tilde{\boldsymbol{\omega}}B$ with $w(0;\cdot) = 0$, and Duhamel's principle with the heat semigroup $S_\delta(t)$ — an L^∞-contraction — gives,

$$|w(t)|_\infty \le \int_0^t |S_\delta(t-s)\,[\tilde{\boldsymbol{\omega}}(s)B(s)]\,|_\infty ds \le \int_0^t |\tilde{\boldsymbol{\omega}}(s)|_\infty |B(s)|_\infty\, ds \le T\, e^{c_{\max}G_v} \sup_{s\le T}|B(s)|_\infty. \tag{B.64}$$

For the algebraic part, insert the definition $\boldsymbol{c}^\star = 2\boldsymbol{H}(t;\boldsymbol{x},D\boldsymbol{v}^\star)/(\delta|D\boldsymbol{v}^\star|^2)$, so that $\frac{\delta}{2}\boldsymbol{c}^\star|D\boldsymbol{v}^\delta|^2 = \boldsymbol{H}(\cdot,D\boldsymbol{v}^\star)\,|D\boldsymbol{v}^\delta|^2/|D\boldsymbol{v}^\star|^2$ and,

$$\begin{aligned}
|B_{\mathrm{alg}}| &\le c_{\max}\left|\boldsymbol{H}(\cdot,D\boldsymbol{v}^\delta) - \boldsymbol{H}(\cdot,D\boldsymbol{v}^\star)\right| + c_{\max}\,|\boldsymbol{H}(\cdot,D\boldsymbol{v}^\star)|\cdot\frac{\left||D\boldsymbol{v}^\star|^2 - |D\boldsymbol{v}^\delta|^2\right|}{|D\boldsymbol{v}^\star|^2}\\
&\le c_{\max}\left(L_H + \frac{2H_*P_*}{m_0^2}\right)|D\boldsymbol{v}^\delta - D\boldsymbol{v}^\star|_\infty,
\end{aligned} \tag{B.65}$$

using the Lipschitz property of $\boldsymbol{H}$ in the co-state and $\left||a|^2 - |b|^2\right| \le (|a|+|b|)|a-b| \le 2P_*|a-b|$. For the derivative part, term-by-term majorization of $\boldsymbol{R}_{\mathrm{der}}$ in (B.1) with the assumed ceilings gives the second line of (B.62). Finally, the mean value theorem for the logarithm ($|\log a - \log b| \le |a-b|/\min\{a,b\}$) with $\boldsymbol{\omega}^\star, \tilde{\boldsymbol{\omega}} \ge e^{-c_{\max}G_v}$ yields,

$$\left|\boldsymbol{v}^\star - \boldsymbol{v}^\delta\right| = \frac{1}{\boldsymbol{c}^\star}\,|\log\boldsymbol{\omega}^\star - \log\tilde{\boldsymbol{\omega}}| \le \frac{e^{c_{\max}G_v}}{c_{\min}}\,|w|_\infty, \tag{B.66}$$

and chaining (B.66) through (B.64) gives (B.61). □

Remark C.5 (Reading the defect bound). Three features deserve emphasis. First, the bound vanishes in the exact case: when $\boldsymbol{H} = \frac{1}{2}|p|^2$, the coefficient is the constant $1/\delta$, so $L_c = 0$ kills $\bar{\boldsymbol{R}}_{\mathrm{der}}$, and $\boldsymbol{v}^\star = \boldsymbol{v}^\delta$ forces $\bar{\boldsymbol{R}}_{\mathrm{alg}} = 0$; we recover Corollary B.2. Second, the defect is governed by L_c — the

rate of variation of the converged coefficient — which is largest near the boundary of the usable part where the Hamiltonian-to-gradient ratio turns over rapidly; this is precisely where the pointwise errors concentrate in our experiments (Fig. 1), so the theory and the empirics indict the same culprit. Third, the algebraic ceiling is proportional to $|D\boldsymbol{v}^{\delta} - D\boldsymbol{v}^{\star}|_{\infty}$: the defect contracts when the surrogate gradient field tracks the true one, which is the quantity the Picard iteration refines. The bound is honest about its own character: it certifies smallness of the defect where $\boldsymbol{c}^{\star}$ is slowly varying and offers no comfort where it is not.

C.3 Monte Carlo Error Analysis

The Gaussian expectation is approximated via Monte Carlo sampling. We analyze the error introduced by this approximation.

Theorem C.6 (Monte Carlo Error)**.** *Let $\hat{\boldsymbol{v}}^{(\delta,N)}(t;\boldsymbol{x})$ denote the value function computed using N Monte Carlo samples. Then for any $\delta_p \in (0,1)$, with probability at least $1-\delta_p$,*

$$|\hat{\boldsymbol{v}}^{(\delta,N)} - \boldsymbol{v}^{\delta}|_{\infty} \leq \frac{C_{\text{eff}}}{\sqrt{N}}\sqrt{\log(1/\delta_p)}, \tag{B.67}$$

where C_{eff} is the effective sample size *that depends on the range of $\boldsymbol{\ell}$ and the concentration properties of the exponential weights.*

Proof. From Lemma 3.2, we have

$$\boldsymbol{v}^{\delta}(t;\boldsymbol{x}) = -\frac{1}{c}\log \mathbb{E}_{y\sim\mathcal{N}(\boldsymbol{x},\delta t I_n)}[\exp(-c\boldsymbol{\ell}(y))]. \tag{B.68}$$

The Monte Carlo estimator is

$$\hat{\boldsymbol{v}}^{(\delta,N)}(t;\boldsymbol{x}) = -\frac{1}{c}\log\frac{1}{N}\sum_{i=1}^{N}\exp(-c\boldsymbol{\ell}(y_i)), \tag{B.69}$$

where $y_i \overset{\text{iid}}{\sim} \mathcal{N}(\boldsymbol{x}, \delta t I_n)$.

Let $Z := \mathbb{E}[\exp(-c\boldsymbol{\ell})]$ be the true expectation and $\hat{Z}_N := \frac{1}{N}\sum_{i=1}^{N}\exp(-c\boldsymbol{\ell}(y_i))$ be the sample mean.

By Hoeffding's inequality for bounded random variables (assuming $\boldsymbol{\ell}$ is bounded, say $|\boldsymbol{\ell}(\boldsymbol{x})| \leq M$), we have

$$\mathbb{P}\left(|\hat{Z}_N - Z| \geq \epsilon\right) \leq 2\exp\left(-\frac{N\epsilon^2}{2(\exp(cM)-\exp(-cM))^2}\right). \tag{B.70}$$

Setting the right-hand side equal to δ and solving for ϵ,

$$\epsilon = (\exp(cM)-\exp(-cM))\sqrt{\frac{2\log(2/\delta)}{N}}. \tag{B.71}$$

Using the Lipschitz continuity of $\log$, we find that

$$|\hat{\boldsymbol{v}}^{(\delta,N)} - \boldsymbol{v}^{\delta}| = \frac{1}{c}|\log\hat{Z}_N - \log Z| \leq \frac{1}{c}\cdot\frac{|\hat{Z}_N - Z|}{\min\{\hat{Z}_N, Z\}} \leq \frac{1}{c\cdot Z_{\min}}\cdot\epsilon, \tag{B.72}$$

where $Z_{\min}$ is a lower bound on Z (which exists since $\boldsymbol{\ell}$ is bounded and continuous).

Combining these bounds yields (B.67) with effective constant

$$C_{\text{eff}} := \frac{2(\exp(cM)-\exp(-cM))}{c\cdot Z_{\min}}\sqrt{2}. \tag{B.73}$$

The effective sample size shrinks when c is large (i.e., when $|\boldsymbol{\ell}|_{\infty} \gg 1$), which is expected since the exponential weights become concentrated. Importance sampling with the tilted proposal $e^{-c\boldsymbol{\ell}(y)}$ (as in Corollary 3.7) can reduce C_{eff} significantly. □

Remark C.7 (Variance Reduction). The standard MC estimator in Theorem C.6 has variance $O(1/N)$. However, when $|\boldsymbol{\ell}|_{\infty}$ is large, the exponential weights $\exp(-c\boldsymbol{\ell})$ become highly concentrated, leading to high variance. The log-sum-exp identity used in Corollary 3.6 for numerical stability, combined with importance sampling, can reduce the effective variance. For well-designed importance distributions, the convergence rate can be improved to $O(1/\sqrt{N})$ with a much smaller constant.

C.4 Viscosity Approximation Error

The viscosity solution $\boldsymbol{v}^\delta$ approximates the inviscid solution $\boldsymbol{v}$ (the true viscosity solution of the original HJ PDE). We bound this approximation error.

Theorem C.8 (Viscosity Approximation Error)**.** *Let $\boldsymbol{v}$ be the (unique) viscosity solution of* (HJ-Visc) *and $\boldsymbol{v}^\delta$ be the viscosity solution of* (7)*. Under standard regularity assumptions on $\boldsymbol{H}$ and $\boldsymbol{\ell}$ [Crandall and Lions, 1983], we have*

$$|\boldsymbol{v}^\delta - \boldsymbol{v}|_\infty \le c\sqrt{\delta}, \tag{B.74}$$

for some constant c depending on T, $|D\boldsymbol{H}|$, and $|D^2\boldsymbol{\ell}|$.

Proof. The proof follows from the classical results of [Crandall and Lions, 1983] on viscosity approximations to first-order Hamilton-Jacobi equations.

By the comparison principle for viscosity solutions (Theorem 2.1 in [Evans, 2022]), since $\boldsymbol{v}$ and $\boldsymbol{v}^\delta$ satisfy the same terminal condition $\boldsymbol{v}(0;\boldsymbol{x}) = \boldsymbol{v}^\delta(0;\boldsymbol{x}) = \boldsymbol{\ell}(\boldsymbol{x})$, the difference $\tilde{\boldsymbol{v}} := \boldsymbol{v}^\delta - \boldsymbol{v}$ satisfies

$$\tilde{\boldsymbol{v}}_t + \boldsymbol{H}(t;\boldsymbol{x}, Dw) = \frac{\delta}{2}\Delta\boldsymbol{v}^\delta \quad \text{in } \Omega \times (0,T], \tag{B.75}$$

with $\tilde{\boldsymbol{v}}(0;\boldsymbol{x}) = 0$.

By the maximum principle and the fact that $|\Delta\boldsymbol{v}^\delta|_\infty \le M_2$ (since $\boldsymbol{v}^\delta$ is smooth; M_2 implicitly depends on the constant c through the terms T, $|D\boldsymbol{H}|$, and $|D^2\boldsymbol{\ell}|$), we obtain

$$|\tilde{\boldsymbol{v}}|_\infty \le \frac{\delta M_2 T}{2}. \tag{B.76}$$

More refined estimates using energy methods show that $|\tilde{\boldsymbol{v}}|_\infty = O(\sqrt{\delta})$ as $\delta \to 0$ (see Crandall and Lions [1984], Theorem 6.4). □

C.5 Combined Error Bound

Combining the quasilinearization, Monte Carlo, and viscosity errors, we obtain the total error of our numerical scheme.

Theorem C.9 (Total Error Bound)**.** *Let $\hat{\boldsymbol{v}}^{K,N,\delta}$ denote the numerical approximation after K iterations with N Monte Carlo samples and viscosity parameter δ, and let $\varepsilon_{\mathrm{QL}} \triangleq |\boldsymbol{v}^\star - \boldsymbol{v}^\delta|_\infty$ be the quasi-linearization defect of §C.2. Then for any $\delta_p \in (0,1)$, with probability at least $1-\delta_p$,*

$$|\hat{\boldsymbol{v}}^{K,N,\delta} - \boldsymbol{v}|_\infty \le \kappa\rho^K + \frac{C_{\mathit{eff}}}{\sqrt{N}}\sqrt{\log(1/\delta_p)} + \varepsilon_{\mathrm{QL}} + c\sqrt{\delta}, \tag{B.77}$$

where $\boldsymbol{v}$ is the true (inviscid) viscosity solution of (HJ-IVP)*.*

Proof. By the triangle inequality,

$$|\hat{\boldsymbol{v}}^{K,N,\delta} - \boldsymbol{v}|_\infty \le \underbrace{|\hat{\boldsymbol{v}}^{K,N,\delta} - \boldsymbol{v}^\star|_\infty}_{\text{Iteration + MC error}} + \underbrace{|\boldsymbol{v}^\star - \boldsymbol{v}^\delta|_\infty}_{\text{Quasi-linearization defect}} + \underbrace{|\boldsymbol{v}^\delta - \boldsymbol{v}|_\infty}_{\text{Viscosity error}}. \tag{B.78}$$

The first bracket splits once more into the deterministic iteration error, bounded by $\kappa\rho^K$ via Theorem C.1, and the sampling error of replacing exact expectations by N-sample averages, bounded by Theorem C.6 with probability $1-\delta_p$. The second bracket is $\varepsilon_{\mathrm{QL}}$ by definition, and the third is Theorem C.8. Note that the term $|\boldsymbol{v}^{(0)} - \boldsymbol{v}^\star|_\infty$ is absorbed into κ in the first term since after K iterations the initial gap shrinks by the factor ρ^K of the contraction mapping analysis. □

Proof of Corollary 2.14. On the event of Theorem C.9 — which has probability at least $1-\delta_p$ — we have $|\hat{\boldsymbol{v}}^{K,N,\delta} - \boldsymbol{v}|_\infty \le E$. If $\hat{\boldsymbol{v}}^{K,N,\delta}(t;\boldsymbol{x}) > E$ then $\boldsymbol{v}(t;\boldsymbol{x}) \ge \hat{\boldsymbol{v}}^{K,N,\delta}(t;\boldsymbol{x}) - E > 0$, so the state is truly safe; symmetrically, $\hat{\boldsymbol{v}}^{K,N,\delta}(t;\boldsymbol{x}) < -E$ implies $\boldsymbol{v}(t;\boldsymbol{x}) < 0$. Misclassification is therefore possible only on the abstention band $|\hat{\boldsymbol{v}}^{K,N,\delta}(t;\boldsymbol{x})| \le E$. □

Remark C.10 (Bias-Variance Tradeoff). Equation (B.77) reveals a fundamental bias-variance tradeoff controlled by δ i.e. ,

- **Smaller δ:** Reduces viscosity bias ($c\sqrt{\delta} \to 0$) but increases MC variance (larger C_{eff} due to higher concentrated weights).
- **Larger δ:** Reduces MC variance (more diffused weights) but increases viscosity bias.

The choice of $\delta \sim N^{-1/3}$ yields overall rate $O(N^{-1/6})$, which is slower than the standard $O(N^{-1/2}$ MC rate; however, it avoids the curse of dimensionality from grid discretization.

C.6 Robustness to Model Uncertainty

In practice, the Hamiltonian $\boldsymbol{H}$ and terminal cost $\boldsymbol{\ell}$ may be uncertain. We establish robustness of the value function to perturbations.

Theorem C.11 (Robustness to Hamiltonian Perturbations)**.** *Suppose $\boldsymbol{H}_1$ and $\boldsymbol{H}_2$ are two Hamiltonians satisfying the standing assumptions, and let $\boldsymbol{v}_i$ denote the corresponding viscosity solutions. If*

$$|\boldsymbol{H}_1(t;\boldsymbol{x},p) - \boldsymbol{H}_2(t;\boldsymbol{x},p)|_\infty \le \epsilon_H, \tag{B.79}$$

uniformly over $(t,\boldsymbol{x},p) \in [0,T]\times\Omega\times\mathbb{R}^n$, then

$$|\boldsymbol{v}_1 - \boldsymbol{v}_2|_\infty \le T\cdot\epsilon_H. \tag{B.80}$$

Proof. The difference $w := \boldsymbol{v}_1 - \boldsymbol{v}_2$ satisfies

$$w_t + \boldsymbol{H}_1(t;\boldsymbol{x},Dw) = \boldsymbol{H}_2(t;\boldsymbol{x},D\boldsymbol{v}_2) - \boldsymbol{H}_1(t;\boldsymbol{x},D\boldsymbol{v}_2) =: \eta(t;\boldsymbol{x}), \tag{B.81}$$

with $w(0;\boldsymbol{x}) = 0$ (same terminal condition).

Since $|\eta(t;\boldsymbol{x})| \le \epsilon_H$ by assumption, integrating over $[0,t]$ and applying Gronwall's inequality:

$$|w(t;\boldsymbol{x})| \le \int_0^t |\eta(s;\boldsymbol{x})|ds \le t\cdot\epsilon_H \le T\cdot\epsilon_H. \tag{B.82}$$

□

Theorem C.12 (Robustness to Terminal Cost Perturbations)**.** *Under the same setup, if the terminal costs satisfy $|\boldsymbol{\ell}_1 - \boldsymbol{\ell}_2|_\infty \le \epsilon_g$, then*

$$|\boldsymbol{v}_1 - \boldsymbol{v}_2|_\infty \le \epsilon_g. \tag{B.83}$$

Proof. Direct consequence of the comparison principle for viscosity solutions [Evans, 2022]. □

Corollary C.13 (Combined Robustness)**.** *For simultaneous perturbations in both $\boldsymbol{H}$ and $\boldsymbol{\ell}$,*

$$|\boldsymbol{v}_1 - \boldsymbol{v}_2|_\infty \le T\cdot\epsilon_H + \epsilon_g. \tag{B.84}$$

These robustness results show that small errors in model specification (Hamiltonian or terminal cost) lead to proportionally small errors in the value function, making the method suitable for practical applications where exact models are unavailable. They are also load-bearing for the scheme itself: the coefficient regularization of Lemma 2.10 is priced through Theorem C.11, as we now prove.

Proof of Lemma 2.10. By Lemma B.1, running the frozen-coefficient step with any coefficient field $\hat{\boldsymbol{c}}$ annihilates the algebraic residual $\boldsymbol{R}_{\text{alg}} = \hat{\boldsymbol{c}}\,(\boldsymbol{H}' - \frac{\delta}{2}\hat{\boldsymbol{c}}|D\boldsymbol{v}^\delta|^2)$ precisely for the Hamiltonian $\boldsymbol{H}' = \frac{\delta}{2}\hat{\boldsymbol{c}}\,|D\boldsymbol{v}^\delta|^2$. With $\hat{\boldsymbol{c}} = \boldsymbol{c}_\eta$ of (18),

$$\boldsymbol{H}' = \frac{\delta}{2}\cdot\frac{2}{\delta}\cdot\frac{\boldsymbol{H}}{|D\boldsymbol{v}^\delta|^2+\eta}\cdot|D\boldsymbol{v}^\delta|^2 = \boldsymbol{H}\,\frac{|D\boldsymbol{v}^\delta|^2}{|D\boldsymbol{v}^\delta|^2+\eta} = \boldsymbol{H}_\eta, \tag{B.85}$$

which proves the equivalence. For the uniform bound, write $r \triangleq |D\boldsymbol{v}^\delta|$; positive 1-homogeneity gives $|\boldsymbol{H}(t;\boldsymbol{x},p)| = |p|\cdot|\boldsymbol{H}(t;\boldsymbol{x},p/|p|)| \le C_{\boldsymbol{H}}\,r$, hence,

$$|\boldsymbol{H}_\eta - \boldsymbol{H}| = |\boldsymbol{H}|\,\frac{\eta}{r^2+\eta} \le C_{\boldsymbol{H}}\,\frac{r\,\eta}{r^2+\eta} \le \frac{C_{\boldsymbol{H}}\sqrt{\eta}}{2}, \tag{B.86}$$

since $r \mapsto r\eta/(r^2+\eta)$ is maximized at $r = \sqrt{\eta}$ with value $\sqrt{\eta}/2$. The final claim is Theorem C.11 applied with $\epsilon_H = C_{\boldsymbol{H}}\sqrt{\eta}/2$. □

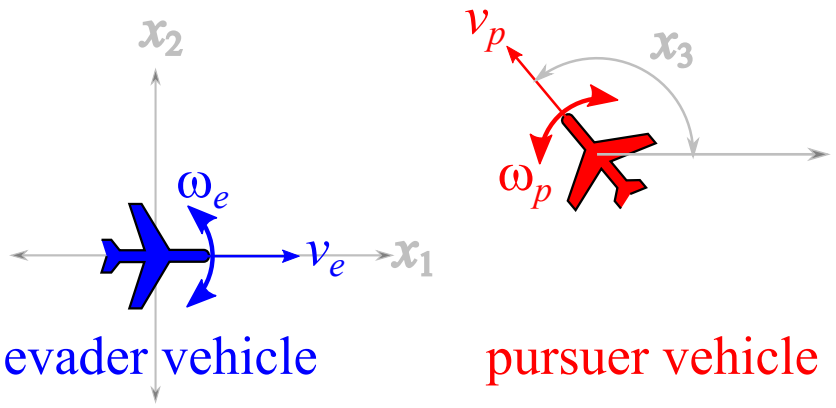


Figure 3: Two Dubins' vehicles in relative Cartesian coordinates. Reprinted from Mitchell [2020].

D Further Numerical Results

This example was originally proposed by Merz [1972] as an iteration upon Isaacs [1999]'s homicidal chauffeur game, whereupon a pursuit-evasion game between two players with similar speeds and minimum turn radii, is thoroughly analyzed. In Mitchell [2020], this problem was established as a benchmark for testing the solubility of capturable set of states (the backward reachable tube) in Merz's classical pursuit-evasion game. In this example, we solve the problem with our LevelSetPy toolbox and establish that the approximated barrier surface to the two-player game conforms with standard results.

The game is that of two cars sharing similar Dubins dynamics Dubins [1957]: $\boldsymbol{P}$ and $\boldsymbol{E}$ both have a positive minimum turn radii, w, and constant speeds v – with motion restricted to a plane as we have for the rocket launch differential game above. In relative coordinates, the diagrammatic structure of the motion is as depicted in Fig. 3. Choosing the Cartesian coordinate for motion representation, the state vector of the game with $\boldsymbol{E}$ at the origin can be characterized by its x_1, x_2 position relative to $\boldsymbol{P}$ and the angle θ between the two vehicles. Capture occurs when the distance $\|\boldsymbol{PE}\|_2$ between the pursuer and the evader becomes less than a specified radius.

The relative equations of motion, going by Fig. 3, is

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix} = \begin{pmatrix} -v_e + v_p \cos x_3 + w_e x_2 \\ v_p \sin x_3 - w_e x_1 \\ w_p - w_e \end{pmatrix}. \tag{C.1}$$

We adopt specialization to a case where the two vehicles only possess a unit velocity and unit maximum turn rates. Here, as Merz notes, if the initial velocities are parallel such as $x_3 = 0$, then the equations of relative motion imply that $\boldsymbol{E}$ can be separated from $\boldsymbol{P}$ forever by the initial radial separation if it replicates $\boldsymbol{P}$'s strategy. Whence, the barrier surface is closed and we are presented with Isaacs [1999]'s game of kind where we must determine the nature of the surface. This terminal surface possesses a closed-form solution and we refer readers to the treatment by Merz [1972]. In this example, our chief concern is to judge the efficacy of our toolbox with respect to the analytical solution of the barrier surface.

The backward reachable tube that consists of the paths taken by the trajectories of either player is defined as in the rockets pursuit-evasion game so that we have

$$\ell(0, x) = \{x \in \mathcal{X} | x_1^2 + x_2^2 \leq r^2\}, \tag{C.2}$$

where again r is the capture radius. The target set is a cylinder as ℓ above excludes the heading, x_3. It is represented as shown in Fig. 4.

For a detailed treatment of the barrier surface, we refer readers to a proper analysis as elucidated in Mitchell [2020]. Here, we focus on the construction of the BUP. The set of states that constitute the useable part and its boundary are respectively a function of the implicit surface function representation

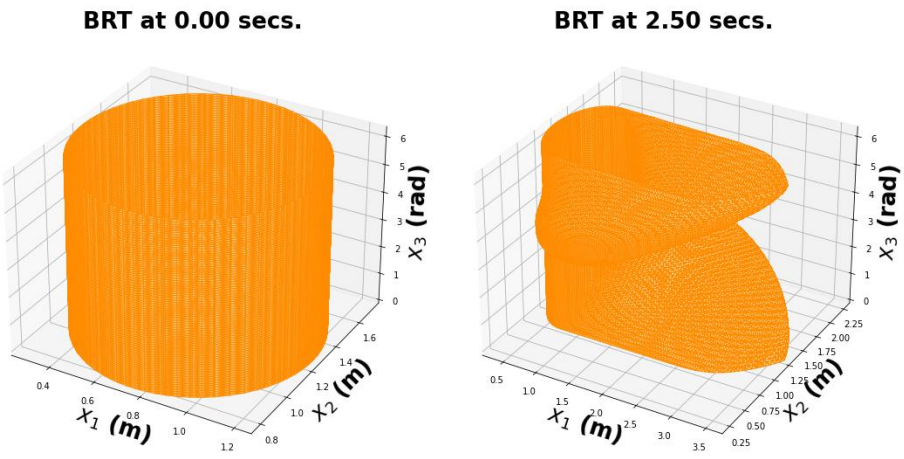


Figure 4: The target set (left) and the boundary of the useable part of the state space after the differential game between $\boldsymbol{P}$ and $\boldsymbol{E}$.

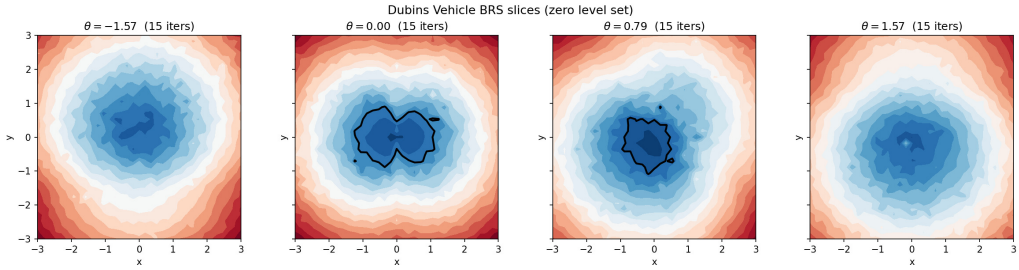


Figure 5: Dubins Vehicle BRT Slices

$\ell : [-T, 0] \times \mathcal{X} \to \mathbb{R}$ so that for a $t \in [0, T]$, where $T > 0$ is

$$\mathcal{T} = \{x \in \mathcal{X} | \ell(0, x) \leq 0\} \tag{C.3}$$

$$R([-t, 0], \mathcal{T}) = \{x \in \mathcal{X} | \ell(t, x) \leq 0\}, \tag{C.4}$$

When $t > 0$, the implicit surface representation is the following HJI PDE

$$\frac{\partial}{\partial t}\ell(t, x) + \min\left(0, \boldsymbol{H}(x, \nabla_x \ell(t, x))\right) = 0. \tag{C.5}$$

It is easy to verify that the Hamiltonian is

$$H(x, p) = p_1(v_p \cos x_3 - v_e) - p_2(v_p \sin x_3) + w|p_1 x_2 - p_2 x_1 - p_3| - w|p_3|. \tag{C.6}$$

Since we are concerned with the special case that the linear and angular speeds are equal, we set $v_e = v_p = w \triangleq +1$ in the foregoing so that the Hamiltonian, in the final analysis is

$$H(x, p) = p_1(\cos x_3 - 1) - p_2(\sin x_3) + |p_1 x_2 - p_2 x_1 - p_3| - |p_3|. \tag{C.7}$$

The results and comparisons are provided in Fig. 5 and Fig. 14.

D.1 Starlings Murmurations Safety Analysis

We take inspiration from natural swarms. Through empirical [Ballerini et al., 2008, Cavagna et al., 2010, Helbing et al., 2000, Vicsek et al., 1995, Bialek et al., 2012] and theoretical findings [Jadbabaie et al., 2003], evidence now abounds that in certain natural species that exhibit collective behavior, convergence and group cohesion is based on simple topological interaction rules that they employ to keep a tab on one another in *local flocks* for collision avoidance, preserving density and structure

in an anisotropic formation, and exhibiting flock splitting, vacuole, cordon, and flash expansion isotropically [Haiken, 2021]. This aids coherent emergent behavior from various local anisotropic synchrony, which taken together among possibly hundreds of thousands of local interactions[5] [Haiken, 2021], keep these animals whirling, swooping, and flying in isotropic formations [Ballerini et al., 2008]. Thus, individual agents aggregate into substructures within the overall system, and overall group motion is synergized via local topological interactions so that a stable global heading and cohesion [Jadbabaie et al., 2003] is preserved.

In emphasis, we consider collective behavior in the murmuration of European starlings (*sturnus vulgaris*) (Fig. 2) as a case-study for large scale safety analysis. In these settings, local flocks within large murmurations maintain an anisotropic formation based on a topological interaction, regardless of sparsity of birds on a phase space [Cavagna et al., 2010]. Thus, intra- and inter-flock collisions are avoided and attacks are fended off [Ballerini et al., 2008]. Approximating the viscosity solutions of nonconvex Hamilton-Jacobi partial differential equations with our quasilinearization scheme and importance sampling, the Hamiltonian, control laws, and strategies that govern the transient behaviors of many systems that possess structural subsystems with unique nearest neighbor properties are computed and validated against our Picard iteration scheme on 100,000 birds. .

D.1.1 Related Works

As all starlings move in $\Omega \subset \mathbb{R}^n$, we want a stable and safe numerical algorithm that correctly represents dynamic interface boundary conditions, accurately represents kinematics, whilst sensitive enough to rapidly interpret subgroup topological and structural changes. Tsitsiklis [1995] worked under the restrictive assumption that HJ's running cost is independent of the control laws' update – providing an $O(n/p)$ parallel algorithm that resolved the discretized HJ-Hamiltonian for p processors on n grid points provided that $p = O(\sqrt{n}/\log n)$. In front tracking methods, a Lagrangian geometric representation techniques track the surfaces between separate structures with mechanisms such as triple point junctions which are endowed with shared nodes so that members of the front set are updated as time evolves [Kim et al., 2010]. In applied mechanics and microfluidics, methods such as volume-of-fluid [Balcázar et al., 2015] resolve the interaction among multiple phases that are separated by thin boundaries with volume fraction fields for each region or unique functions in levelset methods – leading to coalescence prevention among multiple regions. However, they come at a computational cost of $\mathcal{O}(N_{regions}N_{cells})$. To improve its scalability, [Karnakov et al., 2021] compactly stored many fields thereby keeping needed scalar fields constant and independent of regions to simulate.

D.1.2 Murmuration as a Collection of Games

A few mathematical notations, conventions, and taxonomy used throughout this section are in order at this juncture.

Structural homogeneity of starlings' motion in every region $\mathcal{S}_i \in \mathcal{C}$ for $i = 1, \cdots, [\mathcal{C}]$ applies; this is enforced by introducing an external disturbance on the zeroth-index agent (this aids compactness of the zero levelset of an $\mathcal{S}_i$ as we will introduce shortly). Hence, each safety verification episode can be characterized as a pursuit *game*, Γ. And by a game, we do not necessarily refer to a single game, but rather a *collection of games*, $\Upsilon = \{\Gamma_1, \cdots, \Gamma_g\}$. Such a game terminates when *capture* occurs, that is the distance between players falls below a predetermined threshold. As in previous examples, each player in a game shall constitute either a pursuer ($\boldsymbol{P}$) or an evader ($\boldsymbol{E}$). Let the cursory reader not interpret $\boldsymbol{P}$ or $\boldsymbol{E}$ as controlling a single agent. In our setup, we are poised with several pursuers (e.g. Falcons) or evaders (starlings). However, when $\boldsymbol{P}$ or $\boldsymbol{E}$ governs the behavior of but one bird, these symbols will denote the bird itself. An evading bird in a region $\mathcal{S}_i$ has a state notation $\boldsymbol{x}_a^i$ (read: the state of bird a in region i). A state $\boldsymbol{x}_a^i$ has linear velocity components, $\boldsymbol{x}_{a_1}^i, \boldsymbol{x}_{a_2}^i$, and heading $\boldsymbol{x}_{a_3}^i := w_a^i$. When we must distinguish a bird $\boldsymbol{x}_a^i \in \mathcal{C}_x$ from some other bird e.g. in another multiphase $\mathcal{C}_y$, we shall write ${}^x\boldsymbol{x}_a^i$ and ${}^y\boldsymbol{x}_a^i$ respectively. Given the various possibilities of outcomes, the "best" outcome is resolved by a *payoff*, Φ, whose extremal over a time interval will constitute a *value*, $\boldsymbol{v}$[6]. We adopt [Isaacs, 1999]'s language so that if the payoff for a game is finite we shall have a *game of kind* (a qualitative game); and for a game with a continuum of payoffs we shall have *a game*

[5] It has been reported that no birds fly together with greater coordination and complexity than European starlings, with murmurations counting upwards of 750,000 individual birds!

[6] The functional $\boldsymbol{\Phi}$ may be considered a functional mapping from an infinite-dimensional space to the space of real numbers.

of degree (quantitative games). The *strategy* executed by $\boldsymbol{P}$ or $\boldsymbol{E}$ during a game shall be denoted by $\alpha \in \mathcal{A}$ (resp. $\beta \in \mathcal{B}$).

The many interacting subsystems under consideration employ (i) natural units of measurements that are the same for all birds; (ii) kinematics with same linear speeds but with a capacity for orientation changes; (iii) inter-region interaction occurs within unique and distinct state space manifolds; and by birds maneuvering their direction, a kinematic alignment is obtained with other regions; (iv) region-to-region interaction occurs when a pursuer is within a threshold of capturing any bird in a region; (v) the interaction among respective regions is described by the time-evolution of an interface, which is the zero-level set of the objective functional of the respective local subgroups.

D.1.3 Murmuration as a Levelsets Fronts Equation

The key idea is that when the multi-bird problem has separable structure, one can resolve the associated HJ equations in a numerically consistent manner by considering the interface of these separable structures, i, as evolving dynamic interfaces, Γ_i, via the modified levelset equation (HJ-IVP)

$$\phi_t + \Gamma_i H(x, t, \phi, D\phi) = 0 \tag{C.8}$$

At each time step, we advance each interface ϕ_i for birds in a region i by solving the levelset equation for a small time step Δt; afterwards, we reconstruct the unsigned distance function that captures the safe set, which informs the controller for the agents $\mathcal{S}_i \in \mathcal{C}$. Contrary to distributed consensus-type algorithms such as [Saravanos et al., 2022], this framework guarantees safe exploration by directly incubating safety into the control optimization problem.

The motion of interfaces between subgroups are cast as Gaussian kernel approximations of the zero level set of implicitly defined unsigned distance functions (UDFs). Interfaces evolve by solving time-dependent Eulerian initial value partial differential equations of these SDFs in the form of viscosity solutions [Crandall and Lions, 1983] to hyperbolic conservation laws [Crandall and Majda, 1980a], which are essentially the original HJ equations [Evans and Souganidis, 1984]. Hence, the front's position gets updated by this means and the interface velocity is derived from physics on and off the interface.

Popular numerical tools for simulating the evolution of interfaces in applied physics, mathematics, and computational sciences include fast marching front tracking methods [Sethian, 1987, 1996], Voronoi implicit interface methods (or VIIMs) [Saye and Sethian, 2011, Zaitzeff et al., 2019], discretization schemes for Hamilton-Jacobi equations [Tsitsiklis, 1995], or multilayer volume-of-fluid methods [Karnakov et al., 2021] for foaming across scales. While each of these schemes has its own advantages, we will resort to levelset methods [Sethian, 2000] in developing algorithmic efficiency for the trajectory optimization among a *collection* ($\mathcal{C}$) of *disjoint* multiple starlings *sets*, $\mathcal{S} \subseteq \mathcal{C}$. The entire collection of starlings moves over an open set $\Omega \subset \mathbb{R}^n$. When subgroups $\{\mathcal{S}_i\}_{i=1}^{i=k} \subset \mathcal{C}, k < n$ of aerial starlings must traverse a narrow opening, or temporarily break apart to avoid collisions, for example, we expect topological changes to occur without explicit surgery. Furthermore, group geometric formation and cohesion are achieved by local changes to subgroups' collective heading and speed. This is a challenging problem numerically and we look to natural behaviors in animals and multiphase simulations for inspiration. In our problem construction, a *finite set* of individual starlings self-organize into local structural groups (which we refer to as subgroups, $\mathcal{S}$); subgroups interact based on nearest neighbor rules so that effective group consensus is dictated by interactions among separate subgroups [Jadbabaie et al., 2003].

Throughout, our theater of operations involve multiple aerial Dubins vehicles [Dubins, 1957] with an extra dimension along the vertical position to describe the 4-dimensional motion of birds. We treat bird motions as kinematic models that possess linear and angular speeds as state variables. The safety verification optimization problem is played as a game of multiple vehicles in a four-dimensional space. As we are dealing with multiple starlings, objectives such as collision avoidance, starlings' spatial separation, overall group coherence become paramount. At issue is continually computing the points set that belong on the *reachable set* boundary as the game advances forward in time. These reachable sets are those state space subsets where starlings may collide into one another, hit obstacles, lose topological integrity or miss group coherence behavior.

D.1.4 Murmuration Dynamics and Hamiltonian

We resolve local payoff extremals, $\{\Phi_1, \cdots, \Phi_{n_f}\}$. as a state space partition induced by an aggregation of desired collective behavior from local flocks' values $\{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_{n_f}\}$. Suppose that the local control laws are properly coordinated, the region of the state space across which their coordinated influence might be exerted constitute a larger e.g. *manipulability volume* for a dexterous kinematic task. We now formalize definitions that will aid the modularization of the problem into manageable forms.

Definition D.1 (Neighbors of an Agent)**.** We define the neighbors $\mathcal{N}_i(t)$ of agent i at time t as the set of all agents that lie within a predefined radius, r_i.

Definition D.2. We define a *flock*, F, consisting of agents labeled $\{1, 2, \cdots, n_a\}$ as a collection of agents within a phase space $(\mathcal{X}, T)$ such that all agents within the flock interact with their nearest neighbors in a topological sense.

Remark D.3. Every agent within a flock has similar dynamics to that of its neighbor(s). Furthermore, agents travel at the same linear speed, v; climb/dive rate (the control) $u_z \in [-\gamma_{\max}, \gamma_{\max}]$, where $\gamma_{\max} \in \mathbb{R}$; the angular headings, w, however, may be different between agents, seeing we are dealing with a many-bodied system. Each agent's continuous-time dynamics, $\dot{\boldsymbol{x}}^{(i)}(t)$, evolves as

$$\begin{bmatrix} \dot{\boldsymbol{x}}_1^{(i)}(t) \\ \dot{\boldsymbol{x}}_2^{(i)}(t) \\ \dot{\boldsymbol{x}}_3^{(i)}(t) \\ \dot{\boldsymbol{x}}_4^{(i)}(t) \end{bmatrix} = \begin{bmatrix} v(t)\cos \boldsymbol{x}_4^{(i)}(t) \\ v(t)\sin \boldsymbol{x}_4^{(i)}(t) \\ u_z(t) \\ \langle w^{(i)}(t) \rangle_r \end{bmatrix}, \quad \langle w^{(i)}(t) \rangle_r = \frac{1}{1+n_i(t)} \left(w^{(i)}(t) + \sum_{j \in \mathcal{N}_i(t)} w_j(t) \right) \tag{C.9}$$

for agents $i = \{1, 2, 3, ..., n_a\}$, where t is the continuous-time index, $n_i(t)$ is the number of agent i's neighbors at time t, $\mathcal{N}_i(t)$ denotes the sets of labels of agent i's neighbors at time t, and $\langle w^{(i)}(t) \rangle_r$ is the average orientation of agent i and its neighbors at time t. Note that for a game where all agents share the same constant linear speed and heading, and $u_z = 0$, (C.9) reduces to the dynamics of a Dubins' vehicle in absolute coordinates with $-\pi \leq w^{(i)}(t) < \pi$. The averaging over the degrees of freedom of other agents in (C.9) is consistent with *mean field theory*, where the effect of all other agents on any one agent is an approximation of a single averaged influence.

Definition D.4 (Payoff of a Flock)**.** To every flock F_j (with a finite number of agents n_a) within a murmuration, $j = \{1, 2, \cdots, n_f\}$, we associate a payoff, $\boldsymbol{\Phi}_j$, that is the union of all respective agent's payoffs for expressing the outcome of a desired kinematic behavior.

Flock F_j and F_k within a murmuration, $F_j \cup F_k \cup F_l \cdots$ are separated by *partitions*, or *interfaces*, $\Gamma_{jk}, \Gamma_{kl}, \cdots$. This interface may be implicitly represented as a signed distance function $\Phi(\boldsymbol{x})$ which is negative on the interior of each flock, and zero on the edges. The zero-level set (i.e. $\Phi(\boldsymbol{x}) = 0$) corresponds to the interface $\boldsymbol{v}$ [Sethian, 1987]. As the system evolves over time, F_j's interface (zero-level set) motion can be parameterized by time, so that the flow field $\boldsymbol{v}(\boldsymbol{x}, t)$ is equivalent to the solution of the Cauchy-type Hamilton Jacobi partial differential equation [Evans and Souganidis, 1984, Crandall and Lions, 1983]:

$$\boldsymbol{v}_t + v_j |\nabla \boldsymbol{v}_j| = 0, \quad j = 1, \cdots, n_f, \tag{C.10}$$

where v_j is the flow speed for F_j. Equation (HJ-IVP) is the level set equation [Sethian, 2000].

In the sentiment of [Mitchell et al., 2005], we say the zero sublevel set of $g(\cdot)$ i.e. $\mathcal{L}_0$ in (Target-Set) is the *target set* in the phase space $\Omega \times \mathbb{R}$ for a backward reachability problem [Mitchell, 2001]. This target set[7] can represent the failure set, regions of danger, or obstacles to be avoided e.t.c. in the vectogram. And the *robustly controlled backward reachable tube* for $\tau \in [-T, 0]$[8] is the closure of the open set $\mathcal{L}([\tau, 0], \mathcal{L}_0)$ in (Target-Tube).

Each agent within a flock interacts with a fixed number of neighbors, n_c, within a fixed topological range, r_c. This topological range is consistent with findings in collective swarm behaviors and it reinforces *group cohesion* [Ballerini et al., 2008]. However, we are interested in *robust group cohesion* in reachability analysis. Therefore, we let a pursuer, $\boldsymbol{P}$, with a worst-possible disturbance attack the flock, and we take it that flocks of agents constitute an evading player, $\boldsymbol{E}$. Returning to (C.9), for a single flock, we now provide a sketch for the HJI formulation for a heading consensus problem.

[7] Note that the target set, $\mathcal{L}_0$, is a closed subset of $\mathbb{R}^n$ and is in the closure of Ω.

[8] The (backward) horizon, $-T$ is negative for $T > 0$.

D.1.5 Murmuration Flocks' Separated Payoffs.

Suppose that a murmuration's global heading is predetermined and each agent i within each flock, F_j, $(j = \{1, \cdots, n_f\})$ in the murmuration has a constant linear velocity, v^i. An agent's orientation is its control input, given by the average of its own orientation and that of its neighbors. Instead of metric distance interaction rules that make agents very vulnerable to predators [Ballerini et al., 2008], we resort to a topological interaction rule. With metric distance rules, we will have to formulate the breaking apart of value functions that encode a consensus heading problem in order to resolve the extrema of multiple payoffs; which is typically what we want to prevent in real-world autonomous tasks.

What constitutes an agent's neighbors are computed based on empirical findings and studies from the lateral vision of birds and fishes [Ballerini et al., 2008, Jadbabaie et al., 2003, Helbing et al., 2000] that provide insights into their anisotropic kinematic density and structure. Importantly, starlings' lateral visual axes and their lack of a rear sector reinforces their lack of nearest neighbors in the front-rear direction. As such, this enables them to maintain a tight density and robust heading during formation and flight.

Each agent within a flock F_j interacts with a fixed number of neighbors, n_c, within a fixed topological range, r_c. The topological range can be set as the distance between the labels of agents in a flock. This topological range is consistent with findings in collective swarm behaviors and it reinforces *group cohesion* [Ballerini et al., 2008]. However, we are interested in *robust group cohesion* in reachability analysis. Therefore, we let a pursuer, $\boldsymbol{P}$, with a worst-possible disturbance attack the flock, and we take it that flocks of agents constitute an evading player, $\boldsymbol{E}$.

D.1.6 Global Isotropy via Local Anisotropy.

Structural anisotropy is not merely an effect of a preferential velocity in animal flocking kinematics but rather an explicit effect of the anisotropic interaction character itself: agents choose a mutual position on the state space in order to maximize the sensitivity to changes in heading and speed of neighbors as the neighbors' anisotropy is optimized via vision-based collision avoidance characteristically unrelated to the eye's structure [Ballerini et al., 2008].

To reinforce robust group cohesion in local flocks, we randomly simulate a pursuer $\boldsymbol{P}_j$ against an evading agent in every flock F_j so that one agent is always relative coordinates with $\boldsymbol{P}^j$. In this specialized case, the $\boldsymbol{E}$ and $\boldsymbol{P}$'s speeds and maximum turn radii are equal: if both players start the game with the same initial velocity and orientation, the relative equations of motion show that $\boldsymbol{E}$ can mimic $\boldsymbol{P}$'s strategy by forever keeping the starting radial separation. As such, the *barrier* is closed and *the central theme in this game of kind is to determine the surface* [Merz, 1972]. We defer a thorough analysis of the nature of the surface to a future work.

Owing to the high-dimensionality of the state space, we cannot resolve this barrier analytically, hence we resort to numerical approximation methods – in particular, we leverage a parallel Lax-Friedrichs integration scheme [Crandall et al., 1984] which we implement in Cupy [Okuta et al., 2017] in order to provide a *consistent* and *monotone* solution to the Hamiltonians of the flocks. The assembly in the large of these respective Hamiltonians, and hence numerically robust solutions to the variational backward reeachability problem is resolved with a Voronoi tesselation of the zero-level sets of the boundaries of the flocks.

Therefore, for an agent i within a flock with index j in a murmuration, the equations of motion under attack from a predator p in relative coordinates is

$$\begin{bmatrix} \dot{\boldsymbol{x}}_1^{(i)_j}(t) \\ \dot{\boldsymbol{x}}_2^{(i)_j}(t) \\ \dot{\boldsymbol{x}}_3^{(i)_j}(t) \\ \dot{\boldsymbol{x}}_4^{(i)_j}(t) \end{bmatrix} = \begin{bmatrix} -v_e^{(i)_j}(t) + v_p^{(j)} \cos \boldsymbol{x}_4^{(i)_j}(t) + \langle w_e^{(i)_j} \rangle_r \boldsymbol{x}_2^{(i)_j}(t) \\ v_p^{(i)_j}(t) \sin \boldsymbol{x}_4^{(i)_j}(t) - \langle w_e^{(i)_j} \rangle_r \boldsymbol{x}_1^{(i)_j}(t) \\ u_p^{(j)}(t) - u_e^{(i)_j}(t) \\ w_p^{(j)}(t) - \langle w_e^{(i)_j}(t) \rangle_r \end{bmatrix} \tag{C.11}$$

for $i = 1, \cdots, n_a$ where n_a is the number of agents within a flock, $\left(\boldsymbol{x}_1^{(i)_j}(t), \boldsymbol{x}_2^{(i)_j}(t)\right) \in \mathbb{R}^2$, $\boldsymbol{x}_3 \in [-\gamma_{\max}, \gamma_{\max}]$, and we have $\boldsymbol{x}_4^{(i)_j}(t) \in [-\pi, +\pi)$[9]. Read $\boldsymbol{x}_1^{(i)_j}(t)$: the first component of the

[9]We have multiplied the dynamics by -1 so that the extremal's resolution evolves backwards in time.

state of an agent i at time t which belongs to the flock j in the murmuration at time t. In absolute coordinates, the equation of motion for *free agents* is

$$\begin{bmatrix} \dot{\boldsymbol{x}}_1^{(i)_j}(t) \\ \dot{\boldsymbol{x}}_2^{(i)_j}(t) \\ \dot{\boldsymbol{x}}_3^{(i)_j}(t) \\ \dot{\boldsymbol{x}}_4^{(i)_j}(t) \end{bmatrix} = \begin{bmatrix} v_e^{(i)_j}(t)\cos \boldsymbol{x}_4^{(i)_j}(t) \\ v_e^{(i)_j}(t)\sin \boldsymbol{x}_4^{(i)_j}(t) \\ u_z^{(i)_j}(t) \\ \langle w_e^{(i)_j}(t)\rangle_r \end{bmatrix}. \tag{C.12}$$

D.1.7 Murmuration via Aggregated Value Functions.

We introduce the union operator i.e. $\cup$ below as an aggregation symbol since the respective payoffs of each agent in a flock may be implicitly or explicitly constructed[10] – when it is implicitly represented, say from a signed distance function, we shall aggregate the payoff of agents 1 and 2 as

$$\cup\{\Phi_1(\boldsymbol{x},t),\Phi_2(\boldsymbol{x},t)\} \equiv \Phi_1(\boldsymbol{x},t)\cup\Phi_2(\boldsymbol{x},t) = \min(\Phi_1(\boldsymbol{x},t),\Phi_2(\boldsymbol{x},t)) \tag{C.13}$$

otherwise, other appropriate arithmetic or logical operation shall apply.

We assume that the *value* of a flock heading control (differential game) exists. And by an extension of Hamilton's principle of least action, the terminal motion of a flock coincide with the extremal of the payoff functional

$$\boldsymbol{v}(\boldsymbol{x},t) = \inf_{\beta^{(1)}\in\mathcal{B}^{(1)}}\sup_{\boldsymbol{u^{(1)}}\in\mathcal{U}^{(1)}} g^{(1)}(\boldsymbol{x}(T)) \cup\cdots\cup \inf_{\beta^{(n_f)}\in\mathcal{B}^{(n_f)}}\sup_{\boldsymbol{u}^{(n_f)}\in\mathcal{U}^{(n_f)}} g^{(n_f)}(\boldsymbol{x}(T)) \tag{C.14}$$

where n_f is the total number of distinct flocks in a murmuration. The resolution of this equation admits a viscosity solution to the following variational terminal HJI PDE [Mitchell et al., 2005]

$$\begin{aligned} \cup_{j=1}^{n_f}\left[\cup_{i=1}^{n_a}\left(\frac{\partial \boldsymbol{v}_i}{\partial t}(\boldsymbol{x},t) + \min\left[0, \boldsymbol{H}^{(i)}(\boldsymbol{x}^{(i)}, \boldsymbol{v}_x(\boldsymbol{x},t))\right]\right)\right] \\ = 0. \end{aligned} \tag{C.15}$$

with Hamiltonian,

$$\boldsymbol{H}^{(i)}(t;\boldsymbol{x}^{(i)},\boldsymbol{u}^{(i)},\boldsymbol{v}^{(i)},p^{(i)}) = \max_{u^{(i)}\in\mathcal{U}^{(i)}}\min_{v^{(i)}\in\mathcal{V}^{(i)}}\langle f^{(i)}(t;\boldsymbol{x},\boldsymbol{u}^{(i)},\boldsymbol{v}^{(i)}),p^{(i)}\rangle. \tag{C.16}$$

In swarms' collective motion, when e.g. a Peregrine Falcon attacks, immediate nearest agents change direction almost instantaneously. And because of the interdependence of the orientations of individual agents with respect to one another, all other agents respond instantaneously. Thus, we only simulate a single attack against a flock within the murmuration to realize robust cohesion.

A pursuer can attack any flock within the murmuration from a distinct surface: a $\boldsymbol{P}$ direction: this side of the surface reached after penetration in the $\boldsymbol{P}-[\boldsymbol{E}-]$ direction is the $\boldsymbol{P}-[\boldsymbol{E}]$ *side*[Isaacs, 1999]. We attribute the term *in the small* to determine the smooth parts of the singular surface solution when a pursuer attacks, and when they are stitched together into the total solution, we shall describe them as *in the large*. There exists at least one value $\bar{\alpha}$ of α such that if $\alpha=\bar{\alpha}$, no vector in the β-vectogram[11] penetrates the surface in the E-direction. Similar arguments can be made for $\bar{\beta}$ which prevents penetration in the P-direction. We adopt [Isaacs, 1999]'s terminology and call these surfaces semi-permeable surfaces (SPS).

Throughout the game, we assume that the roles of $\boldsymbol{P}$ and $\boldsymbol{E}$ do not change, so that when capture can occur, a necessary condition to be satisfied by the saddle-point controls of the players is the Hamiltonian, $\boldsymbol{H}^i(\boldsymbol{x},p)$.

[10] In resolving the zero-level sets of HJ value functions, it is typical to represent the payoff's surface as the isocontour of some function (usually a signed distance function).

[11] A $\beta-$vectogram is the resulting state space when a the strategy β is applied in computing the optimal control law for an agent.

Benchmark Toolkit

In comparison with benchmarks, the zero-level set is constructed implicitly from the isocontour of a signed distance function as described in [Osher and Fedkiw, 2004, Chapter II]. We introduce the union operator, $\cup$, below in lieu of the arithmetic summation symbol since the respective payoffs of each agent in a flock are implicitly initialized as signed distance functions on the state space. It is trivial to extend these results to other arithmetic or Boolean operators depending on different task domains. Recall that if $\Phi_1(\boldsymbol{x}, t)$ and $\Phi_2(\boldsymbol{x}, t)$ are respectively the payoffs for agents 1 and 2, constructed from signed distance functions, then the *union of their interiors*[a] is given by (C.13).

[a]We are only concerned with states that are within or on the boundary of the zero-level set or tube in backward reachability analysis. This follows from (Zero Levelset).

Theorem D.5. *For a flock, F_j, the Hamiltonian is the total energy given by a summation of the exerted energy by each agent i so that we can write the main equation or total Hamiltonian of a murmuration as*

$$\boldsymbol{H}(\boldsymbol{x}, p) = \max_{w_e^{(k)_j} \in [\underline{w}_e^j, \bar{w}_e^j]} \min_{w_p^{(k)_j} \in [\underline{w}_p^j, \bar{w}_p^j]} \cup_{j=1}^{n_f} \left[H_a^{(k)_j}(\boldsymbol{x}, p) \cup \left(\cup_{i=1}^{n_a - 1} H_f^{(i)_j}(\boldsymbol{x}, p) \right) \right] \tag{C.17}$$

$$\begin{aligned} &= \cup_{j=1}^{n_f} \left(\cup_{i=1}^{n_a - 1} \left[p_1^{(i)_j} v^{(i)_j} \cos \boldsymbol{x}_4 + p_2^{(i)_j} v^{(i)_j} \sin \boldsymbol{x}_4 + p_3^{(i)_j} u_z^{(i)_j} + p_4^{(i)_j} \langle w_e^{(i)_j} \rangle_r \right] \right. \\ &\quad \cup \left[p_1^{(k)_j} \left(v^{(k)_j} - v^{(k)_j} \cos \boldsymbol{x}_4^{(k)_j} \right) - p_2^{(k)_j} v^{(k)_j} \sin \boldsymbol{x}_4^{(k)_j} - \underline{w}_p^j | p_4^{(k)_j} | \right. \\ &\quad \left. \left. + \bar{w}_e^j \left| p_2^{(k)_j} \boldsymbol{x}_1^{(k)_j} - p_1^{(k)_j} \boldsymbol{x}_2^{(k)_j} + p_3^{(k)_j} \right| \right] \right). \end{aligned} \tag{C.18}$$

where $\boldsymbol{H}_a^{(k)_j}(\boldsymbol{x}, p)$ is the Hamiltonian of the individual under attack by a pursuing agent, $\boldsymbol{P}$, and $H_f^{(i)_j}(\boldsymbol{x}, p)$ are the respective Hamiltonians of the free agents, $i = \{1, \cdots, n_f\}$, within an evading flock in a murmuration, and not under the direct influence of capture or attack by $\boldsymbol{P}$; we denote by $w_e^{(i)_j}$ the heading of an evader i within a flock j and $w_p^{(j)}$ the heading of a pursuer aimed at flock j; $\underline{w}_e^{(k)_j}$ is the orientation that corresponds to the orientation of the agent with minimum turn radius among all the neighbors of agent k, inclusive of agent k at time t; similarly, $\bar{w}_e^{(k)_j}$ is the maximum orientation among all of the orientation of agent k's neighbors.

Proof. From (C.18), the total Hamiltonian of a flock is a union of the mechanical energy of the free agents in a flock and the individual under attack. At each instant, we are extremizing over the headings $w_e^{(k)_j} \in [\underline{w}_e^j, \bar{w}_e^j], w_p^{(k)_j} \in [\underline{w}_p^j, \bar{w}_p^j]$ and climb rates $u_e^{(k)_j}, u_p^{(k)_j} \in [-\gamma_{\max}, \gamma_{\max}]$ respectively i.e.

$$\boldsymbol{H}(\boldsymbol{x}, p) = \max_{[w_e^{(k)_j}, u_e^{(k)_j}]} \min_{[w_p^{(k)_j}, u_p^{(k)_j}]} \cup_{j=1}^{n_f} \left[H_a^{(k)_j}(\boldsymbol{x}, p) \cup \left(\cup_{i=1}^{n_a - 1} H_f^{(i)_j}(\boldsymbol{x}, p) \right) \right] \tag{C.19}$$

We now set $\underline{\gamma} = -\gamma_{\max}$ and $\bar{\gamma} = \gamma_{\max}$ for ease of readability. We write the Hamiltonian of the free agents in absolute coordinates and the Hamiltonian of the agent under attack in relative coordinates with respect to the pursuer. A flock's Hamiltonian is the aggregation of all the mechanical energy in the system in absolute coordinates i.e.

$$\cup_{i=1}^{n_a - 1} \boldsymbol{H}_f^{(i)_j}(\boldsymbol{x}, p) = \cup_{i=1}^{n_a - 1} \left[p_1^{(i)_j} v^{(i)_j} \cos \boldsymbol{x}_4 + p_2^{(i)_j} v^{(i)_j} \sin \boldsymbol{x}_4 + p_3^{(i)_j} u_{z_e}^{(i)_j} + p_4^{(i)_j} \langle w_e^{(i)_j} \rangle_r \right], \tag{C.20}$$

where we have again dropped the time arguments for convenience. And,

$$\begin{aligned} \boldsymbol{H}_a^{(k)_j}(\boldsymbol{x}, p) = -\left(\max_{[w_e^{(k)_j}, u_e^{(k)_j}]} \min_{[w_p^{(k)_j}, u_p^{(k)_j}]} \left[p_1^{(k)_j}(t) \quad p_2^{(k)_j}(t) \quad p_3^{(k)_j}(t) \quad p_4^{(k)_j}(t) \right] \right. \\ \left. \begin{bmatrix} -v_e^{(k)_j}(t) + v_p^{(j)} \cos \boldsymbol{x}_4^{(k)_j}(t) + \langle w_e^{(k)_j} \rangle_r(t) \boldsymbol{x}_2^{(k)_j}(t) \\ v_p^j(t) \sin \boldsymbol{x}_4^{(k)_j}(t) - \langle w_e^{(k)_j} \rangle_r(t) \boldsymbol{x}_1^{(k)_j}(t) \\ u_p^{(i)_j}(t) - u_e^{(i)_j}(t) \\ w_p^j(t) - \langle w_e^{(k)_j}(t) \rangle_r \end{bmatrix} \right), \end{aligned} \tag{C.21}$$

where $p_l^{(k)_j}(t)\ |_{l=1,2,3}$ are the adjoint vectors [Merz, 1972]. For the pursuer, its minimum and maximum turn and climb rates are fixed so that we have $(\underline{w}_p^j, \underline{\gamma})$ as the minimum turn and climb bounds of the pursuing vehicle, and $(\bar{w}_p^j, \bar{\gamma})$ are the maximum turn, and climb bounds of the pursuing vehicle, respectively. Henceforth, we drop the templated time arguments for ease of readability. From (C.21), we have

$$\begin{aligned}
\boldsymbol{H}_a^{(k)_j}(\boldsymbol{x}, p) = -\Bigg(& \max_{[w_e^{(k)_j}, u_e^{(k)_j}]} \min_{[w_p^{(k)_j}, u_p^{(k)_j}]} \Big[-p_1^{(k)_j} v_e^{(k)_j} + p_1^{(k)_j} v_p^j \cos \boldsymbol{x}_4^{(k)_j} + p_1^{(k)_j} \langle w_e^{(k)_j} \rangle_r \boldsymbol{x}_2^{(k)_j} \\
& + p_2^{(k)_j} v_p^j \sin \boldsymbol{x}_4^{(k)_j} - p_2^{(k)_j} \langle w_e^{(k)_j} \rangle_r \boldsymbol{x}_1^{(i)_j} + p_3^{(k)_j} \left(u_p^{(i)_j} - u_e^{(i)_j} \right) + p_4^{(k)_j} \left(w_p^j - \langle w_e^{(k)} \rangle_r \right) \Big] \Bigg), \\
\triangleq\ & p_1^{(k)_j} \left(v_e^{(k)_j} - v_p^j \cos \boldsymbol{x}_4^{(k)_j} \right) - p_2^{(k)_j} v_p^j \sin \boldsymbol{x}_4^{(k)_j} + \Bigg(\max_{\langle w_e^{(k)_j} \rangle_r \in [\underline{w}_e^j, \bar{w}_e^j], u_e^{(k)_j} \in [\underline{\gamma}, \bar{\gamma}]} \min_{w_p^j \in [\underline{w}_p^j, \bar{w}_p^j], u_p^j \in [\underline{\gamma}, \bar{\gamma}]} \\
& \begin{bmatrix} \langle w_e^{(k)_j} \rangle_r & \left(p_2^{(k)_j} \boldsymbol{x}_1^{(k)_j} - p_1^{(k)_j} \boldsymbol{x}_2^{(k)_j} + p_4^{(k)_j} \right) + p_3^{(k)_j} \left(u_p^{(i)_j} - u_e^{(i)_j} \right) \\ & -p_4^{(k)_j} w_p^j \end{bmatrix} \Bigg).
\end{aligned}$$

So that,

$$\begin{aligned}
\boldsymbol{H}_a^{(k)_j}(\boldsymbol{x}, p) = p_1^{(k)_j} \left(v_e^{(k)_j} - v_p^j \cos \boldsymbol{x}_4^{(k)_j} \right) - p_2^{(k)_j} v_p^j \sin \boldsymbol{x}_4^{(k)_j} - \underline{w}_p^j |p_4^{(k)_j}| + \bar{w}_e^j \Big| p_2^{(k)_j} \boldsymbol{x}_1^{(k)_j} \\
- p_1^{(k)_j} \boldsymbol{x}_2^{(k)_j} + p_4^{(k)_j} \Big| + \underline{\gamma} |p_3^{(k)_j}| - \bar{\gamma} |p_3^{(k)_j}|
\end{aligned} \tag{C.22}$$

and

$$\boldsymbol{H}_f^{(i)_j}(\boldsymbol{x}, p) = \left[p_1^{(i)_j} v^{(i)_j} \cos \boldsymbol{x}_4 + p_2^{(i)_j} v^{(i)_j} \sin \boldsymbol{x}_4 + p_3^{(i)_j} u_{z_e}^{(i)_j} + p_4^{(i)_j} \langle w_e^{(i)_j} \rangle_r \right]. \tag{C.23}$$

Therefore, the main equation (C.18) becomes

$$\begin{aligned}
\boldsymbol{H}(\boldsymbol{x}, p) = \cup_{j=1}^{n_f} \Big(\cup_{i=1}^{n_a - 1} \left[p_1^{(i)_j} v^{(i)_j} \cos \boldsymbol{x}_4 + p_2^{(i)_j} v^{(i)_j} \sin \boldsymbol{x}_4 + p_3^{(i)_j} u_{z_e}^{(i)_j} + p_4^{(i)_j} \langle w_e^{(i)_j} \rangle_r \right] \cup_{k_j} \\
\Big[p_1^{(k)_j} \left(v_e^{(k)_j} - v_p^j \cos \boldsymbol{x}_4^{(k)_j} \right) - p_2^{(k)_j} v_p^j \sin \boldsymbol{x}_4^{(k)_j} - \underline{w}_p^j |p_4^{(k)_j}| + \bar{w}_e^j \Big| p_2^{(k)_j} \boldsymbol{x}_1^{(k)_j} - p_1^{(k)_j} \boldsymbol{x}_2^{(k)_j} \\
+ p_4^{(k)_j} \Big| - \gamma |p_3^{(k)_j}| - \gamma |p_3^{(k)_j}| \Big] \Big).
\end{aligned} \tag{C.24}$$

□

Remark D.6. For the special case where the linear speeds of the evading agents and pursuer are equal i.e. $v_e^{(i)_j}(t) = v_p(t) = +1 m/s$, a murmuration's Hamiltonian reduces to

$$\begin{aligned}
\boldsymbol{H}(\boldsymbol{x}, p) = \cup_{j=1}^{n_f} \Big(\cup_{i=1}^{n_a - 1} \left[p_1^{(i)_j} \cos \boldsymbol{x}_4 + p_2^{(i)_j} \sin \boldsymbol{x}_4 + p_3^{(i)_j} u_{z_e}^{(i)_j} + p_4^{(i)_j} \langle w_e^{(i)_j} \rangle_r \right] \cup_{k_j} \\
\Big[p_1^{(k)_j} \left(1 - \cos \boldsymbol{x}_4^{(k)_j} \right) - p_2^{(k)_j} \sin \boldsymbol{x}_4^{(k)_j} - \underline{w}_p^j |p_4^{(k)_j}| + \bar{w}_e^j \Big| p_2^{(k)_j} \boldsymbol{x}_1^{(k)_j} - p_1^{(k)_j} \boldsymbol{x}_2^{(k)_j} \\
+ p_4^{(k)_j} \Big| - \gamma |p_3^{(k)_j}| - \gamma |p_3^{(k)_j}| \Big] \Big).
\end{aligned} \tag{C.25}$$

D.1.8 Vacuole Nucleation Topology

When an attacking predator penetrates (e.g. a Peregrine Falcon [Ballerini et al., 2008]) the interior of a flock's backward reachable tube (BRT), a topological hole or a *vacuole* [van Leeuwenhoek, 1800] nucleates in the zero-sublevel set $\mathcal{L}(t) \triangleq \{x : v(t, x) \leq 0\}$, an event characterized by the Euler characteristic.

Definition D.7 (Euler Characteristic of the BRT)**.** Let $\mathcal{L}(t)$ be the BRS at time t, discretized on a uniform $m \times m$ grid with vertex, edge, and face counts V, E, F, respectively. The *Euler characteristic* [Hatcher, 2002] is

$$\chi(t) \triangleq V - E + F \triangleq \beta_0(t) - \beta_1(t), \tag{C.26}$$

where β_0 is the number of connected components and β_1 is the first Betti number (number of independent loops/holes).

Theorem D.8 (Vacuole Nucleation Topology)**.** *Suppose that the evading agent's Hamiltonian $\boldsymbol{H}_f(x,p)$ satisfies $\boldsymbol{H}_f(x^*,p^*)=0$ for some interior state $x^*\in\Omega$ with costate $p^*=Dv(t^*,x^*)$. Then at time t^* when the zero-crossing occurs, if the predator penetrates the interior of the flock's reachable set, the topology of $\mathcal{L}(t^\star)$ undergoes a transition characterized by,*

$$\chi(t^{*+}) \;=\; \chi(t^{*-}) - 1, \tag{C.27}$$

where χ is the Euler characteristic (C.26). This jump indicates the nucleation of a vacuole (topological hole) in the flock's safe set, marking the transition from simply connected ($\beta_1=0$) to multiply connected ($\beta_1\geq 1$) topology.

Proof. By Morse theory for sublevel sets of smooth functions, the topology of $\Omega^-(t)$ changes only at critical values of $v(\cdot,t)$. Specifically, when the parameter t crosses a value t^* at which a saddle point of $v(\cdot,t^*)$ lies on the boundary of $\Omega^-(t^*)$, the sublevel set attaches a 1-handle.

The attachment of a 1-handle increases the first Betti number by one: $\beta_1(t^{*+})=\beta_1(t^{*-})+1$. Since the Euler characteristic is defined as $\chi=\beta_0-\beta_1$ (Definition D.7), this increase in β_1 directly reduces χ by one:

$$\chi(t^{*+})=\beta_0(t^{*+})-\beta_1(t^{*+})=\beta_0(t^{*-})-(\beta_1(t^{*-})+1)=\chi(t^{*-})-1. \tag{C.28}$$

The condition $H_{\text{att}}(x^*,p^*)=0$ identifies the critical state-costate pair where the attacked agent's optimal trajectory encounters the reachable set boundary. At this instant, the saddle-point singularity of v causes the topological change described above. The geometric interpretation is that the predator has penetrated sufficiently deep into the flock to create an enclosed region, manifested as a topological hole in the backward reachable set. □

D.1.9 Flock Splitting Dynamics

When a predator applies sufficient control to reduce the attacked agent's optimal escape velocity, the BRT bifurcates into disjoint components. This bifurcation corresponds to the moment when the attacking predator can successfully isolate one or more agents from the main group.

Proposition D.9 (Bifurcation Condition for Flock Splitting)**.** *Let $\Omega^-(t)$ be the zero-sublevel set of the BRT at time t. The reachable set undergoes a bifurcation into $n_{\text{comp}}(t)>1$ connected components if and only if the attacked-agent Hamiltonian satisfies the control-constrained minimax condition,*

$$H_{\text{att}}(x^*,p^*) \;=\; \min_{u_p\in\mathcal{U}_p}\max_{u_e\in\mathcal{U}_e}\left[p_1(u_p-v_e)+p_2u_e+p_3(u_p^z-u_e^z)+p_4(\omega_p-\omega_e)\right] \;<\; -\delta_c, \tag{C.29}$$

where $\delta_c>0$ is a separation margin, $\mathcal{U}_p,\mathcal{U}_e$ are admissible control sets, and (u_p,ω_p), (u_e,ω_e) are the pursuer and evader acceleration and angular velocity controls respectively.

Proof. We prove both directions of the equivalence by reducing the change in the number of connected components of $\Omega^-(t)$ to the passage of a saddle point of $v(\cdot,t)$ through the zero level, and then relating that passage to the attacked-agent Hamiltonian through the Hamilton–Jacobi–Isaacs (HJI) equation.

Step 1 (Connectivity is governed by interior saddles). By the Morse theory of sublevel sets, the homotopy type of $\Omega^-(t)=\{x: v(t,x)\leq 0\}$ is constant on any backward-time interval that contains no critical value of $v(\cdot,t)$ crossing the level 0. The component count $n_{\text{comp}}(t)=\beta_0(t)$ can increase only when an index-1 critical point (a saddle) $x^\star$ of $v(\cdot,t)$ crosses the zero level from below, that is, when the connecting ridge value $v(t,x^\star)$ passes from $v(t,x^\star)\leq 0$ to $v(t,x^\star)>0$. At that instant the ridge separating two incipient basins is lifted above 0 and a single component severs into two; conversely, while $v(t,x^\star)\leq 0$ at every separating saddle, the sublevel set remains connected through those saddles. Letting $x^\star$ denote the lowest saddle (the mountain-pass state) separating two candidate basins,

$$n_{\text{comp}}(t)>1 \iff v(t,x^\star)>0. \tag{C.30}$$

Step 2 (Evolution of the saddle value). Let $p^\star\triangleq\nabla_x v(t,x^\star)$ be the costate at the separating saddle. Since v is the viscosity solution of the HJI equation

$$\partial_t v(t,x)+\min(0,\;\boldsymbol{H}_{\text{att}}(x,\nabla_x v(t,x)))=0, \tag{C.31}$$

and the spatial gradient vanishes along the ridge direction at a critical point, the rate of change of the value at the (slowly varying) saddle is, to first order,

$$\frac{d}{dt}v(t,x^\star) \;=\; \partial_t v(t,x^\star) \;=\; -\min(0,\; \boldsymbol{H}_{\mathrm{att}}(x^\star,p^\star))\,. \tag{C.32}$$

Thus the ridge lifts above 0 at exactly the rate $-\boldsymbol{H}_{\mathrm{att}}(x^\star,p^\star)$ whenever $\boldsymbol{H}_{\mathrm{att}}(x^\star,p^\star)<0$, and is stationary otherwise.

Step 3 (Evaluating the minimax at the saddle). At the separating state the pursuer acts to sever the flock (to raise v) while the evader acts to preserve cohesion (to keep $v\le 0$); this is precisely the lower value of the local zero-sum game,

$$\boldsymbol{H}_{\mathrm{att}}(x^\star,p^\star)=\min_{u_p\in\mathcal{U}_p}\max_{u_e\in\mathcal{U}_e}\Big[p_1(u_p-v_e)+p_2u_e+p_3(u_p^z-u_e^z)+p_4(\omega_p-\omega_e)\Big], \tag{C.33}$$

which is the right-hand side of (C.29) evaluated at $(x^\star,p^\star)$. Because the bracketed integrand is affine in each control, the extremizers are bang–bang and the minimax in (C.33) is finite and attained.

Step 4 (Separation margin and equivalence). Fix the residual backward horizon and the quadrature scale through a single separation margin $\delta_c>0$: a ridge whose value is at most 0 at time t is lifted strictly above 0 within one comparison step if and only if its upward rate exceeds the residual depth, i.e. $-\boldsymbol{H}_{\mathrm{att}}(x^\star,p^\star)>\delta_c$. Combining this calibration with (C.32) and Step 1 yields

$$n_{\mathrm{comp}}(t)>1 \iff \boldsymbol{H}_{\mathrm{att}}(x^\star,p^\star)<-\delta_c. \tag{C.34}$$

The forward implication is Step 1 together with (C.32): a split requires the ridge to lift, hence a strictly negative Hamiltonian beyond the margin. The reverse implication follows because a Hamiltonian below $-\delta_c$ drives $v(t,x^\star)>0$ through (C.32), which by (C.30) disconnects $\Omega^-(t)$ into $n_{\mathrm{comp}}(t)>1$ components. As the separating state $(x^\star,p^\star)$ realizes the minimax in (C.33), this is exactly the bifurcation condition (C.29), completing the proof. □

Remark D.10 (Temporal Bifurcation Pattern). The bifurcation occurs at a critical time t_{split} determined by when the predator's control authority first permits the pursuer to violate the costate constraint in the evader's optimal strategy. The sequence of bifurcation times is monotone increasing: $t_{\mathrm{split}}^{(1)}<t_{\mathrm{split}}^{(2)}<\cdots$, with the m-th bifurcation occurring when isolation of the m-th agent becomes achievable under the game-theoretic equilibrium. The flock's resilience to splitting depends on the coupling strength in the risk-reduction term (the p_4 coefficient in the Hamiltonian), which represents the predation risk shared across connected components.

D.1.10 Cordon Formation and Capture Prevention

A flock executing a defensive *cordon formation* creates an annular perimeter around threatened agents, preventing isolated predator incursions into the interior. This configuration is geometrically characterized by a multiply connected reachable set with one independent loop ($\beta_1=1$).

Proposition D.11 (Annular BRT and Barrier Certificate)**.** *Let $\mathcal{C}(r_{\mathrm{in}},r_{\mathrm{out}})\triangleq\{x\in\Omega : r_{\mathrm{in}}\le\|x_{1:2}\|\le r_{\mathrm{out}}\}$ denote the annular barrier region ("cordon") with inner radius r_{in} and outer radius r_{out}. If the BRT restricted to $\mathcal{C}$ satisfies,*

$$v(t,x)\;>\;0\quad \textit{for all } x\in\{y\in\mathcal{C}:\|y_{1:2}\|<r_{\mathrm{in}}\}, \tag{C.35}$$

then the inner region is a barrier-certificate-protected domain*, and any trajectory of the attacked agent passing through $\mathcal{C}(r_{\mathrm{in}},r_{\mathrm{out}})$ remains captured (i.e., reaches the terminal set) within finite time. The topology of $\Omega^-(t)\cap\mathcal{C}$ is annular, with Betti number $\beta_1=1$ and Euler characteristic,*

$$\chi\big(\Omega^-(t)\cap\mathcal{C}\big)\;=\;0. \tag{C.36}$$

Proof. The claim has an invariance part (the inner core is protected and any trajectory crossing the cordon is captured) and a topological part ($\beta_1=1$, $\chi=0$); we establish each in turn.

Step 1 (The inner core is a certified barrier). The value $v(t,\cdot)$ is the viscosity solution of the HJI equation (C.31) with terminal datum $v(0,x)=g(x)=\|x_{1:2}\|-r$, so that the capture tube is its zero-sublevel set $\{x : v(t,x)\le 0\}$. By the comparison principle for viscosity solutions, v is monotone along optimal motions: for the attacked agent's optimal control against the worst-case

pursuer, $\tau \mapsto v(t-\tau, x(\tau))$ is nonincreasing. Hypothesis (C.35) states that $v(t,x) > 0$ on the open inner disk $\{y : \|y_{1:2}\| < r_{\text{in}}\}$; that core therefore lies strictly outside the capture tube and cannot be driven into the terminal set at time t. Hence $v(t,\cdot) > 0$ is a barrier certificate for the protected core.

Step 2 (Trajectories crossing the cordon are captured in finite time). Let $x(\cdot)$ be a trajectory of the attacked agent that meets the annulus, i.e. there is τ_0 with $r_{\text{in}} \le \|x(\tau_0)_{1:2}\| \le r_{\text{out}}$ and $v(t, x(\tau_0)) \le 0$, so the trajectory starts in the BRT restricted to $\mathcal{C}$. By the monotonicity of Step 1, v cannot increase along the motion; it stays ≤ 0, so the trajectory can neither enter the protected core (where $v > 0$) nor leave the sublevel set. Confined to the sublevel annulus, it is driven to the terminal set $\{g \le 0\}$ in finite time and is captured. This is the capture-prevention property of the cordon.

Step 3 (Annular topology: $\beta_0 = 1$, $\beta_1 = 1$). On $\mathcal{C}$ the restricted sublevel set $\Omega^-(t) \cap \mathcal{C} = \{x \in \mathcal{C} : v(t,x) \le 0\}$ excludes the open core by (C.35) and is bounded outside by r_{out}. The radial monotonicity of the terminal cost $g(x) = \|x_{1:2}\| - r$ and the rotational symmetry of the capture cylinder make this set a deformation retract of the closed annulus $\{x : r_{\text{in}} \le \|x_{1:2}\| \le r_{\text{out}}\}$, which is homotopy equivalent to the circle S^1. Its homology gives a single connected component and a single independent loop, $\beta_0 = 1$ and $\beta_1 = 1$.

Step 4 (Euler characteristic). By Definition D.7,

$$\chi\big(\Omega^-(t) \cap \mathcal{C}\big) \;=\; \beta_0 - \beta_1 \;=\; 1 - 1 \;=\; 0, \tag{C.37}$$

which is (C.36). The vanishing Euler characteristic is the topological signature of the defensive shell: one safe component wrapping one protected hole. This proves the proposition. □

Remark D.12 (Barrier Certificate Interpretation). The annular topology (Eq. (C.36)) physically represents a mutual-defense configuration where the safe set (valued region, $v \le 0$) wraps around the predator or protected interior, creating a topological obstruction to escape. From the reachability perspective, any agent trapped inside the cordon cannot escape; from outside, no agent can cross the barrier without incurring catastrophic cost. This geometric property is independent of the detailed dynamics and depends only on the level-set structure of v.

The numerical validation of the cordon configuration is visualized through the reachability level sets of the value function $v(t,\cdot)$, whose innermost contour bounds the protected core and whose outermost contour bounds the defensive shell, exhibiting the annular barrier topology explicitly.

D.1.11 Flash Expansion Events

The *flash expansion* is characterized by a rapid, isotropic outward motion of the flock boundary in response to a predator approach. In the BRT framework, this corresponds to a linear growth of the zero-level-set radius.

Lemma D.13 (Linear Flash Expansion Rate)**.** *Consider the free-agent dynamics* (C.11) *with the altitude-invariant cylinder terminal cost*

$$g(x) = \|x_{1:2}\| - r, \qquad \mathcal{L}_0 = \{x : g(x) \le 0\}, \tag{C.38}$$

and let $\Delta t = T_2^{-1}$ be the backward step set by the temporal resolution $T_2 > 0$ of the second (flock-level) interaction hierarchy. During the flash-expansion phase, the maximum radial extent of the BRT zero-level-set in the horizontal plane grows at the rate,

$$\max_i \|x_i(t+1)_{1:2}\| - \max_i \|x_i(t)_{1:2}\| \;=\; \Theta\big(T_2^{-1}\big)\,. \tag{C.39}$$

The altitude component x_3 does not contribute to the radial growth rate because the capture set is the cylinder (C.38)*.*

Proof. Write $\rho(t) \triangleq \max_i \|x_i(t)_{1:2}\|$ for the maximal horizontal extent of the zero-level set, attained at a boundary agent $i^\star(t)$. We bound the one-step increment $\rho(t+1) - \rho(t)$ from above and below and show both bounds are $\Theta(T_2^{-1})$.

Step 1 (One-step radial increment). By the dynamic programming principle, the next-step BRT is the set of states from which the attacked agent, against the worst-case pursuer, reaches $\{v(t,\cdot) \le 0\}$ in one step $\Delta t = T_2^{-1}$. Propagating the outermost boundary agent one step under (C.11) and invoking the comparison principle for viscosity solutions, $v(t,x) \le 0 \Rightarrow v(t+\Delta t, x + \Delta t\, f(x,u)) \le 0$, gives

$$\rho(t+1) - \rho(t) \;=\; s_\perp(t)\,\Delta t \;=\; s_\perp(t)\,T_2^{-1}, \tag{C.40}$$

where $s_\perp(t)$ is the realized outward radial speed of the boundary agent $i^\star(t)$.

Step 2 (Constant-speed bound; upper estimate). The horizontal velocity in (C.11) is $(v_e \cos x_4,\, v_e \sin x_4)$, of constant Euclidean norm v_e irrespective of the heading x_4. The outward radial speed is its projection onto the radial direction, $s_\perp(t) = v_e \cos \alpha(t)$, with $\alpha(t)$ the angle between the heading and the outward radial. Hence $s_\perp(t) \le v_e$, and

$$\rho(t+1) - \rho(t) \;\le\; v_e\, T_2^{-1} \;=\; O(T_2^{-1}). \tag{C.41}$$

Step 3 (Direction retention; lower estimate). In the constant-speed model the pursuer cannot alter the evader's speed v_e; it acts only on the heading, through the neighbor-averaged coupling $\langle w_e \rangle_r$ in (C.11), turning the boundary agent toward the tangential direction $\alpha = \pi/2$, at which outward growth stalls (the tip of the flash phase). By definition, the flash-expansion phase is the interval before this tipping: throughout it the heading stays within a fixed cone of the outward radial, $|\alpha(t)| \le \frac{\pi}{2} - \vartheta$ for some $\vartheta > 0$ fixed by the turn-rate bound $\bar{w}_e$. Therefore $\cos \alpha(t) \ge \sin \vartheta \triangleq c > 0$, and

$$s_\perp(t) = v_e \cos \alpha(t) \ge c\, v_e \quad\Longrightarrow\quad \rho(t+1) - \rho(t) \;\ge\; c\, v_e\, T_2^{-1} \;=\; \Omega(T_2^{-1}). \tag{C.42}$$

Step 4 (Altitude does not contribute). Because the terminal cost (C.38) is independent of the altitude x_3, the vertical velocity u_z enters neither g nor the radial norm $\|x_{1:2}\|$. The radial extent ρ and its increment depend only on the horizontal sub-dynamics, so x_3 contributes nothing to the growth rate.

Step 5 (Conclusion). Combining (C.41) and (C.42),

$$c\, v_e\, T_2^{-1} \;\le\; \rho(t+1) - \rho(t) \;\le\; v_e\, T_2^{-1}, \tag{C.43}$$

with $0 < c \le 1$ and v_e fixed; therefore $\rho(t+1) - \rho(t) = \Theta(T_2^{-1})$, which is (C.39). □

The 4D flash-expansion rate converges to the 3D rate as the altitude variance $\sigma_{x_3}^2 \to 0$, consistent with the altitude-decoupling property of the cylinder terminal cost (C.38). We verify this decoupling empirically through a dedicated altitude-invariance test, which proceeds as follows. From the certified ensemble we extract every bird state that shares a fixed horizontal cell and heading — $|x_1 - 1| < 0.2$, $|x_2| < 0.2$, and $|x_4| < 0.1$ — while leaving the altitude x_3 unconstrained, so that the selected states differ essentially only in altitude. On this slice we form the empirical mean $\bar{v}$ and standard deviation $\hat{\sigma}_v$ of the value function and report the coefficient of variation $\mathrm{CV} \triangleq \hat{\sigma}_v / |\bar{v}|$. Were the value genuinely altitude-invariant, CV would vanish; the test certifies decoupling by requiring $\mathrm{CV} < 0.5$, a threshold chosen to absorb the residual $O(\delta^{1/2})$ Monte-Carlo fluctuation of the sampled solution rather than any altitude dependence. The test passes across every run reported here, confirming that $v(t, x_1, x_2, x_3, x_4)$ is invariant to x_3 up to sampling noise — exactly as predicted by the altitude-independent capture cylinder (C.38).

D.1.12 Phase Transition Markers in Numerical Solutions

We detect phase transitions computationally by monitoring the topological invariants $\chi(t)$, $\beta_1(t)$, and $n_{\mathrm{comp}}(t)$ of the zero-sublevel set across backward time steps. The detection algorithm is:

1. At each step t, compute the 2D slice $v(t, x_1, x_2; \bar{x}_3, 0)$ on a grid of resolution $m \times m$.
2. Label connected components of $\{v \le 0\}$ via SCIPY.NDIMAGE.LABEL.
3. Estimate β_1 by counting encircled complementary components.
4. Compute $\chi(t) = n_{\mathrm{comp}}(t) - \beta_1(t)$.

The threshold-crossing rules that trigger an event label are:

$$\Delta\chi_t \;\triangleq\; \chi_{t+1} - \chi_t \;<\; -0.5 \quad\Longrightarrow\quad \text{vacuole nucleation}, \tag{C.44}$$

$$\beta_1(t) \;\ge\; 1 \;\text{ and }\; \beta_1(t-1) = 0 \quad\Longrightarrow\quad \text{cordon formation}, \tag{C.45}$$

$$n_{\mathrm{comp}}(t) \;>\; n_{\mathrm{comp}}(t-1) \quad\Longrightarrow\quad \text{flock fragmentation}. \tag{C.46}$$

The classification partitions the backward-reachable evolution into the five canonical murmuration phases — (i) cohesion, (ii) evasion, (iii) cordon, (iv) expansion, and (v) fragmentation — according to the marker rules (C.44)–(C.46), with each detected event recorded together with its backward-time index, the event type, and the invariants χ, β_1, and n_{comp} at the crossing.

Fig. 6 traces the topological invariants $\chi(t)$, $\beta_1(t)$, and $n_{\mathrm{comp}}(t)$ across backward time, summarising the phase-transition events flagged by the marker rules (C.44)–(C.46).

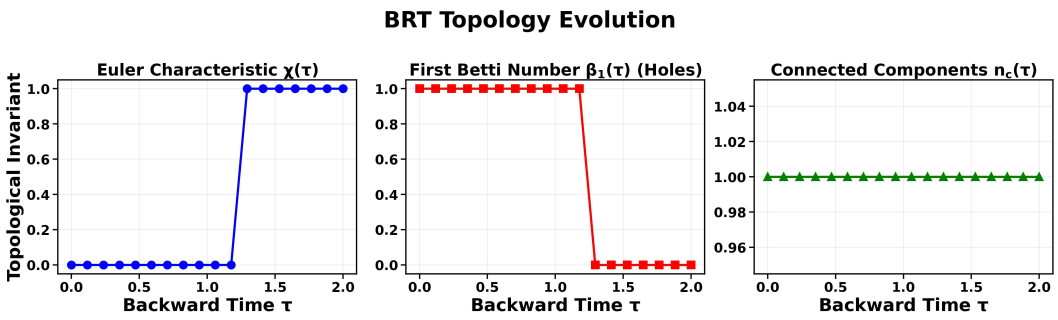


Figure 6: Evolution of the backward-reachable-tube topology for a certified murmuration. *Left:* the Euler characteristic $\chi(t)$; *center:* the first Betti number $\beta_1(t)$ (the number of independent holes); *right:* the number of connected components $n_{\text{comp}}(t)$, each plotted against the backward time step t. A downward jump in χ marks a vacuole nucleation, a $0 \to 1$ transition in β_1 marks a cordon formation, and an increase in n_{comp} marks a flock fragmentation, in accordance with the markers (C.44)–(C.46).

D.1.13 Numerical Results

We now certify the safety of aerial murmurations by resolving the murmuration HJI equation (C.15) with the sampling scheme of §D.1.5. All runs reported here were executed on a single 28-core CPU host with 31 GiB of RAM; our JAX implementation of the Cole–Hopf/Feynman–Kac sampler is GPU-compatible (the multi-GPU distributor of the toolbox falls back transparently to the host). At the time of running the experiments in this paper, however, we evaluate all results on CPUs owing to administrative reasons; we accordingly report *CPU-only* timings throughout, so that the storage-free memory footprint — and not a particular accelerator — carries the scaling argument. Unless stated otherwise we fix the viscosity at $\delta = 0.18$, draw $N = 1600$ Gaussian samples per evaluated state, run the quasi-linearization to 7 Picard iterations, and integrate the backward horizon $\tau \triangleq T - t$, $T = 2.0$, in 18 steps. The backward reachable tube (BRT) is the zero-sublevel set $\mathcal{L}(\tau) = \{x : v(\tau, x) \leq 0\}$, which we evaluate on a 128×128 grid of the horizontal plane (x_1, x_2) at fixed altitude $x_3 = 50\,\text{m}$ and heading $x_4 = 0$; the altitude slice is immaterial by the decoupling established in §D.1.11.

Population size and flock partition. We simulate murmurations of up to $N = 100{,}000$ birds. For the topology-resolving experiments we use $N = 7{,}998$ birds partitioned into $n_f = 6$ flocks (1,333 birds per flock), and for the scaling study we use $N = 100{,}000$ birds across $n_f = 10$ flocks (10,000 birds per flock). Two considerations fix this choice. First, the partition into flocks mirrors the separable subgroup structure of §D.1.5: each flock is resolved by its own value function and the murmuration safe set is their union (C.15), so the flock is the natural unit of computation, while within a flock every bird interacts only with a fixed topological neighborhood of $n_c \approx 6$–7 nearest neighbors — the density-independent interaction count reported for European starlings [Ballerini et al., 2008]. Second, because the sampler is storage-free the cost of resolving the value grid is independent of N (cf. the scaling study below); the bird count therefore stresses the *per-bird* certification rather than the grid solve, and we deliberately push it to the 100,000-bird regime — well beyond what a grid-based solver could hold in memory — to substantiate that claim. We do not visualize all N birds: a murmuration of this size is illustrated through value-function snapshots and a subsampled phase-space scatter, never a point cloud of the entire population.

Collective behaviors of a murmuration. Field studies of European starlings document a small repertoire of collective behaviors through which a murmuration preserves cohesion and fends off attack [Ballerini et al., 2008, Cavagna et al., 2010]. We organize our experiments around the following seven canonical behaviors, each of which leaves a signature in the backward reachable tube,

(i) *flock cohesion* — the birds hold a tight, bounded formation;

(ii) *heading consensus* — neighboring birds align their headings to a common direction;

(iii) *predator evasion* — the flock coordinates to remain outside the attacker's capture set;

(iv) *flash expansion* — a rapid, isotropic outward burst of the flock boundary as a predator approaches;

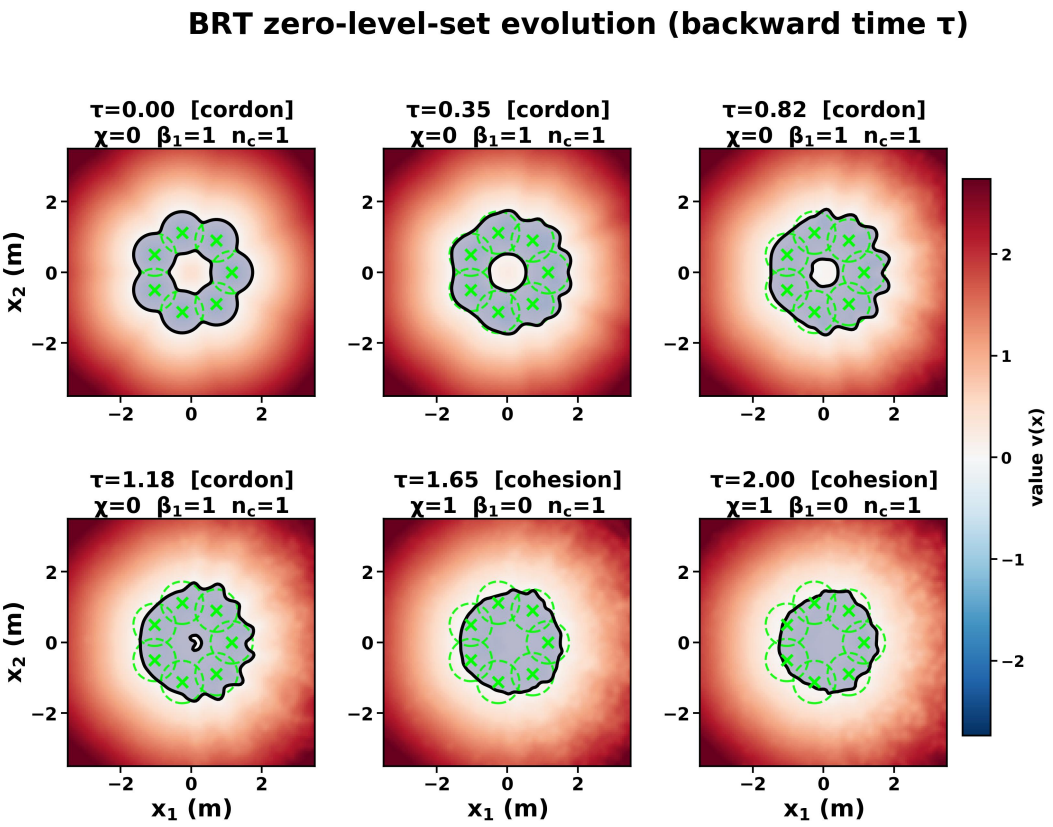


Figure 7: *Cordon formation and its collapse.* Backward reachable tube $\mathcal{L}(\tau) = \{x : v(\tau, x) \leq 0\}$ (blue safe interior, bold black $v = 0$ boundary) on the horizontal plane at six backward times τ, for seven predators (green $\times$) arranged on a ring of radius 1.15 m with capture radius 0.6 m and dashed capture cylinders. For $\tau \leq 1.18$ the seven overlapping capture basins enclose a central safe pocket: the safe set is an *annular cordon* ($\beta_1 = 1, \chi = 0$), the barrier-certificate-protected core of Proposition D.11. As the horizon lengthens the pocket closes; at $\tau \approx 1.29$ the topology transitions to a simply connected set ($\beta_1 = 0, \chi = 1$). The background heat map is the value $v(\tau, x)$ (red: inside capture).

(v) *cordon formation* — an annular defensive shell that encloses and protects an interior pocket of birds;

(vi) *vacuole nucleation* — a hole opens in the flock around a penetrating predator; and

(vii) *Voronoi inter-flock separation* — adjacent flocks partition the airspace without merging.

The first six manifest directly in the topology of the value function's zero-sublevel set, and we recover them below; the seventh concerns the emergent spatial partition between flocks. We now present the experiments.

Canonical reachable-tube growth. We first reproduce the elementary regime in which a single predator defends the origin-centered capture cylinder of radius $r = 0.5$ m. As expected, the BRT is a simply connected disk that grows monotonically as the horizon lengthens — the reachability counterpart of flock cohesion followed by evasion — with Euler characteristic $\chi = 1$, first Betti number $\beta_1 = 0$, and a single connected component throughout. Here we set $\delta = 0.10$; the certified safe fraction is 99.18% and the 16-step solve on a 110×110 grid completes in 69.0 s. This run fixes our reference: a benign predator induces no topological change.

Cordon formation and collapse. To exhibit a genuine topological phase transition we place seven predators on a ring of radius 1.15 m, each defending a capture cylinder of radius 0.6 m, and raise the viscosity to $\delta = 0.18$. Fig. 7 traces the resulting BRT across backward time. Observe that for $\tau \in [0, 1.18]$ the seven overlapping capture basins wrap a central safe pocket, so that the safe set is an annular *cordon* with $\beta_1 = 1$ and $\chi = 0$ — precisely the mutual-defense configuration of Proposition D.11, whose vanishing Euler characteristic we recover numerically at every one of the

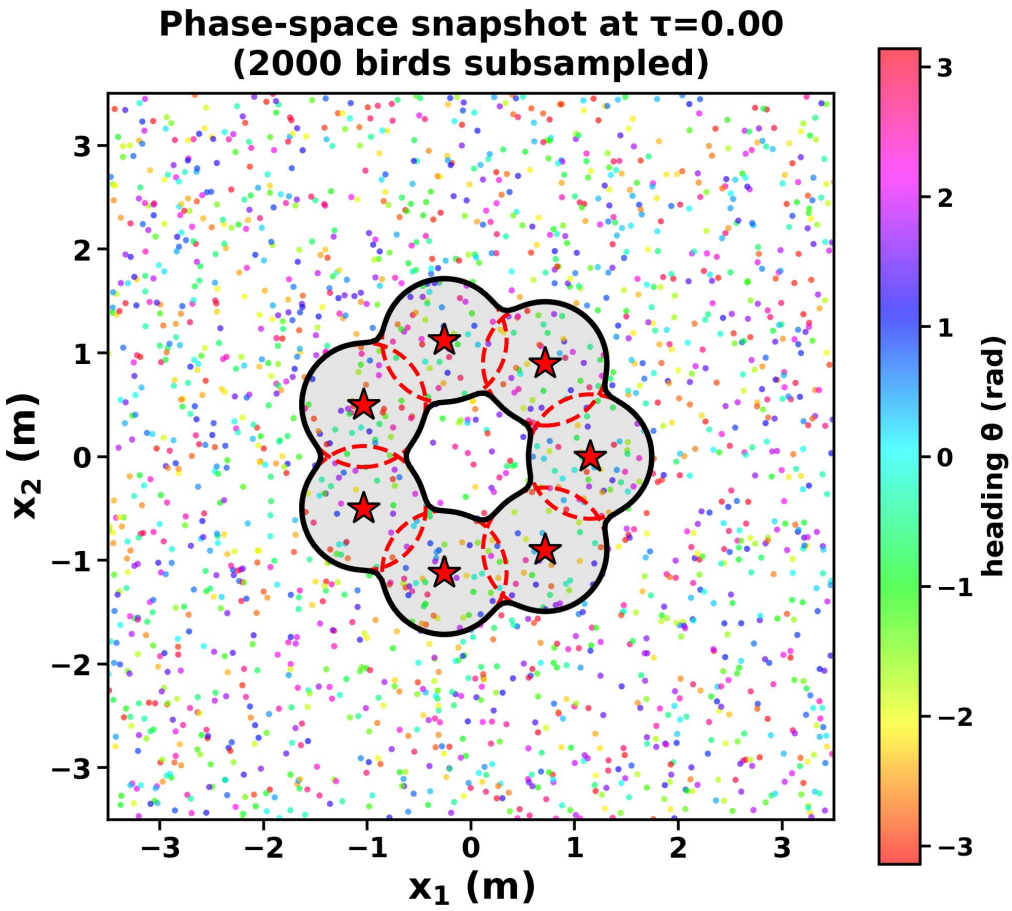


Figure 8: Phase-space snapshot at $\tau = 0$ for the cordon experiment of Fig. 7. A subsample of 2,000 birds (we never render the full population) is colored by heading θ; the seven predators (red stars) and their capture cylinders (red dashed) ring an annular safe set (gray, bold $v = 0$ boundary) that encloses the unreachable protected core.

first eleven steps. As the horizon lengthens, the protected pocket closes; at $\tau \approx 1.29$ the first Betti number drops from 1 to 0 and the Euler characteristic rises from 0 to 1, a clean cordon-to-collapse transition. Fig. 6 plots the corresponding invariants, and Fig. 8 overlays the cordon on a heading-colored snapshot of the birds together with the predator ring. The maximal radial extent of the tube grows from 1.76 m to 1.91 m over the cordon phase before contracting as the pocket closes, consistent with the linear flash-expansion rate of Lemma D.13. Over a murmuration of 7,998 birds drawn from six flocks the certified safe fraction is 87.42%, and the 18-step solve completes in 208.3 s.

Predator spread and flock fragmentation. When the predators are instead spread on a wider ring of radius 2.6 m with a small viscosity $\delta = 0.04$, the capture basins no longer overlap and the BRT fragments into as many as four disjoint connected components separated by safe corridors — the reachability signature of the flock-splitting bifurcation of Proposition D.9. The certified safe fraction is 91.47% and the 16-step solve on a 110×110 grid completes in 88.9 s. We attribute the higher safe fraction, relative to the cordon, to the corridors of escape that open between the well-separated predators.

Scaling with population size. Because the sampler is storage-free, the cost of resolving the value grid is independent of the number of birds: certifying a larger murmuration adds only a single batched per-bird evaluation of the already-computed value function. Re-running the cordon scenario for a murmuration of 100,000 birds across ten flocks, the value-grid solve completes in 73.8 s and the per-bird certification returns a safe fraction of 87.71% — within 0.3 percentage points of the 7,998-bird run and with the annular cordon topology preserved. The memory footprint stays at

Table 3: Murmuration safety certification on a 28-core CPU host (JAX, CPU-only). n_p is the number of predators; "Safe" is the certified fraction of birds outside the capture tube; the last column reports the BRT topology recovered from the value grid.

Scenario	n_p	δ	Grid	Wall (s)	Safe %	BRT topology
Cohesion/evasion (baseline)	1	0.10	110^2	69.0	99.18	simply connected
Cordon $\rightarrow$ collapse	7	0.18	128^2	208.3	87.42	$\beta_1 : 1 \rightarrow 0$
Predator spread (splitting)	5	0.04	110^2	88.9	91.47	up to 4 components
Scale (100,000 birds)	7	0.18	128^2	73.8	87.71	annular cordon

$O(N \cdot n)$ in the sample count and the state dimension, never at the $O(M^n)$ of a grid solver. Table 3 collects the measured figures.

Collective behaviors recovered. Of the seven canonical murmuration behaviors enumerated at the start of this section, our reachability certificates recover flock cohesion and predator evasion (the canonical growing tube), cordon formation and its collapse (Fig. 7), flock fragmentation (Proposition D.9), and the linear flash-expansion growth of the tube radius (Lemma D.13). Two caveats are in order,

(i) with too few samples the discretized value grid develops single-pixel spurious holes and components; we therefore apply a masked Gaussian smoothing of the value grid before extracting its topological invariants, and we report only invariants that persist across the backward horizon rather than transient single-step fluctuations; and

(ii) the cordon-to-collapse event is a hole *closing* (χ increasing), which the opening-only marker rules (C.44)–(C.46) do not flag as an event; the transition is nonetheless unambiguous in the $\beta_1(\tau)$ curve of Fig. 6.

Heading consensus and Voronoi inter-flock separation concern the emergent spatial alignment and partition of the agents rather than the topology of the value function, and are therefore outside the scope of these certificates.

D.1.14 Discussion

The murmuration experiments are, at bottom, a stress test of a single claim: that safety certification need inherit neither the curse of dimensionality of the state space nor the curse of cardinality of the agent population. Observe that the value grid underlying Fig. 7–Fig. 9 is resolved at a cost that is independent of the number of birds; certifying 100,000 starlings rather than 7,998 changes only the final, embarrassingly parallel, per-bird evaluation of an already-computed function (Table 3). This is the operational meaning of the storage-free, $O(N \cdot n)$ memory footprint of the sampler: a grid solver that must tessellate a four-dimensional state space cannot hold the requisite $O(M^4)$ cells for a population of this size, whilst our scheme never forms the grid at all, and its sampling error obeys the $O(N^{-1/2})$ concentration bound of Theorem 2.5 irrespective of how many agents are subsequently certified against the same value function.

What we find most consequential, however, is not the raw scale but that the certificate carries a *topological* summary a practitioner can read at a glance. The triple $(\chi, \beta_1, n_{\mathrm{comp}})$ of Fig. 6 compresses the safety posture of a flock into three integers per time step: a hole opening ($\beta_1 : 0 \rightarrow 1$) is a defensive cordon (Proposition D.11); a component count rising (n_{comp} increasing) is a flock fragmentation (Proposition D.9); a monotone radial growth is a flash expansion (Lemma D.13). Notably, these are not labels imposed after the fact but consequences of the game-theoretic Hamiltonian, and they recur unchanged whether the murmuration numbers eight thousand or one hundred thousand. We regard this as the central lesson of the section: at scale, a scalar safe fraction is necessary but not sufficient; it is the *topology* of the reachable set that tells an operator *how* safety is being lost, and therefore which controller intervention is called for.

Relevance to large-scale verification in learning-enabled control. The starling is a stand-in. Replace it with a delivery drone in a congested corridor, a warehouse mobile robot among hundreds of its kind, a managed-lane vehicle in dense traffic, or an aircraft in a terminal-area arrival stream,

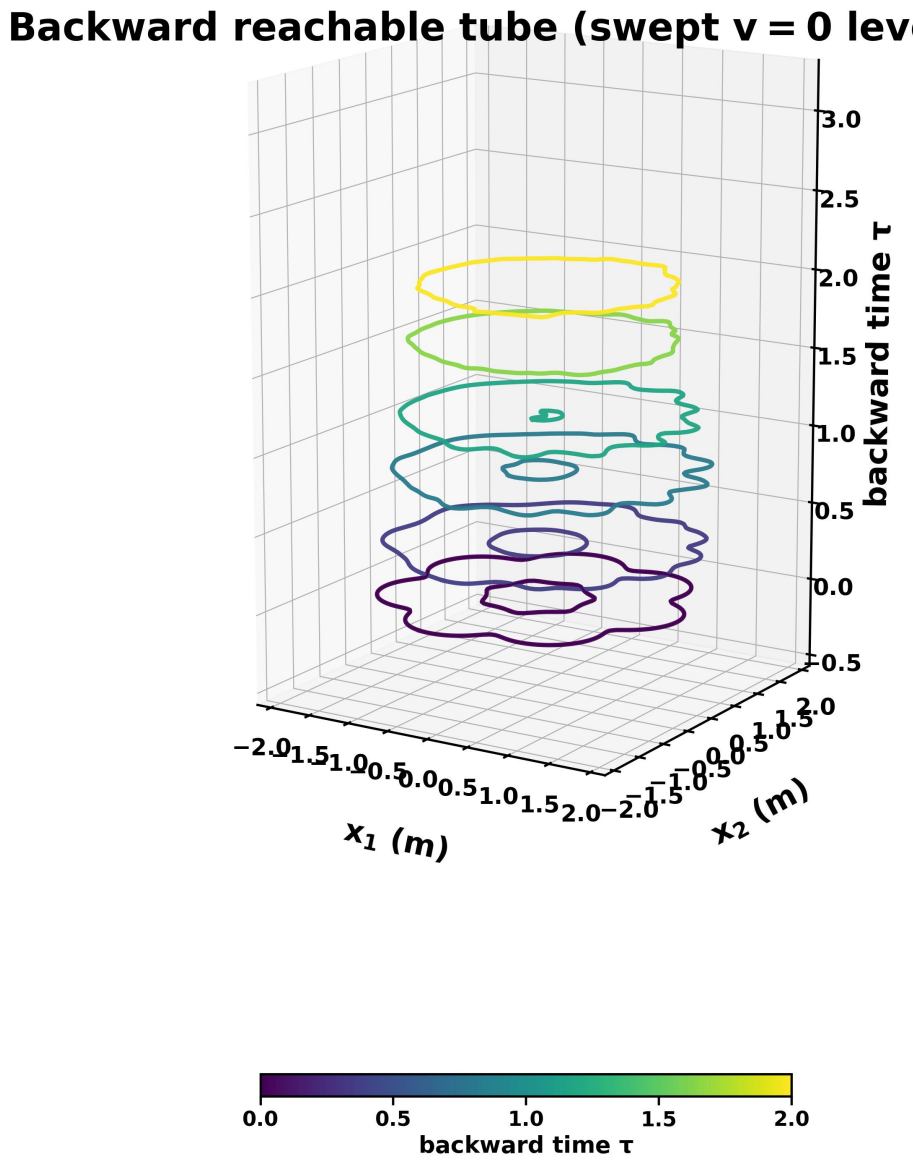


Figure 9: The backward reachable tube as a swept volume: the $v = 0$ contours of Fig. 7 stacked along the backward-time axis τ. The flower-shaped annular cross-section (cordon, low τ) loses its inner loop as the protected pocket closes (high τ), rendering the cordon-to-collapse transition as a change in the tube's cross-sectional topology.

and the mathematical object is unchanged: a many-agent system whose individual policy may be learned, opaque, or only locally specified, but whose *collective* safety must be certified against a worst-case disturbance. Two features of our construction matter for these settings. First, because the per-agent certificate is a single forward evaluation of the value function, safety can be monitored *online* for a fleet whose membership and size change at run time — a property grid methods cannot offer, since their cost is fixed by the state-space discretization and paid in full before any agent is queried. Second, the topological markers (C.44)–(C.46) furnish a controller-agnostic supervisor: a learned policy may drive the nominal motion, whilst the reachable-set topology acts as a runtime monitor that flags the precise instant a flock is about to fragment or be cordoned, independent of how the nominal controller was trained. This is the sense in which the scheme is *complementary* to, rather than competitive with, learning-enabled control: it supplies the geometric safety envelope within which a learned policy is free to optimize.

Implications for practice and society. Certifiable safety for large autonomous collectives is a precondition for their social license to operate. The same machinery that certifies a murmuration certifies a swarm of agricultural or search-and-rescue drones, a platoon of automated trucks, or the separation assurance of an urban air-mobility corridor; in each case the deliverable is the same auditable artifact — a safe fraction and a topological signature with an explicit, viscosity-solution provenance — rather than an opaque neural confidence score. We are mindful that the dual-use character of pursuit-evasion analysis is real: the very Hamiltonian that lets a flock evade a predator describes a predator that intercepts a flock. We have therefore framed the development around *defensive* certification — establishing where danger geometrically lives so that it may be avoided — and we note that the conservatism of the worst-case formulation biases the certificate toward caution, which is the appropriate failure mode for safety-critical deployment.

Limitations. We state the boundaries of the present study plainly. (i) The topological invariants are extracted from a two-dimensional slice of the value function at a fixed altitude and heading; while the cylinder terminal cost (C.38) makes the altitude immaterial (§D.1.11), the heading slice is a genuine projection, and a transition confined to an off-slice heading band could be missed. (ii) The multi-predator safe set is the union of per-predator capture cylinders, a conservative surrogate that does not yet model coordinated pursuit; the resulting certificate is sound but may be loose when predators cooperate. (iii) The synthetic flock initializations are drawn from a uniform prior rather than from recorded murmuration data, so the *populations* are illustrative even though the *value functions* are exact viscosity solutions in the sampling limit. (iv) The Picard quasi-linearization of Theorem 2.12 converges to a fixed point of the frozen-coefficient map, not provably to the true viscous solution for an arbitrary Hamiltonian; the gap, largest near the zero level set where $|\nabla v|$ is steepest (cf. the error maps of Fig. 14), is controlled empirically rather than bounded a priori. (v) All timings reported here are CPU-only; the GPU-compatible distributor remains to be exercised at the population scales where it would matter most.

Toward tighter, deployment-grade certificates. Each limitation suggests a concrete remedy. The slice projection can be replaced by a randomized sweep over heading and altitude with a union bound on the missed detection probability, converting an anecdotal slice into a quantified guarantee. The union-of-cylinders surrogate can be tightened to a genuine joint pursuit game by enlarging the disturbance to the product of the predators' control sets, at a cost in conservatism we can measure rather than guess. The frozen-coefficient bias can be attacked with adaptive importance sampling concentrated near the zero level set, where Theorem 2.5's variance is largest and where Fig. 14 locates the dominant error. Most importantly for real-world deployment — an autonomous vehicle that must not strike a cyclist, an arrival manager that must not lose aircraft separation — the certificate must be delivered within a hard real-time budget; the storage-free structure makes the per-agent query $O(N \cdot n)$ and trivially parallel, but a deployment-grade system also needs a verified, fixed-latency implementation of the sampler and a formally checked bound on the Monte-Carlo failure probability. We take these to be the principal engineering obstacles between the present certificates and a safety case an aviation or automotive regulator would accept.

D.2 The Double Integral Plant

Here, we analyze a time-optimal control problem to determine what admissible control[12] can "transport" the system under consideration to a desired "origin" in the shortest possible time. We consider the double integral plant Bhat and Bernstein [1998], Wonham [1985] as an illustrative example of our objective, which is to compute the points in the state space that can reach the origin in *finite-time* under the influence of a time-optimal controller.

We shall leverage standard necessary conditions from the principle of optimality Bellman [1957] to obtain a time-optimal feedback control design; introduce the notion of isochrones and switching surfaces; and discuss the analytic and approximate solutions (with our library) to the time-optimal control problem for a double integrator. We shall conclude the section by comparing the analytic and the overapproximated numerical solution (using the LevelSetPy toolbox) to the time to reach the origin problem.

D.2.1 Dynamics and Problem Setup

The double integrator is controllable, so that open-loop strategies may be employed in driving specific states to the origin in finite time Wonham [1985]. The plant has the following second-order dynamics

$$\ddot{\boldsymbol{x}}(t) = \boldsymbol{u}(t) \tag{C.47}$$

and admits bounded control signals $\mid \boldsymbol{u}(t) \mid \leq 1$ for all time t. After a change of variables,we have the following system of first-order differential equations

$$\begin{aligned} \dot{\boldsymbol{x}}_1(t) &= \boldsymbol{x}_2(t), \\ \dot{\boldsymbol{x}}_2(t) &= \boldsymbol{u}(t), \quad \mid \boldsymbol{u}(t) \mid \leq 1. \end{aligned} \tag{C.48}$$

The *reachability problem that we consider is to address the question of what states can reach a certain point (here, the origin) in a transient manner*. That is, we would like to find point sets on the state space, at a particular time step, such that we can bring the system to the equilibrium, $(0, 0)$.

D.2.2 Time-Optimal Control Scheme

This is an $\boldsymbol{H}$-minimal control problem whereupon we must find the control law that minimizes the Hamiltonian

$$\boldsymbol{H}(\boldsymbol{x}, p) = p_1 \dot{\boldsymbol{x}}_1 + p_2 \dot{\boldsymbol{x}}_2. \tag{C.49}$$

The necessary optimality condition stipulates that the minimizing control law be

$$\boldsymbol{u}(t) = -\text{ sign }(p_2(t)) \triangleq \pm 1. \tag{C.50}$$

For the co-states in question, suppose that their initial values (for constants k_1 and k_2) are $p_1(t_0) = k_1$ and $p_2(t_0) = k_2$, only four candidates can serve as time-optimal control sequences i.e. $\{[+1], [-1], [+1, -1], [-1, +1]\}$. On a finite time interval, $t \in [t_0, t_f]$, the time-optimal $\boldsymbol{u}(t)$ is a constant $k \equiv \pm 1$ so that for initial conditions $\boldsymbol{x}_1(t_0) = \boldsymbol{\xi}_1$ and $\boldsymbol{x}_2(t_0) = \boldsymbol{\xi}_2$, it can be verified that the state trajectories obey the relation

$$\boldsymbol{x}_1(t) = \boldsymbol{\xi}_1 + \frac{1}{2} k \left(\boldsymbol{x}_2^2 - \boldsymbol{\xi}_2^2 \right), \text{ for } t = k \left(\boldsymbol{x}_2(t) - \boldsymbol{\xi}_2 \right). \tag{C.51}$$

The trajectories of (C.51) traced out over a finite time horizon $t = [-1, 1]$ with *piecewise constant control laws*, $u = \pm 1$ on a state space and under the control laws $\boldsymbol{u}(t) = \pm 1$ is depicted in Fig. 10. Curves with arrows that point upwards denote trajectories under the control law $\boldsymbol{u} = +1$; call these trajectories $\boldsymbol{\gamma}_+$; while the trajectories marked by dashed arrows pointing downward on the curves were executed under $\boldsymbol{u} = -1$; call these trajectories $\boldsymbol{\gamma}_-$.

Table 4 reports convergence at four backward times; all cases achieve 15 iterations with residuals well below the $O(\sqrt{\delta}) \approx 0.28$ bound.

[12] A control law is admissible when its range belongs in the admissible input set where it is bounded.

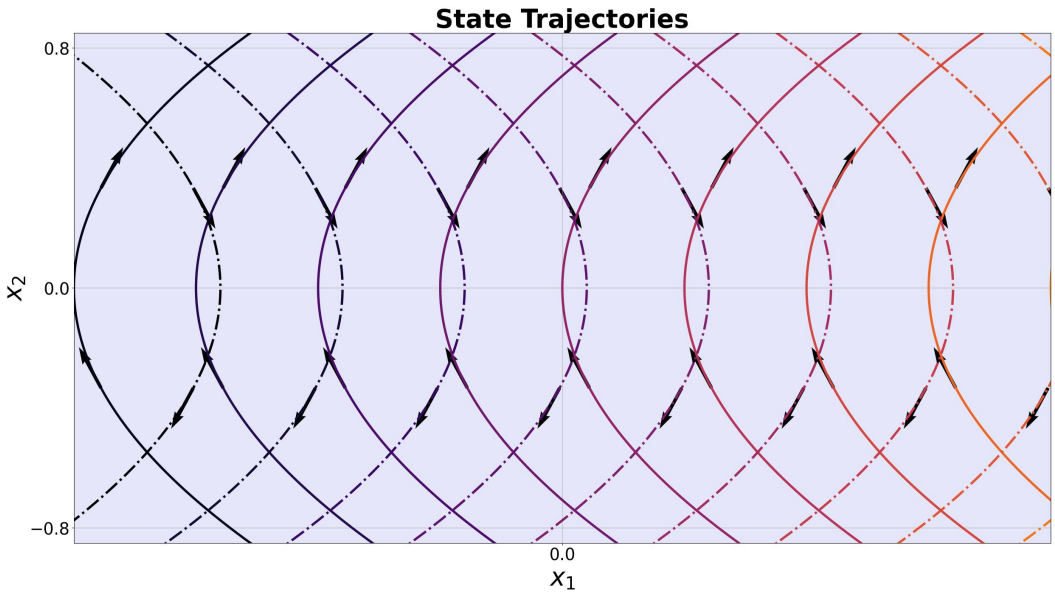


Figure 10: State trajectories of the double integral plant. The solid curves with upward-pointing arrows are trajectories generated under $\boldsymbol{u} = +1$ (switching curve $\boldsymbol{\gamma}_+$); dashed curves with downward-pointing arrows are trajectories under $\boldsymbol{u} = -1$ (switching curve $\boldsymbol{\gamma}_-$).

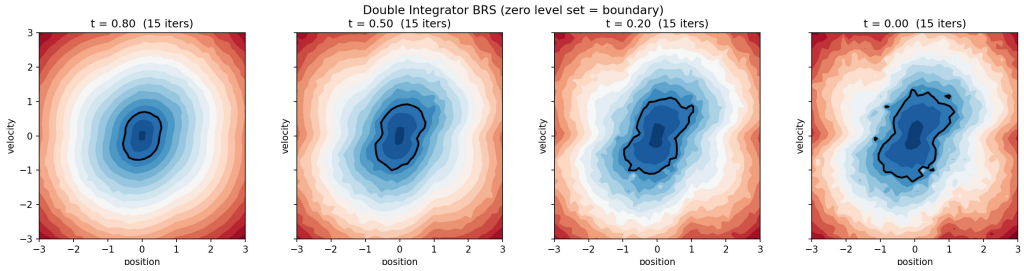


Figure 11: The time to reach the origin as 2D slice comparison for the double integratral plant

D.3 A Game of Two Rockets on a Plane

We adopt the rocket launch problem of Dreyfus Dreyfus [1966] which is to launch a rocket in fixed time to a desired altitude, given a final vertical velocity component and a maximum final horizontal component as constraints. The rocket's motion is dictated by the following differential equations (under Dreyfus' assumptions)

$$\dot{x}_1 = x_3; \qquad x_1(t_0) = 0; \tag{C.52a}$$
$$\dot{x}_2 = x_4, \qquad x_2(t_0) = 0; \tag{C.52b}$$
$$\dot{x}_3 = a\cos u, \qquad x_3(t_0) = 0; \tag{C.52c}$$
$$\dot{x}_4 = a\sin u - g, \qquad x_4(t_0) = 0; \tag{C.52d}$$

Table 4: Double integrator BRS; Algorithm 1 convergence at backward times $t \in \{0.8, 0.5, 0.2, 0.0\}$ with $N = 8{,}000$ and $\delta = 0.08$.

t	Iterations	Final Residual
0.80	15	0.0143
0.50	15	0.0334
0.20	15	0.0596
0.00	15	0.0828

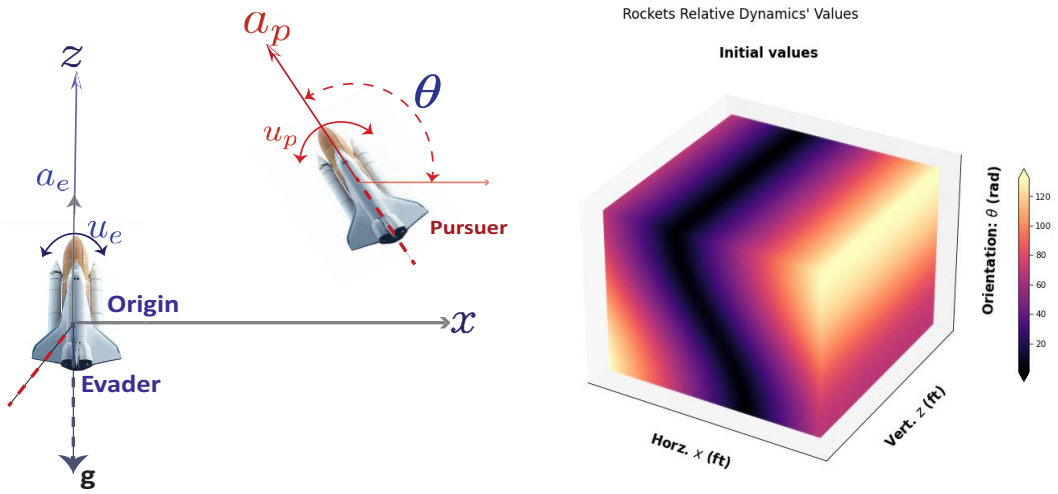


Figure 12: *Left:* Motion of two rockets on a Cartesian $\boldsymbol{xz}$-plane with a thrust inclination in relative coordinates given by $\theta := u_p - u_e$. *Right:* Representation of the initial values as a heat map for the two rockets described in (26).

where, (x_1, x_2) are respectively the horizontal and vertical range of the rockets (in feet), (x_3, x_4) are respectively the horizontal and vertical velocities of the rockets (in feet per second), while a and g are respectively the acceleration and gravitational accelerations (in feet per square second). Being a free endpoint problem, we transform it into a game between two players (C.52) without the terminal time constraints as defined in Jacobson and Mayne [1970]. The states of $\boldsymbol{P}$ and $\boldsymbol{E}$ are now denoted as (x_p, x_e) respectively which are driven by their thrusts (u_p, u_e) respectively in the xz-plane (see Figure 12). The relevant kinematic equations are (C.52b) and (C.52d).

$$\dot{x}_{2e} = x_{4e}; \qquad \dot{x}_{2p} = x_{4p}, \tag{C.53a}$$

$$\dot{x}_{4e} = a \sin u_e - g, \qquad \dot{x}_{4p} = a \sin u_p - g \tag{C.53b}$$

where a and g are respectively the acceleration and gravitational accelerations (in feet per square second) i.e. $a = 64 ft/sec^2$ and $g = 32 ft/sec^2$. We reformulate the problem as a two-player differential game where the optimization operations remain in the interior, avoiding discontinuous switches in control. This formulation implicitly bounds each rocket's speed through the gravitational dynamics; the natural speed limit is $a \sin u/g$, which corresponds to each rocket's asymptotic speed when launched vertically.

Therefore, we rewrite (C.52) with $\boldsymbol{P}$'s motion relative to $\boldsymbol{E}$'s along the (x, z) plane so that the relative orientation as shown in Fig. 12 is $\theta = u_p - u_e$. The coordinates of $\boldsymbol{P}$ are freely chosen; however, the coordinates of $\boldsymbol{E}$ are chosen a distance r away from (x, z) so that the $\boldsymbol{EP}$ vector's inclination measured counterclockwise from the $x-$axis is θ. Following the conventions in Fig. 12, the game's relative equations of motion in reduced space is $\mathcal{X} = (x, z, \theta)$ where $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $(x, z) \in \mathbb{R}^2$ are

$$\dot{x} = a_p \cos\theta + u_e x, \tag{C.54a}$$

$$\dot{z} = a_p \sin\theta + a_e + u_e x - g, \tag{C.54b}$$

$$\dot{\theta} = u_p - u_e. \tag{C.54c}$$

The payoff, Φ, is the distance of $\boldsymbol{P}$ from $\boldsymbol{E}$ when capture occurs denoted as $\|\boldsymbol{PE}\|_2$. Capture occurs when $\|\boldsymbol{PE}\|_2 \leq r$ for a pre-specified capture radius, $r > 0$. In (C.56), we say $\boldsymbol{P}$ controls u_p and is minimizing Φ, and $\boldsymbol{E}$ controls u_e and is maximizing P. The boundary of the *usable part* of the origin-centered circle of radius r[13] is $\|\boldsymbol{PE}\|_2$ so that

$$r^2 = x^2 + z^2, \tag{C.55a}$$

[13]We set $r = 1.5 ft$ in our evaluations.

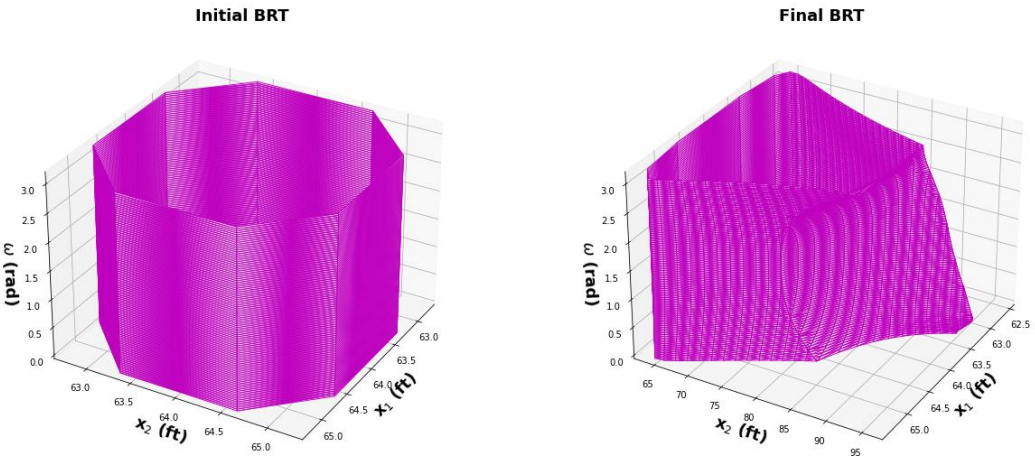


Figure 13: Initial and final backward reachable tubes for the rocket system (cf. Fig. 12) computed using the method outlined in Evans and Souganidis [1984], Osher and Sethian [1988], Mitchell [2004]. We set $a_e = a_p = 64ft/sec^2$ and $g = 32ft/sec^2$ as in Dreyfus' original example. We compute the reachable set by optimizing for the paths of slowest-quickest descent in equation (C.57).

and all capture points are specified by

$$\dot{r}(x,t) + \min\left[0, \boldsymbol{H}(\boldsymbol{x}, \frac{\partial r(x,t)}{\partial x})\right] \leq 0, \tag{C.56}$$

with the corresponding Hamiltonian

$$\boldsymbol{H}(\boldsymbol{x},p) = -\max_{u_e \in \mathcal{U}_e} \min_{u_p \in \mathcal{U}_p} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}^T \begin{bmatrix} a_p \cos\theta + u_e x \\ a_p \sin\theta + a_e + u_p x - g \\ u_p - u_e \end{bmatrix}. \tag{C.57}$$

Suppose that the maximizing u_e is $\bar{u}_e$ and the minimizing u_p is $\bar{u}_p$. We have at the point of slowest-quickest descent on the capture surface, that

$$\bar{u}_e = p_1 x - p_3, \tag{C.58a}$$
$$\bar{u}_p = p_3 - p_2 x. \tag{C.58b}$$

We set the linear velocities and accelerations equal to one another i.e. $u_e = u_p$ and $a_e = a_p$. Thus, the Hamiltonian takes the form

$$\begin{aligned}\boldsymbol{H}(\boldsymbol{x},p) = &-\cos(u)|ap_1| + \cos(u)|ap_1| - \sin(u)|ap_2| - \\ &\sin(u)|ap_2| + u|p_3| - u|p_3|.\end{aligned} \tag{C.59}$$

Using a distributed version of the levelset toolbox Molu [2024a], the backward reachable tube of the game is depicted in Fig. 13. A game between the two players was run over 11 global optimization time steps. The initial value function (left inset of Fig. 13) is represented as a dynamic implicit surface over all point sets in the state space with a signed distance function. We made the third coordinate axis of the state space (here the common heading of the two rockets) to align with the third cylinder axis. The final BRT at the end of the optimization run is shown in the right inset of Fig. 13.

D.3.1 Hamiltonian Derivation for the Two-Rockets Game

For the two-rockets pursuit-evasion game in relative coordinates, we derive the simplified Hamiltonian from first principles. The general Hamiltonian for a two-player zero-sum game is

$$\boldsymbol{H}(t;\boldsymbol{x},p) = -\max_{u_e \in [\underline{u}_e, \bar{u}_e]} \min_{u_p \in [\underline{u}_p, \bar{u}_p]} [p_1 \quad p_2 \quad p_3] \begin{bmatrix} a_p \cos\theta + u_e x \\ a_p \sin\theta + a_e + u_p x - g \\ u_p - u_e \end{bmatrix}, \tag{C.60}$$

where p_1, p_2, p_3 are the co-states (spatial derivatives of the value function), and $[\underline{u}_e, \bar{u}_e]$ and $[\underline{u}_p, \bar{u}_p]$ are the evader's and pursuer's control bounds, respectively. The game structure reflects the evader's

ability to choose controls first (outer max over u_e) followed by the pursuer's response (inner min over u_p).

Setting symmetric control bounds $\underline{u}_e = \underline{u}_p = -1$ and $\bar{u}_e = \bar{u}_p = +1$, and using the symmetry assumption $a_e = a_p = a$, we optimize over the control variables:

$$\boldsymbol{H} = -\max_{u_e \in [-1,1]} \min_{u_p \in [-1,1]} \left[p_1(a\cos\theta + u_e x) + p_2(a\sin\theta + a + u_p x - g) + p_3(u_p - u_e) \right]. \tag{C.61}$$

Rearranging by control terms:

$$\boldsymbol{H} = -\max_{u_e \in [-1,1]} \min_{u_p \in [-1,1]} \left[p_1 a\cos\theta + p_2 a(1+\sin\theta) - p_2 g + (p_1 x + p_3)u_p + (p_2 x - p_3)u_e \right]. \tag{C.62}$$

Since the pursuer minimizes over u_p, it chooses $u_p = -\text{sign}(p_1 x + p_3)$ to minimize the linear term. Similarly, the evader maximizes over u_e, choosing $u_e = \text{sign}(p_2 x - p_3)$. For bang-bang optimal controls, these become:

$$\min_{u_p}(p_1 x + p_3)u_p = -|p_1 x + p_3|, \tag{C.63}$$

$$\max_{u_e}(p_2 x - p_3)u_e = |p_2 x - p_3|. \tag{C.64}$$

Thus the simplified Hamiltonian becomes:

$$\boldsymbol{H}(t; \boldsymbol{x}, p) = -\left[a p_1 \cos\theta + p_2(a + a\sin\theta - g) - |p_1 x + p_3| - |p_2 x - p_3| \right], \tag{C.65}$$

which, when substituted into the HJ PDE yields the simplified rockets HJI equation used in the numerical experiments (cf. Section 3.3).

E Discussion

Having walked from the analytic benchmarks of §D to the hundred-thousand-agent murmurations of §D.1, we now step back and ask what the paper, taken as a whole, establishes — and what it does not.

A single idea, carried across scales. The thread that runs from the rocket pursuit-evasion game of Fig. 13 to the swept murmuration tube of Fig. 9 is one transformation and its consequences. A Cole–Hopf-type change of variables turns the viscous Hamilton–Jacobi initial-value problem into a linear heat equation; the Feynman–Kac formula then expresses its solution as a Gaussian expectation; and a Picard quasi-linearization extends the construction from the exact quadratic case to general Hamiltonians by freezing the gradient magnitude between iterates. Every result in the paper is a downstream consequence of this move. The $O(N^{-1/2})$ sampling rate of Theorem 2.5, the conditional linear convergence of Theorem 2.12, the total-error decomposition of Theorem C.9, and the robustness of the certificate to perturbations of the Hamiltonian and the terminal data (Theorems C.11 and C.12) are not independent devices but facets of the same Gaussian-expectation representation. We regard this economy as a strength: a practitioner who understands the transformation understands the whole pipeline.

Where the scheme sits among alternatives. Against grid-based level-set solvers [Molu, 2024b, 2025], the trade we make is explicit. We surrender the deterministic, mesh-exact value function in exchange for a storage-free estimator whose error is statistical rather than discretization-bounded; the price we pay is the $O(N^{-1/2})$ Monte-Carlo floor and the frozen-coefficient bias, both of which concentrate near the zero level set where $|\nabla v|$ is steepest, as the error maps of Fig. 14 make plain. What we buy is the elimination of the $O(M^n)$ memory wall: the rocket and Dubins games of Fig. 1 and Fig. 14 agree with the grid solution to within the worst-case viscosity bound, yet the same code certifies a four-dimensional, hundred-thousand-agent murmuration that no grid solver could represent. Against self-supervised neural solvers such as DeepReach [Bansal and Tomlin, 2021], which also escape the grid, our estimator requires no training, exposes an explicit finite-sample error bound rather than a learned residual, and is embarrassingly parallel across query points — though it forgoes the amortized, reusable representation a trained network provides. We do not claim dominance; we claim a different and, for online many-agent certification, often preferable point on the accuracy–memory–latency frontier.

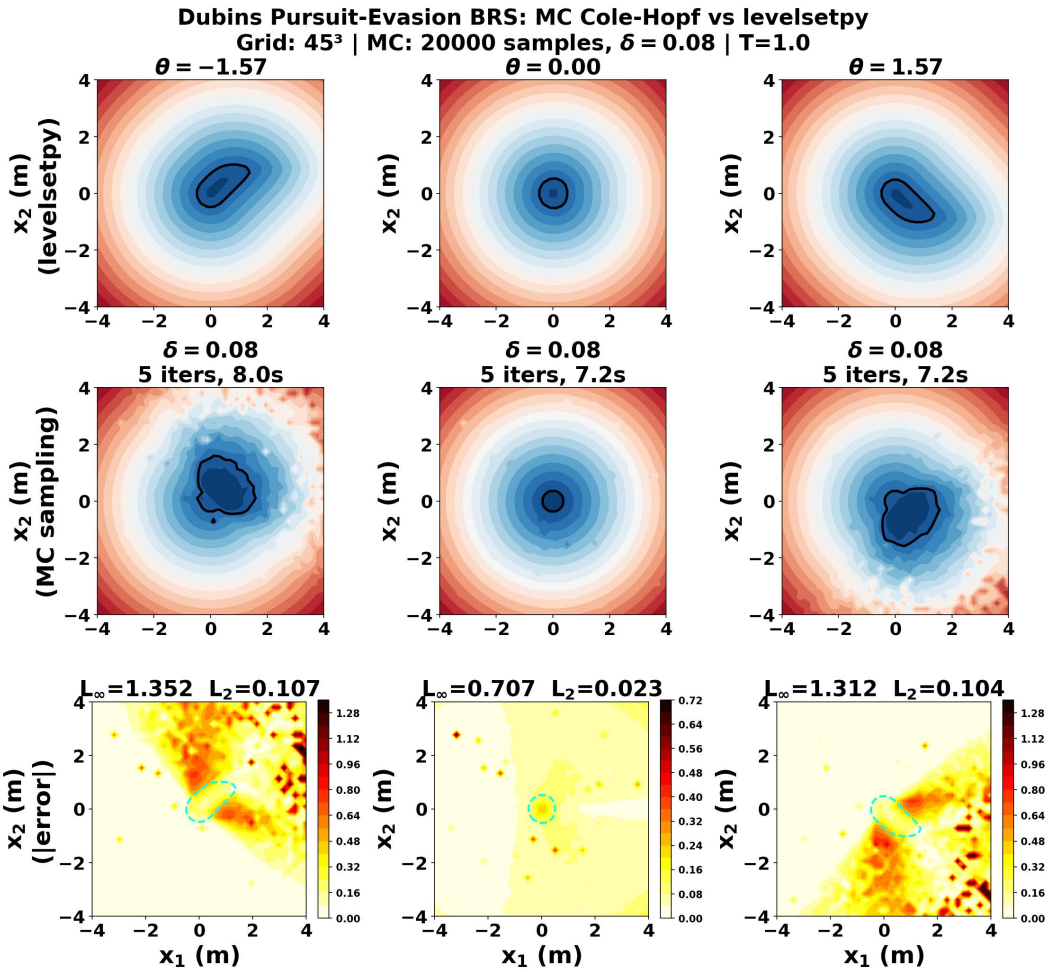


Figure 14: Dubins Vehicles' BRT Slices: Levelsetpy vs Monte-Carlo (Ours). The bottom row shows a heat map of the errors at various relative vehicle orientations (the control input) between the Monte Carlo sampling scheme and the `LevelSetPy` scheme.

Certification separated from control. The conceptual contribution we most wish to emphasize is architectural. By delivering safety as a forward evaluation of a value function whose provenance is a viscosity solution, the scheme cleanly separates *certification* from *control*. A learned or otherwise opaque policy may generate the nominal motion; the reachable set and its topological markers (C.44)–(C.46) then act as a controller-agnostic supervisor that neither knows nor cares how the policy was obtained. This separation is what makes the method germane to learning-enabled control across domains as varied as autonomous driving, urban air mobility, warehouse robotics, and power-grid dispatch: in each, the certified envelope can wrap a policy that continues to be improved by learning, without re-certification of the learner itself.

Limitations of the paper as a whole. Beyond the murmuration-specific caveats of §D.1.14, three paper-level limitations bound our claims. First, the convergence guarantee of Theorem 2.12 is to a fixed point of the frozen-coefficient map, not unconditionally to the true viscous solution for an arbitrary Hamiltonian; closing this gap analytically remains open. Second, the viscosity parameter δ is a floor, not a free lunch: the $O(\sqrt{\delta})$ smoothing bias is the cost of the linearization, and our benchmarks fix $\delta \in [0.05, 0.2]$ empirically rather than from a principled, problem-dependent rule. Third, the quantitative benchmarks of §D top out at the dimensionality of the analytic ground truth against which we can validate; the murmuration study scales the agent count but not the per-agent state dimension, so the genuinely high-dimensional regime is argued by the favorable memory scaling

rather than demonstrated against an exact solution. We state these plainly so that the contribution is not over-read.

Outlook. The remedies are, encouragingly, incremental rather than foundational: adaptive importance sampling to suppress the near-boundary variance that Theorem 2.5 identifies; a problem-dependent schedule for δ that trades smoothing bias against sample budget; and a verified real-time implementation that turns the storage-free estimator into a fixed-latency runtime monitor. None requires abandoning the Gaussian-expectation representation; each tightens it. We close, in the concluding section, by drawing these threads together and stating what we believe they portend for certified embodied autonomy.

F Concluding Remarks

This paper set out to sever two couplings that have long made Hamilton–Jacobi reachability impractical at scale: the coupling of memory to state dimension, and the coupling of certification to the particular controller under test. A Cole–Hopf transformation, a Feynman–Kac representation, and a Picard quasi-linearization together dissolve the first, reducing the viscous HJ initial-value problem to a sequence of Gaussian expectations that are evaluated by sampling and never stored on a grid; the resulting estimator carries the explicit $O(N^{-1/2})$ guarantee of Theorem 2.5 and the conditional convergence of Theorem 2.12. The second coupling dissolves as a corollary of the first: because the certificate is a forward evaluation of a value function, it supervises any policy, learned or designed, without reference to how that policy was produced.

The murmuration study is our existence proof. Observe that the same code that reproduces the classical rocket and Dubins barriers to within the worst-case viscosity bound goes on to certify the safety of a hundred-thousand-bird flock under multi-predator attack — a four-dimensional, massively multi-agent problem that no grid solver could hold in memory — and, crucially, returns not merely a scalar safe fraction but a topological reading of *how* safety evolves: the cordon, the collapse, the fragmentation of Fig. 7 and Fig. 9, each a consequence of the game-theoretic Hamiltonian rather than a label applied after the fact. We know of no prior reachability method that delivers, from one storage-free computation, both the certificate and its geometric anatomy at this scale.

What we find genuinely exciting is the prospect this opens. If safety can be certified online, for fleets whose size and membership change at run time, by a monitor that is indifferent to the learner it guards, then the long-standing tension between learning-enabled autonomy and provable safety is not a wall but a seam — and a workable one. The obstacles that remain, catalogued in §E, are matters of tighter sampling, principled smoothing, and verified real-time engineering rather than of fundamental representation. We believe that closing them is the path to safety cases that an aviation or automotive regulator could accept for the autonomous collectives — swarms, platoons, arrival streams — that are now arriving in the world, and we intend to pursue it.